

RENEET 2026



MAHA MARATHON

Complete

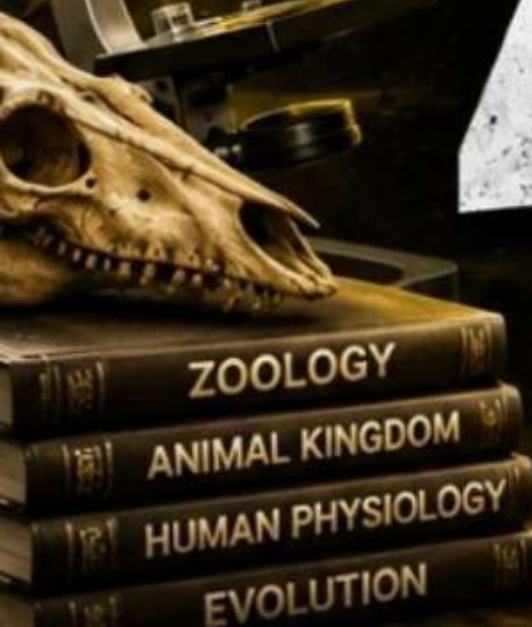
ZOOLOGY

Oneshot

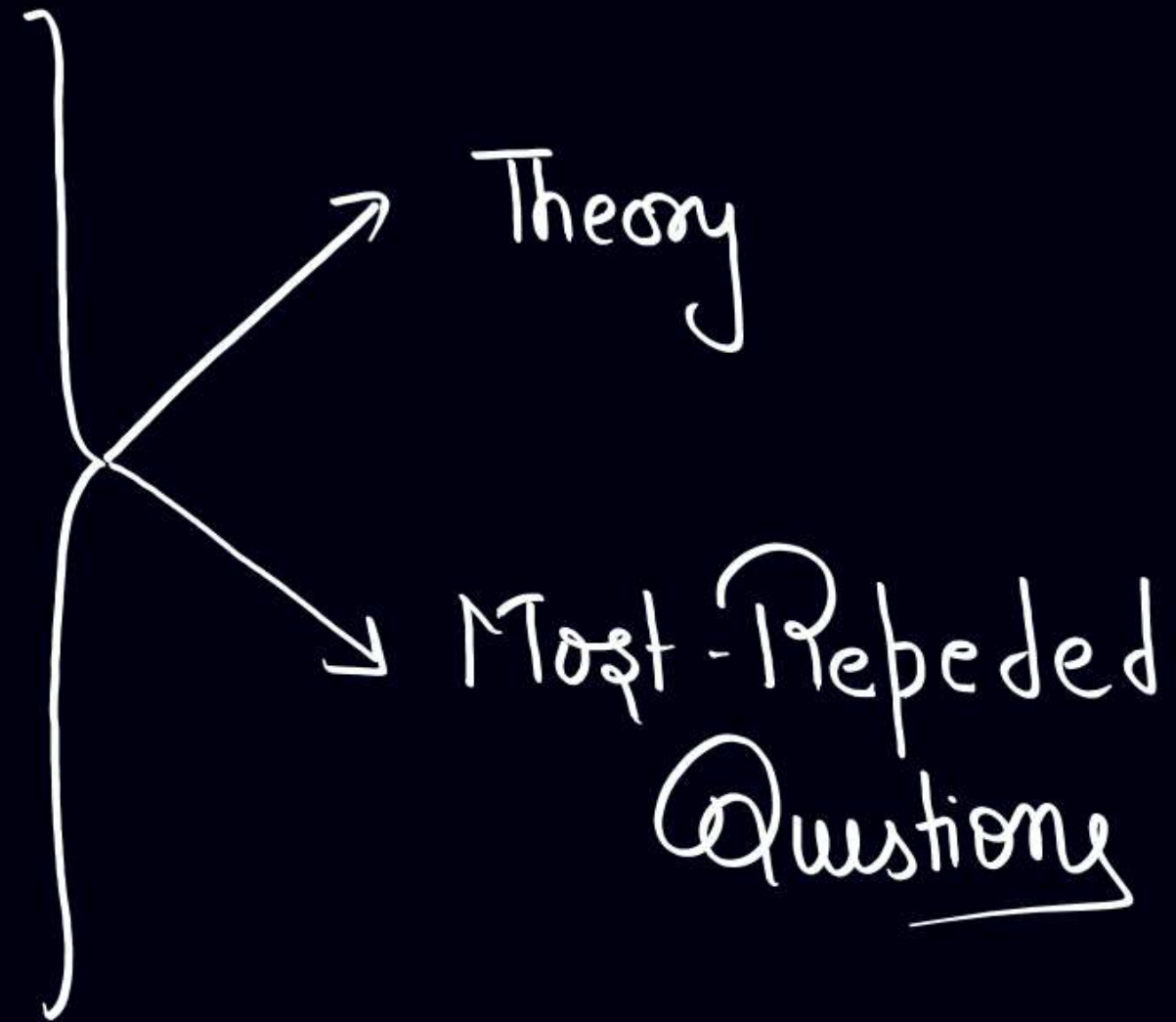


MD Sir

1
PART



Human Physiology
+
Biomolecules
+
Animal Kingdom
+
Frog



Breathing and Exchange of Gases

Respiratory organs

Mechanisms of breathing vary among different groups of animals depending mainly on their habitats and levels of organisation

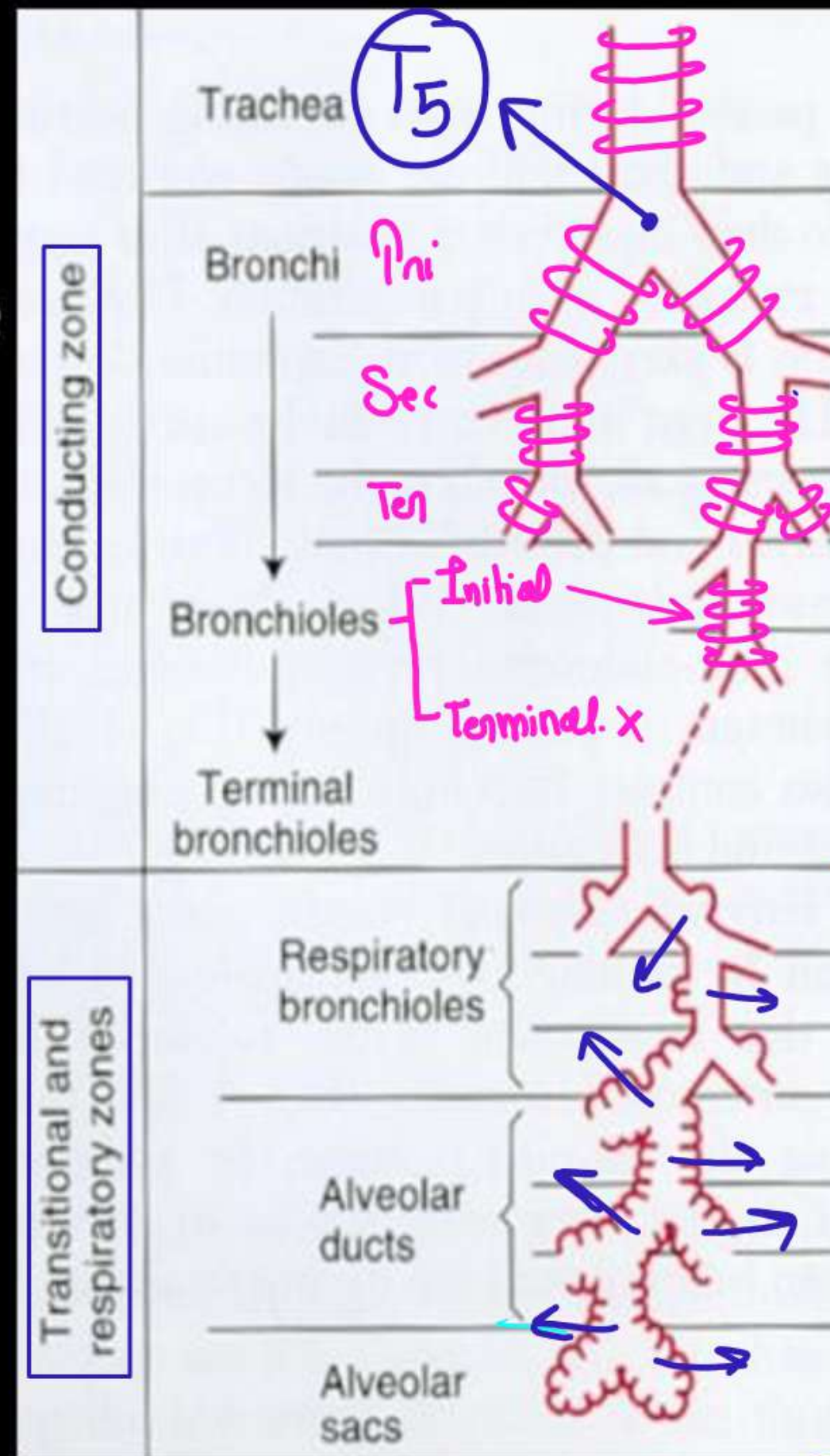
Examples of Animals	Respiratory organs
Sponges, coelenterates, flatworms	Body surface
Earthworms	Moist cuticle
Insects	Network of tubes (tracheal tubes)
Aquatic arthropods, molluscs, Fish	Special Vascularised structures called Gills (Branchial respiration)
Terrestrial forms (amphibians, reptiles, birds and mammals)	Vascularised bags called Lungs (Pulmonary respiration).
Amphibians like Frogs	Moist skin (cutaneous respiration)

Function of Conducting zone-

1. Conduction of air from Ex. Nostrils to Respiratory Zone
2. Clear conducting air from foreign particle
3. Humidification of air
4. Maintain temp. of conducting air according to body temp.

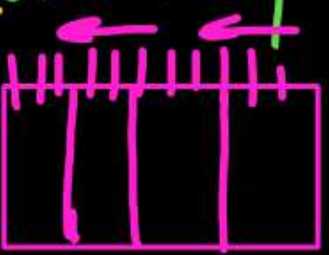
Respiratory Zone -

1. Diffusion of gases ✓



Epithelium in -
Conducting Zone →

- Simple Ciliated Columnar Epithelial Tissue.



Respiratory Zone

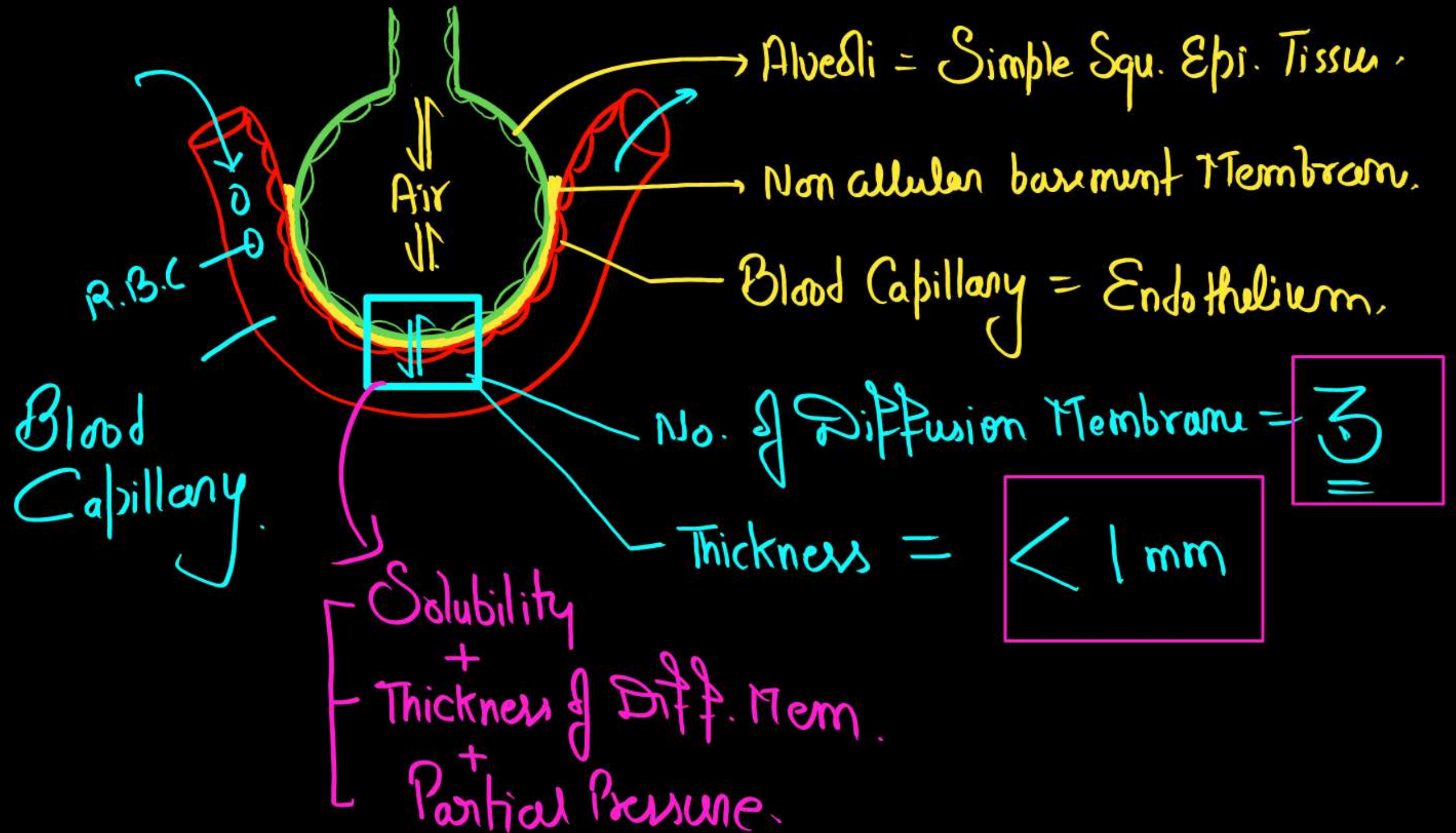
- Simple Squamous Epi. Tissue.



C-Shape Hyaline Cartilage
↓
Trachea to Initial bronchiole
Incomplete Dorsally

Structure of Alveoli

Thin, irregular-walled and vascularised bag-like structures



Inspiration (Active)

Contraction in Diaphragm
(↑ - Thoracic Volu. → A-P axis)



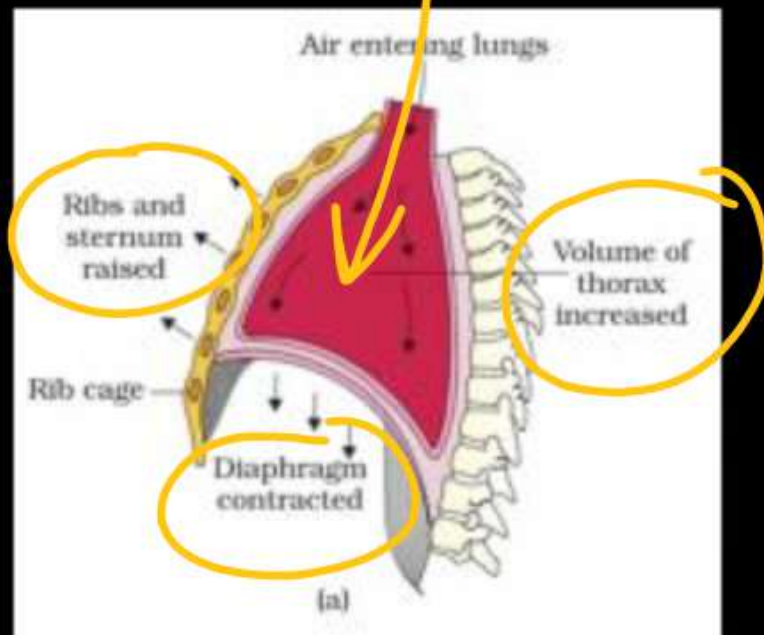
Contraction in E.I.C.M
(↑ - Thoracic Volu. - D-V axis)



↑ - Total Thoracic Volu.

↓ - Intrapulmonary Pressure

↓ - Atm. Air enters in lungs



Expiration (Passive)

Relaxation of Diaphragm
(↓ - Thoracic Volu.)



Relaxation of E.I.C.M
(↓ - Thoracic Volu.)



↓ - Total Volume



↑ - Intrapulmonary Pressure

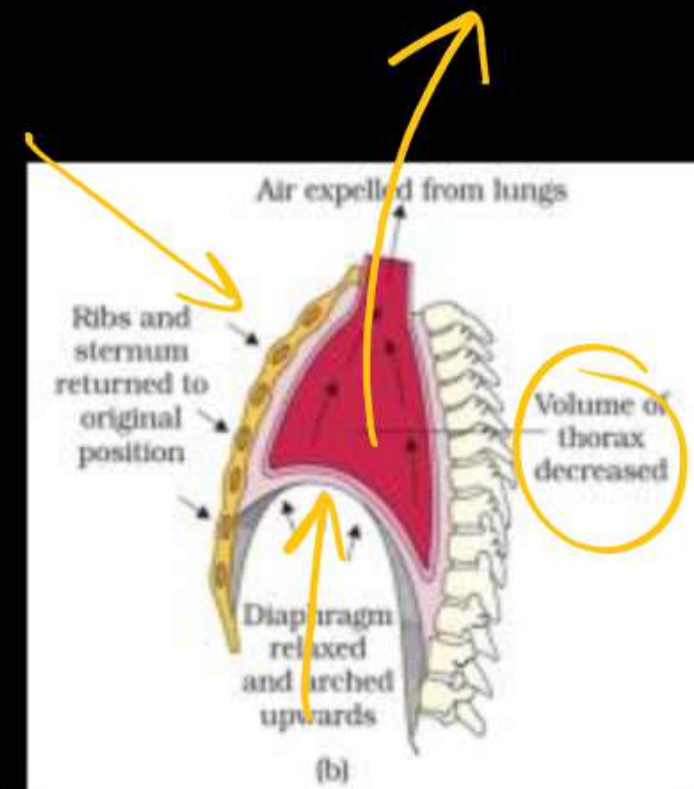
(Expiration)

For full Expiration

Contraction

D-I.C.M ✓

2) Abdominal Mus ✓



Respiratory Volumes and Capacities

= Spirometer

1 Q = 100%

Lung volumes

Spirogram

Pulmonary Capacities

I.R.V = 2500 - 3000 ml

Inspiratory reserve volume

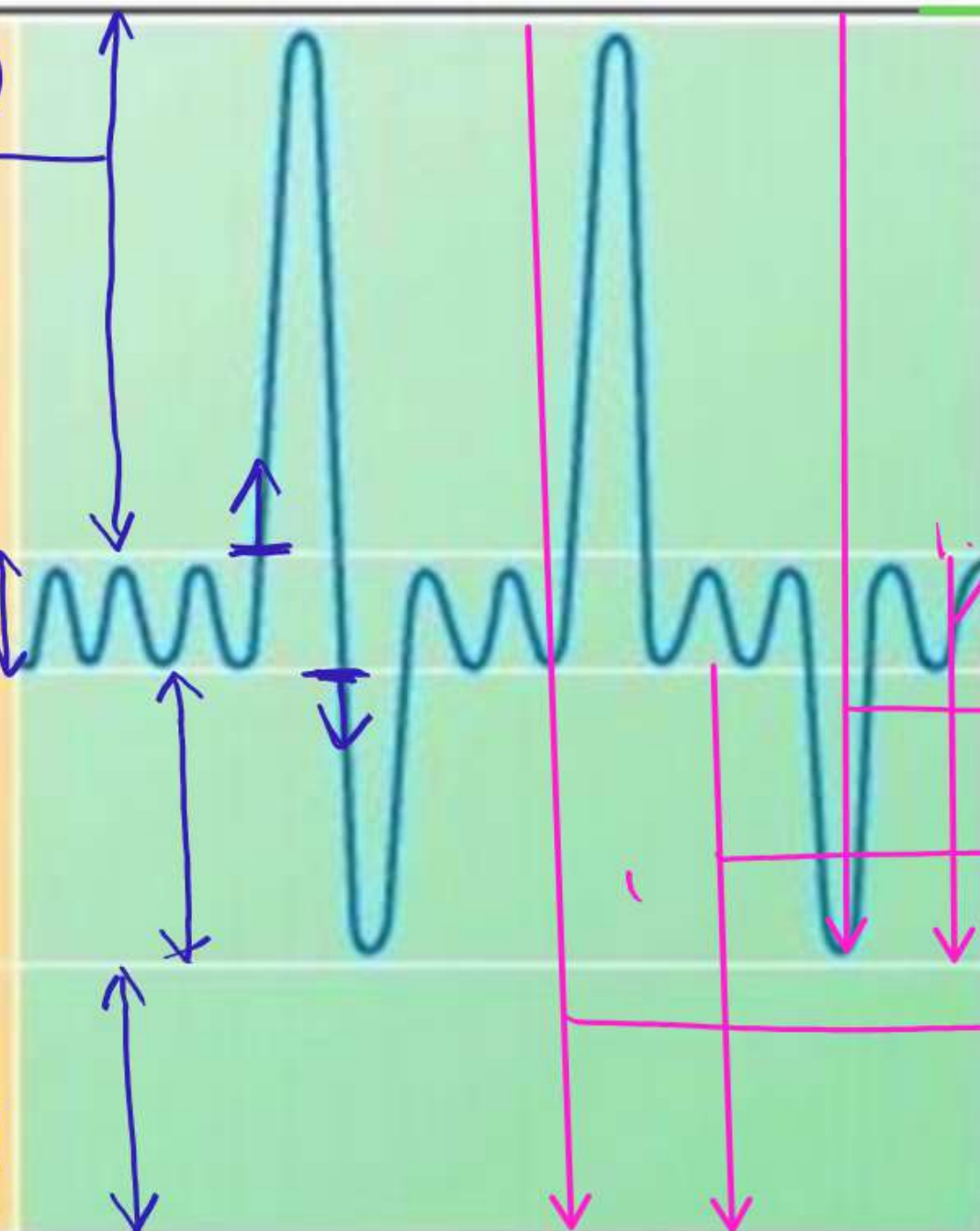
Tidal volume T.V. = 500 ml

Expiratory reserve volume

ERV = 1000 - 1100 ml

Residual volume

R.V. = 1100 - 1200 ml



I.C. = T.V + IRV = 3500 ml

EC = T.V + ERV = 1600 ml

V.C. = T.V + ERV + IRV = 4600

F.R.C. = ERV + RV = 2300

T.L.C. = $\underbrace{T.V + IRV + ERV + RV}_{V_C} = 5800$

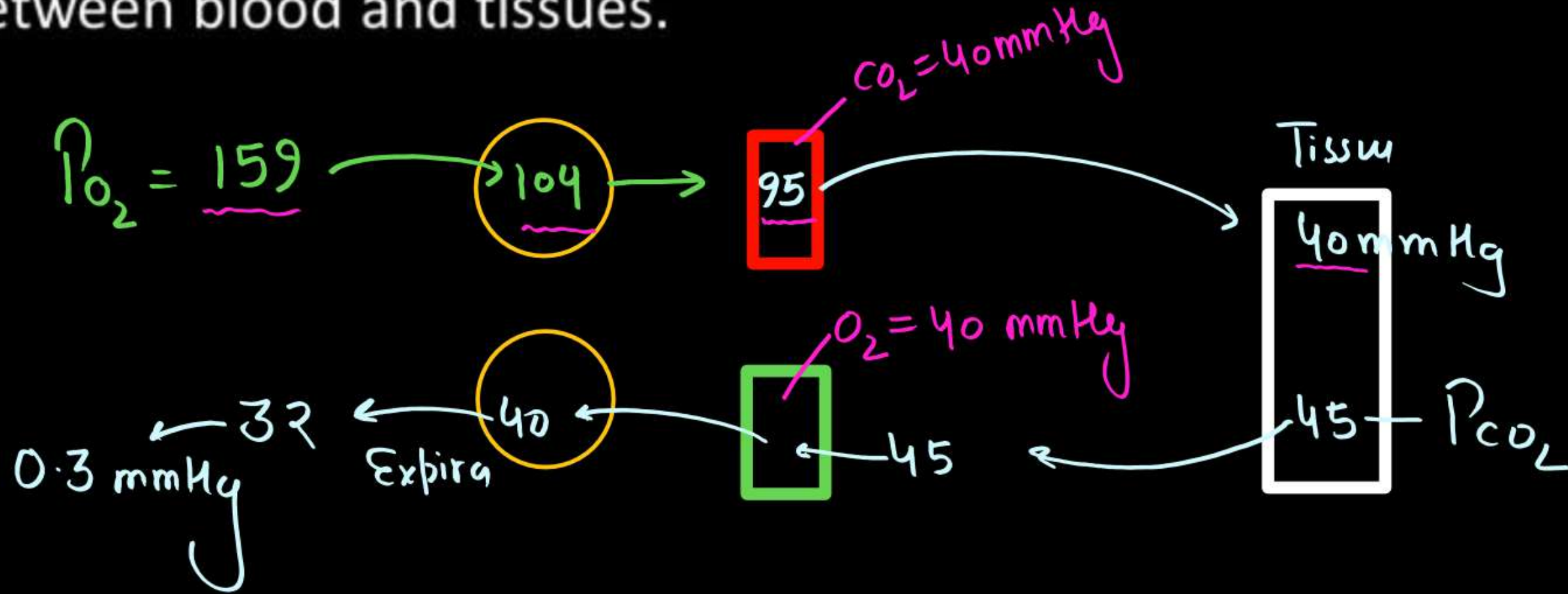
Note –

- Spirometry cannot be used to measure the residual volume or any capacities which incorporate the residual volume such as functional residual capacity (FRC) and total lung capacity (TLC).
- Pulmonary Ventilation -
$$= T.V \times \text{Rate of Breathing}$$
$$= 500 \times 12-16/\text{minute}$$
$$= \underline{6000-8000 \text{ ml}/\underline{\text{minute}}}$$

Exchange of Gases

Q//

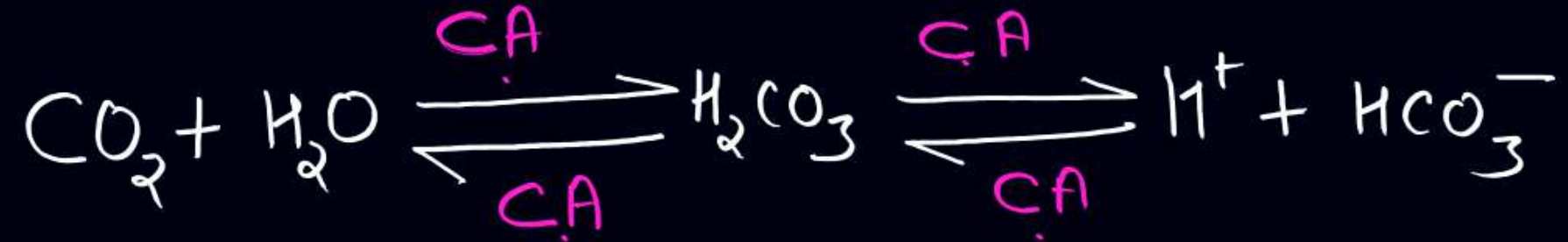
Alveoli are the primary sites of exchange of gases. Exchange of gases also occur between blood and tissues.



Respiratory Gas	Atmospheric Air	Alveoli	Blood (Deoxygenated)	Blood (Oxygenated)	Tissues
O_2	159	104	40	95	40
CO_2	0.3	40	45	40	45

vvfmp

(A) = Near tissue.
- Loading of CO_2
→ R-Shift (Bohr's Eff.) = Dissociation of Hb-O_2



← (B) = Near lungs (unloading of CO_2)
- L-Shift = Haldane Eff. (Association of Hb-O_2)

Co-factor
(Zn^{++})

Carbonic Anhydrase } R.B.C
Plasma

Oxygen Haemoglobin Dissociation Curve

Sigmoid Curve / Bohr's Effect

vvvvmp
@

Favourable

Unfavourable

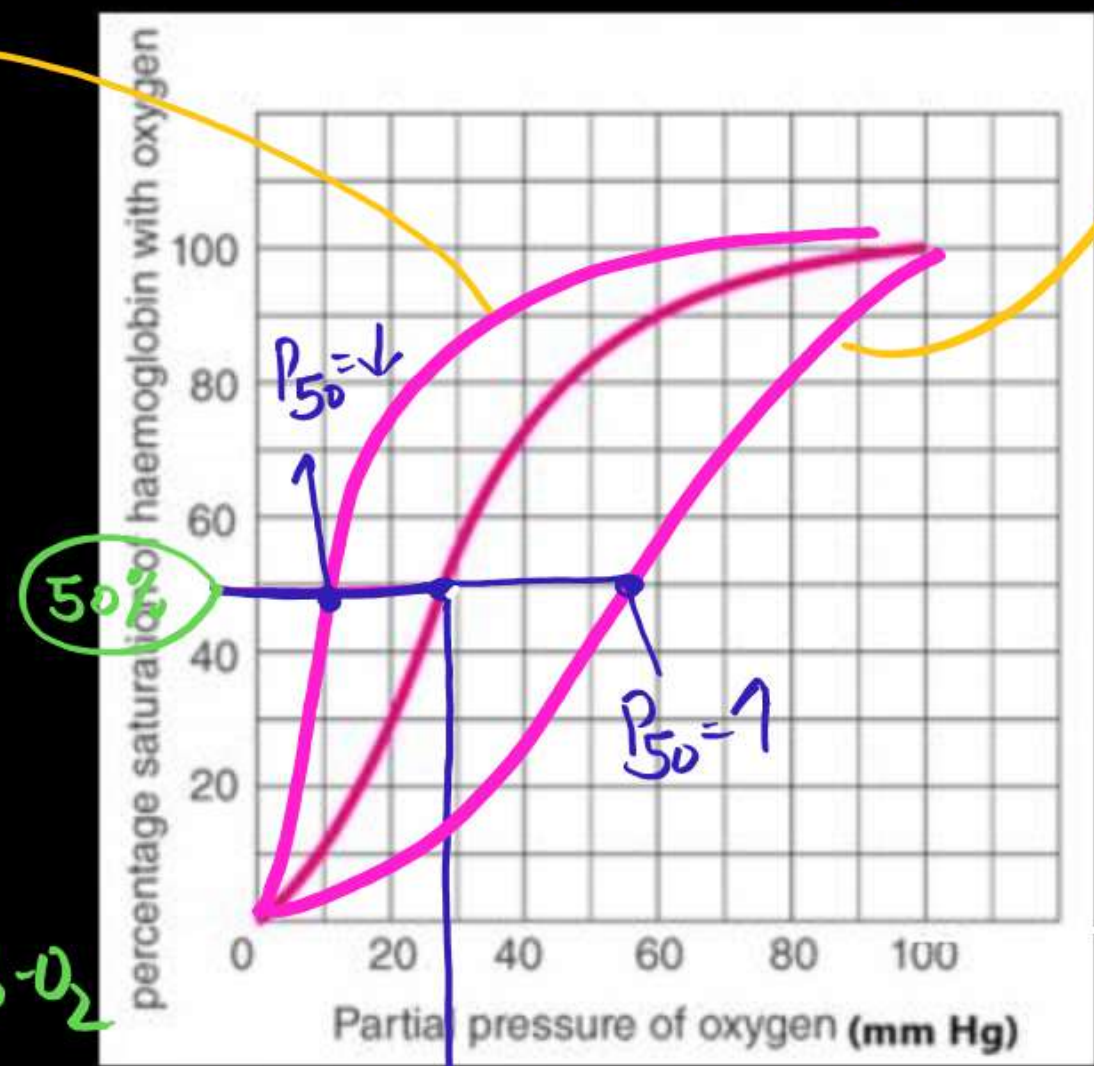
Left Shift
(Haldem Eff.)

$P_{O_2} \uparrow$ & $P_{CO_2} \downarrow$

$H^+ \downarrow \rightarrow \uparrow pH$

$\uparrow$ Association of $Hb-O_2$

$P_{50} \downarrow$ Temp. $\downarrow$
(Near lungs)



Right - Shift (Bohr's Eff.)

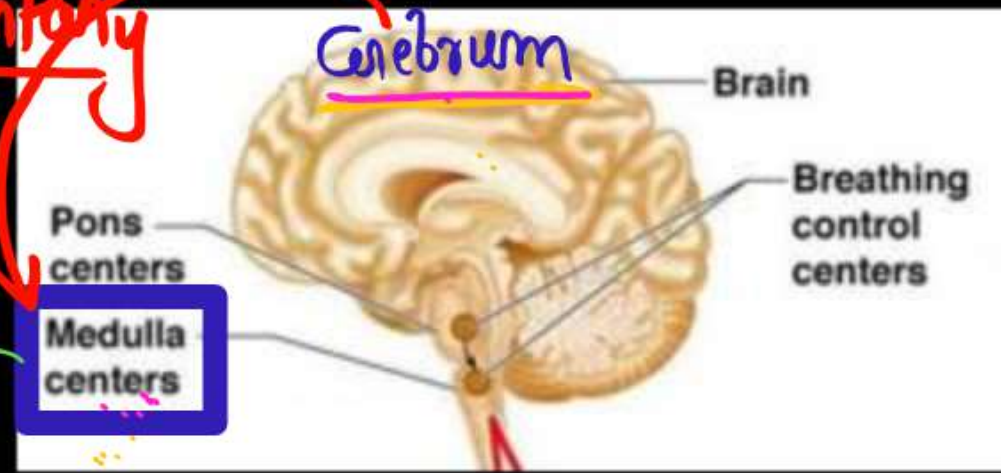
$P_{O_2} \downarrow$ & $P_{CO_2} \uparrow$

$\uparrow H^+ \rightarrow pH \downarrow$

Dissociation of $Hb-O_2$

$P_{50} = \uparrow$
Near Tissue
Temp = $\uparrow$

Nervous Regulation of Breathing-



Control only
Involuntary Action

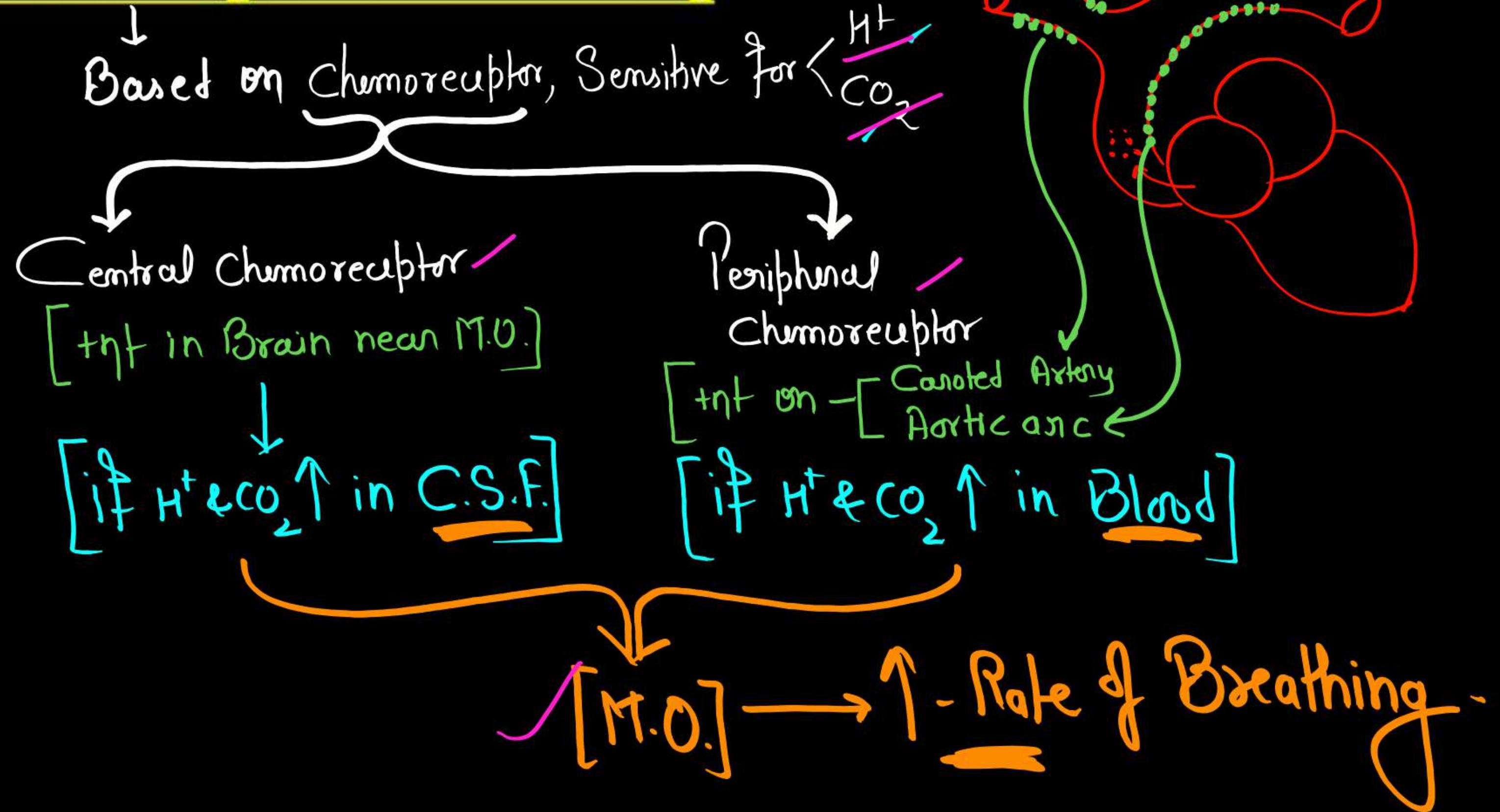
V/Vmb

M.O.
↓
(Respiratory Rhythm Center)

Control = Contraction
of [EICM
Diaphragm] } Normal Breathing

Pons → Pneumotaxic Center } Switch off → Stop inspiration $\approx 1-4$ sec.
↓ - Duration of Inspiration
↑ - Rate of Breathing

Chemical Regulation of Breathing



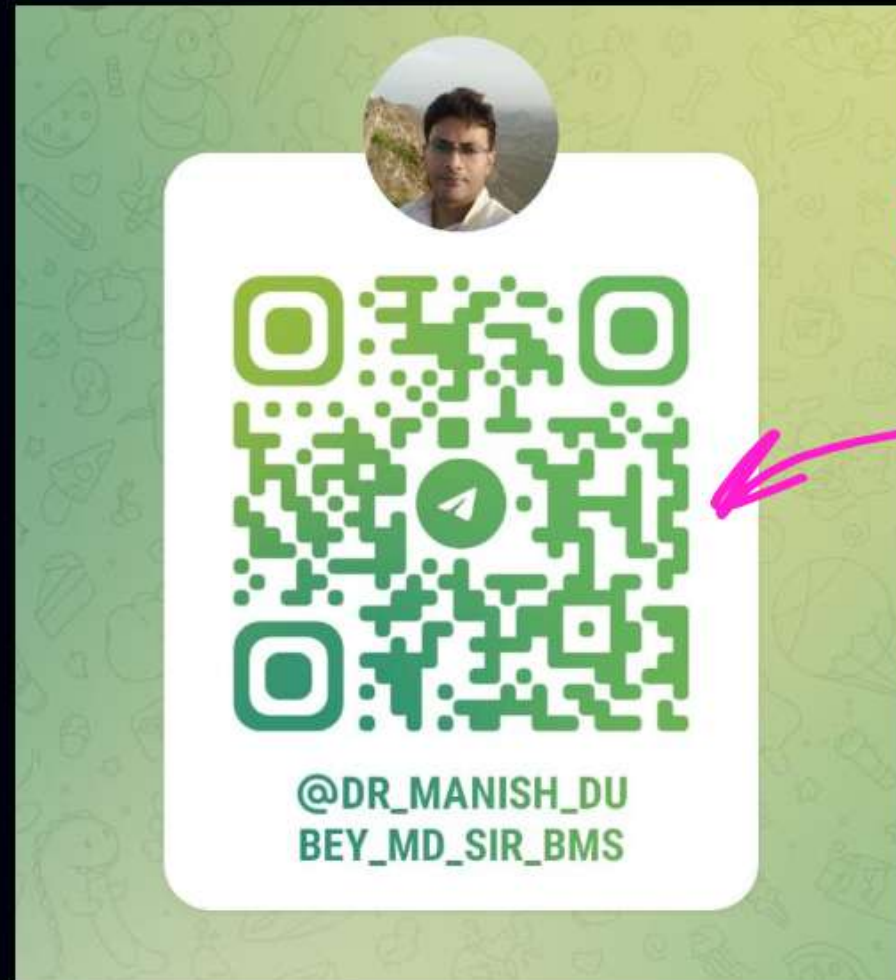
DISORDERS OF RESPIRATORY SYSTEM

(1) Asthma is a difficulty in breathing causing wheezing due to inflammation of bronchi and bronchioles.

(Allergy)

(2) Emphysema is a chronic disorder in which alveolar walls are damaged due to which respiratory surface is decreased. One of the major causes of this is cigarette smoking.

3) - Occupational Respiratory Disorders → Inflammation leading to Fibrosis.
[Proliferation of Fibrous tissue]
(Lungs Damage)



PDF

Body Fluids and circulation

18.1 BLOOD

Origin = Mesodermal

Blood is a special connective tissue consisting of a fluid matrix, plasma, and formed elements.

[Composition of Blood]

[Plasma = 55%]

- 1) - Water = 90-92% NCERT
- 2) - Inorganic Sub = 1%
- 3) - Organic Sub. = 8-9% NCERT
 - Protein = 6-8%

[Formed elements = 45%]
(Blood Cells)

Erythrocyte
(R.B.C)

5-5.5
million/mm³

Leucocyte
(W.B.C)

6000-8000
/mm³

Thrombocyte
(Platelets)

1.5-3.5 Lakh
/mm³

Leucocytes (WBC) –

Number of WBC – $6000 - 8000/\text{mm}^3$

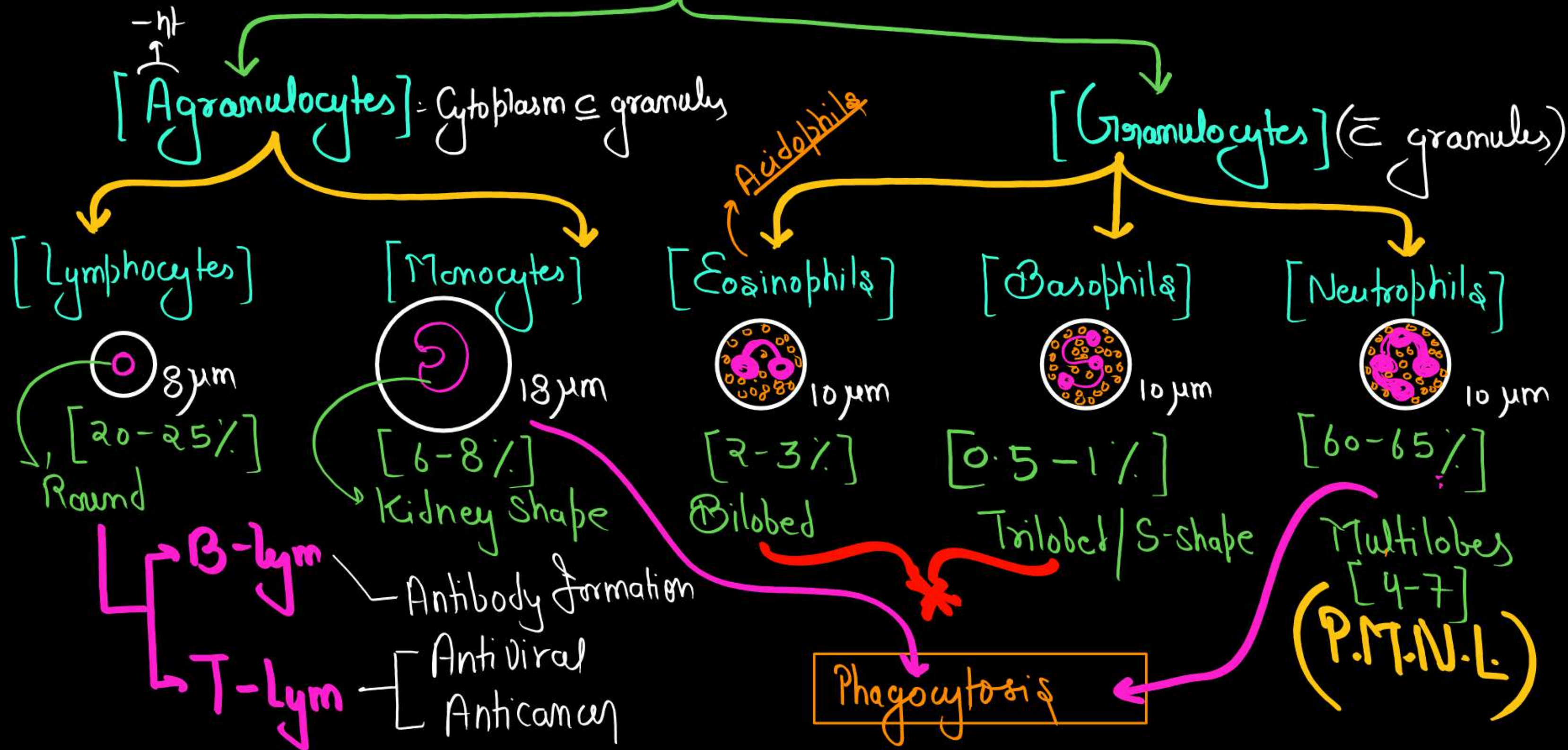
Leucocytosis = $\uparrow$ - No. of W.B.C $\Rightarrow$ During infection ✓

Leukaemia = $\uparrow$ - No. of W.B.C $\Rightarrow$ During Blood Cancer ✓

Diapedesis = Amoeboidal Movement of W.B.C.

Life span = hours ———— Year

Types of Leucocytes (WBC)





[Eosinophils]



1) → Antiallergic

2) - Anti-parasitic

Cranules Contain
lysosome like Enzymes

Resist
fungal infection

Cranules Contain
Antiallergic Enzyme -
Histaminase

Associated With
Allergic Reaction

[Basophils] = Blood

Mast cells

Conn. Tissue

Cranules Contain

Heparin → Anticoagulant

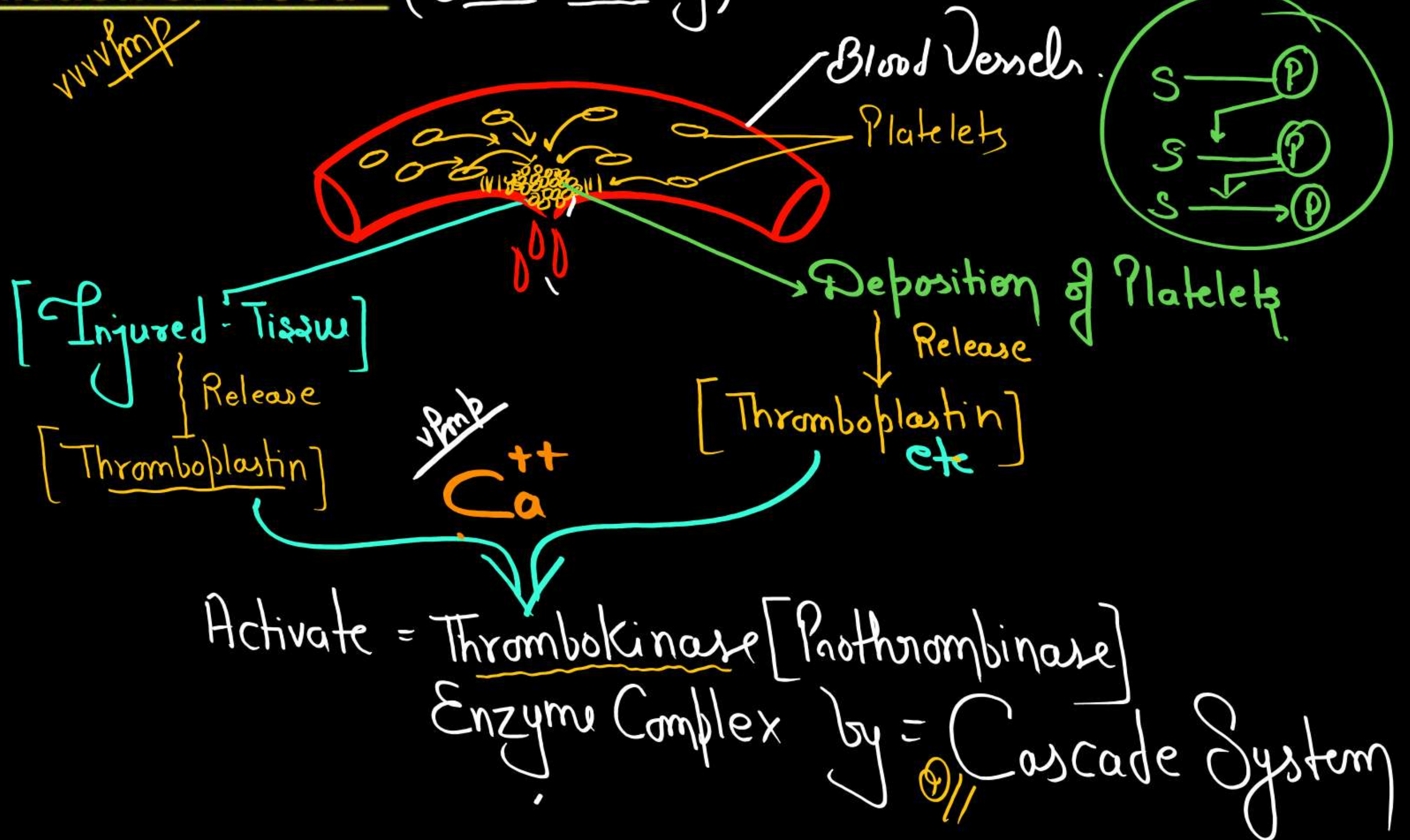
Histamine → Inflammatory Reaction.

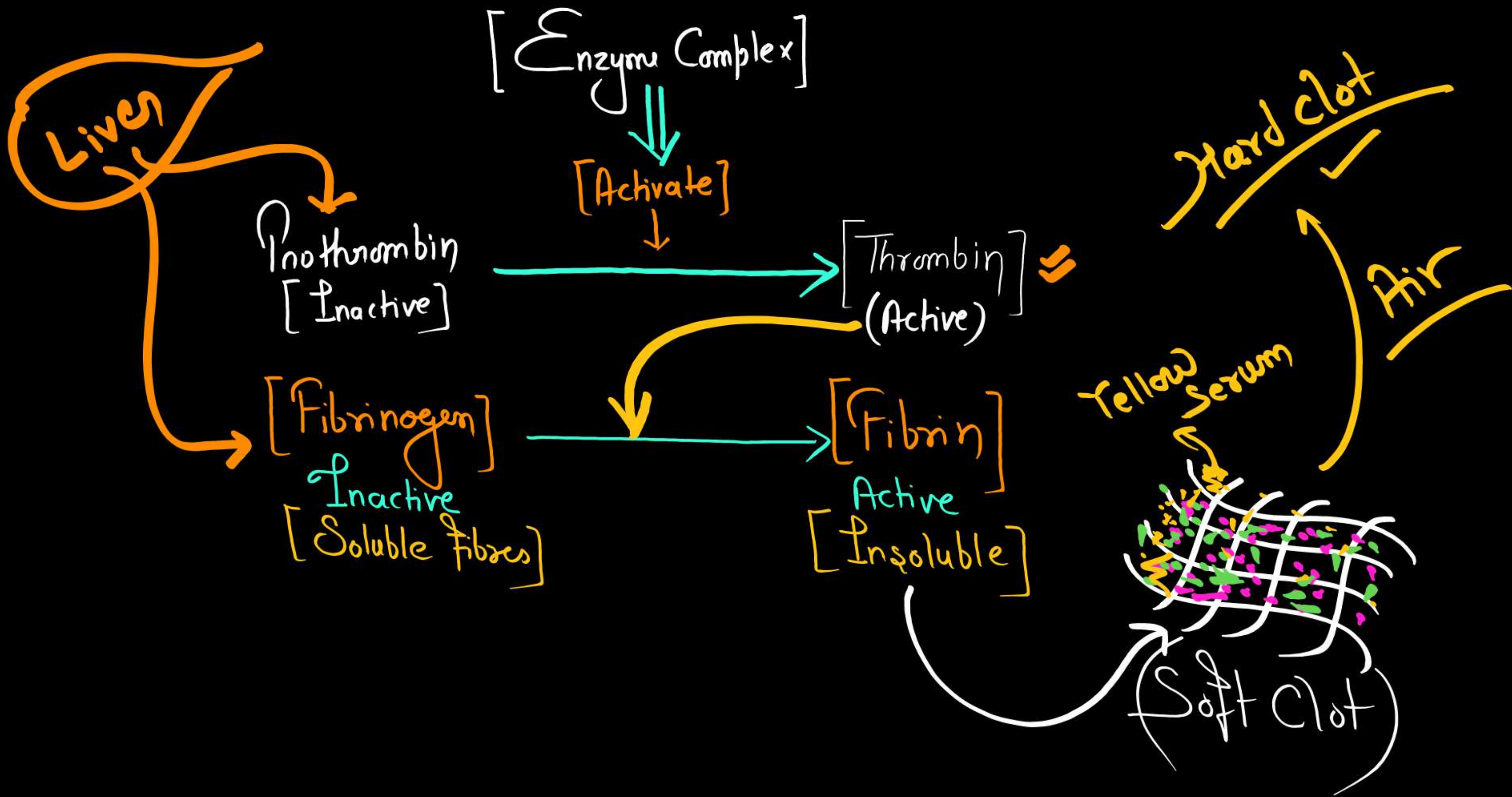
Serotonin → Vasoconstrictor.
etc

Involve in Inflammatory
Reaction

Coagulation of Blood - (Blood Clotting)

~~vvvvvmp~~

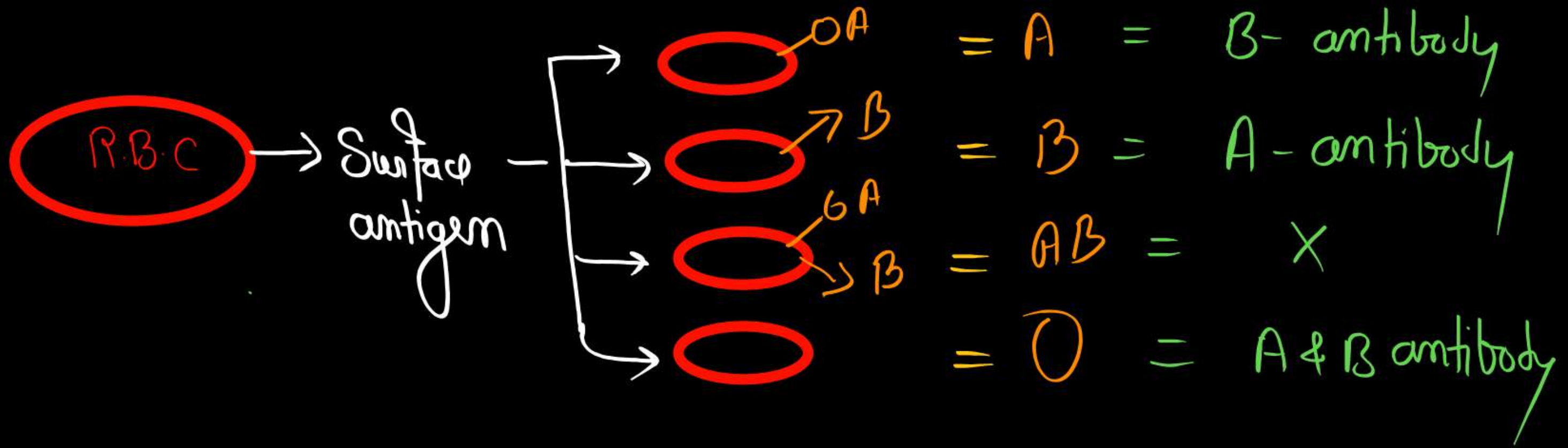




Blood Groups

(1) ABO grouping-

- ABO grouping is based on the presence or absence of **two surface antigens** (chemicals that can induce immune response) on the RBCs namely **A and B**.
- Similarly, the plasma of different individuals contain two natural antibodies (proteins produced in response to antigens)



[Donor] → Antigen → Antibody ← [Recipient]

(Mismatching)

Donor	Antigen		Antibody	Recipient
BY	(Ag)		Ab	BY
A	A		b	A
B	B		a	B
AB	AB		X	AB
O	X		ab	O

(U.D)

(both antigen - nt)

U.R
↓
(both Antibody - nt)

(2) Rh Grouping

- Rh antigen similar to one present in Rhesus monkeys (hence Rh), is also observed on the surface of RBCs of majority (nearly 80 per cent) of humans.

Human population — $\rightarrow$ if Rh antigen +ve on R.B.C. = $Rh^{+ve} = \underline{80\%}$
 $\rightarrow$ if Rh antigen -ve on R.B.C. = $Rh^{-ve} = \underline{20\%}$

Note = Naturally antibody -nt in case of
Rh blood Group.

Rh incompatibility (mismatching)

A special case of Rh incompatibility (mismatching) has been observed between the Rh-ve blood of a pregnant mother with Rh+ve blood of the foetus.

Pregnant Mother = Rh-ve
Foetus = Rh+ve

→ 1st baby = Normal
But other show = jaundice & Anaemia
[Erythroblastosis]
(Rhogum)

Note → This can be avoided by administering Anti-Rh antibody to mother just after the delivery of 1st child.

CONDUCTION OF HEART BEAT (Myogenic Heart)

Origin of Heart beat $\Rightarrow$ Modified Cardiac Muscles

$\Downarrow$
[Nodal Tissue]

Types of Nodal Tissue $\rightarrow$

- 1) $\rightarrow$ S.A. Node (Pace Maker) $\rightarrow$ Largest, Responsible for
- 2) $\Rightarrow$ A.V. Node (Pace Setter) $\rightarrow$ Potential Max^m Rhythmicity of Heart
- 3) $\Rightarrow$ Bundle of His (Bundle of Purkinje fibres / A.V Bundle)
- 4) $\Rightarrow$ Purkinje fibres

$\rightarrow$ Show = Autorhythmicity

1) - S.A. Node

- * Position = Upper Right Corner of R-Atrium
- * Potential = 70-75 beat/minute

2) - A.V. Node =

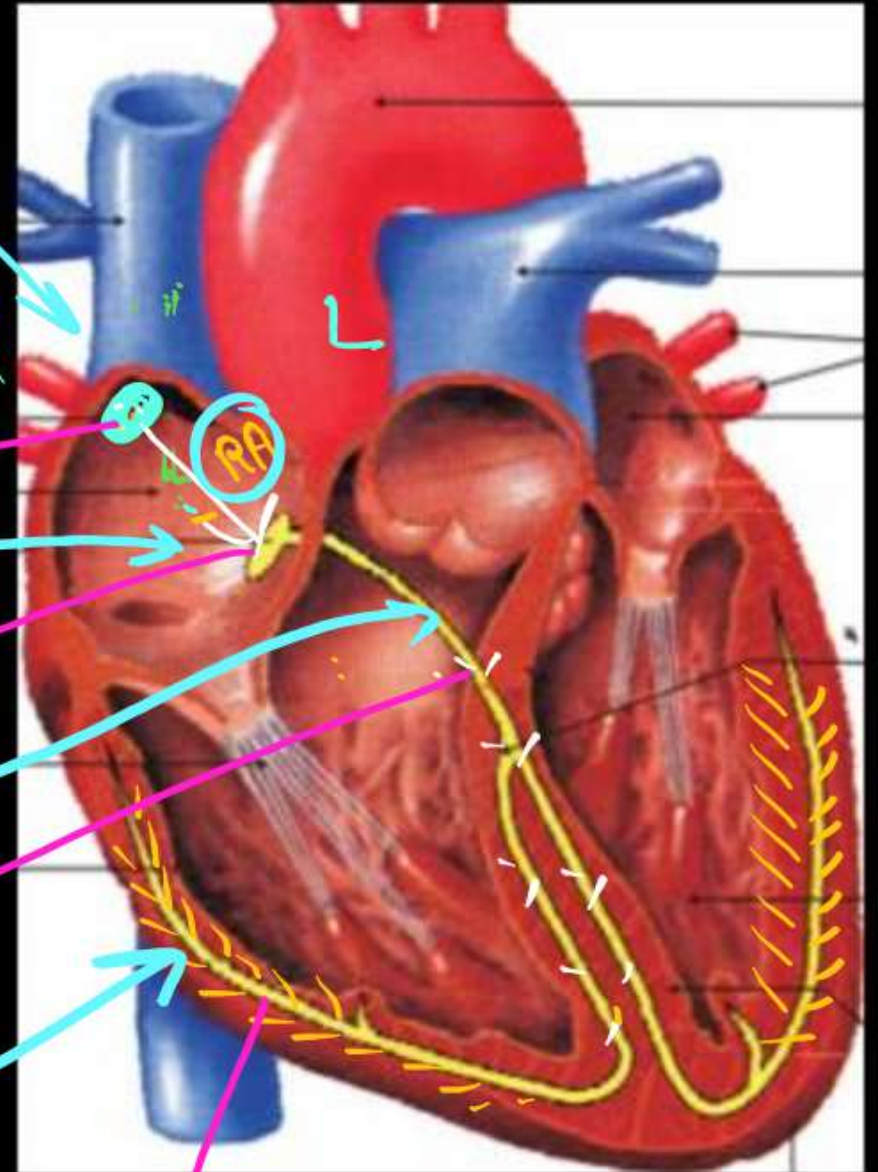
- * Position = Lower Left Corner of R-Atrium
- * Potential = 60-65 impulse or beat/minute

3) - Bundle of His

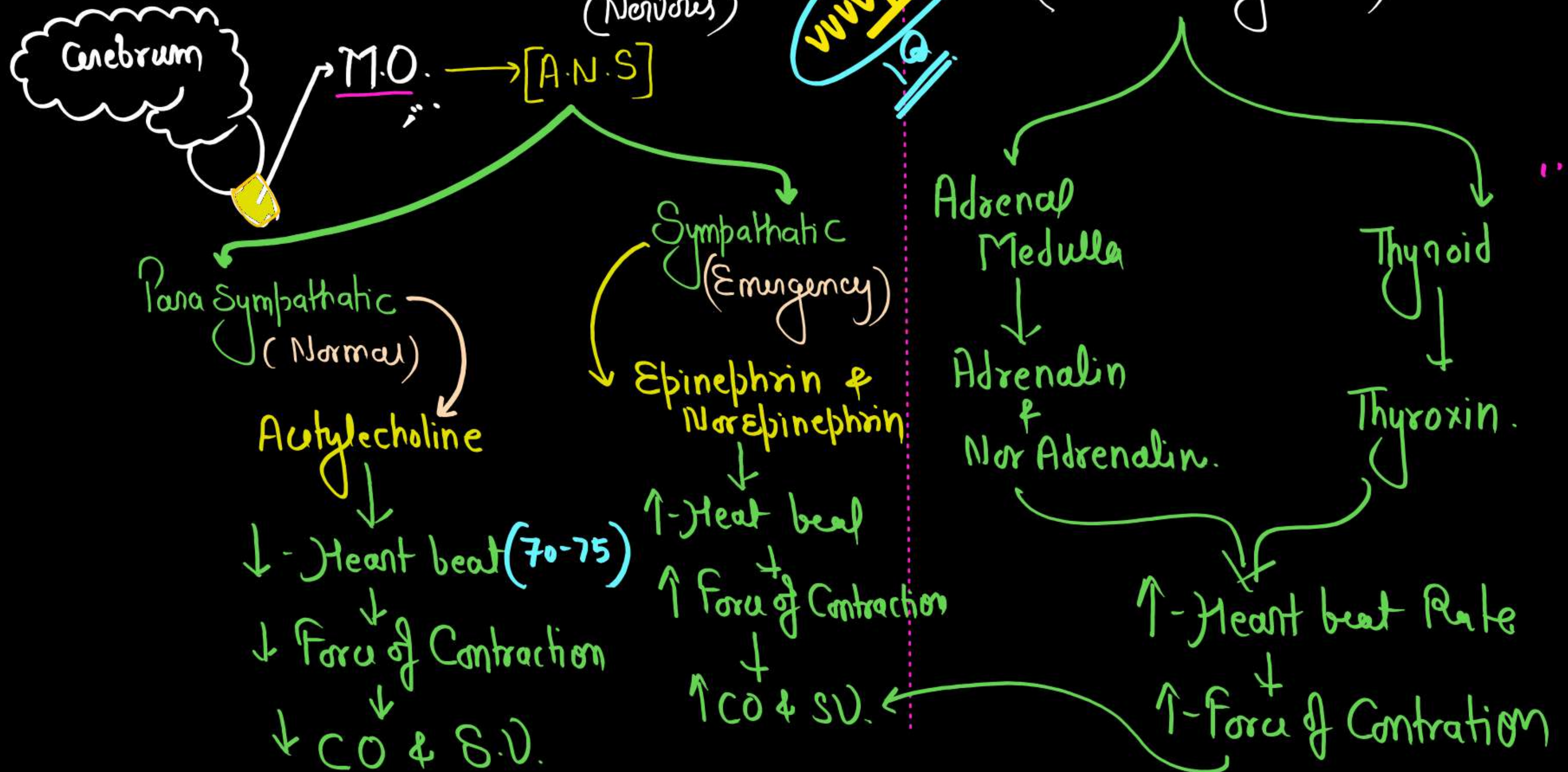
- * Position = InterVentricular Septum
- * Potential = 40-45 impulse/min.

4) - Purkinje fibres

- * Position = Wall of Ventricle
- * Potential = 20-25 impulse/min.



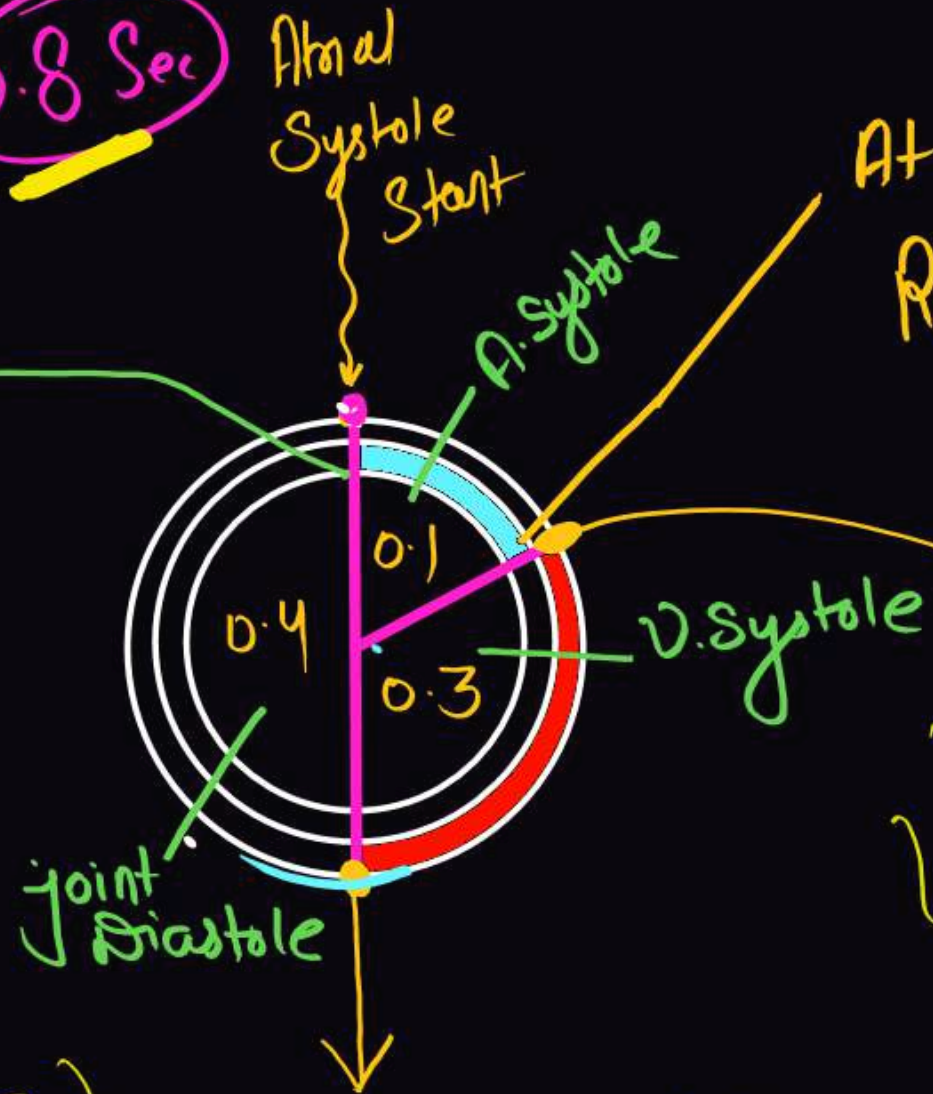
REGULATION OF CARDIAC ACTIVITY



Cardiac Cycle

0.8 Sec

At the end of joint diastole $\Rightarrow$
70% ($\frac{2}{3}$) Ventricle
fill by blood Passively
in 0.4 Sec



At the end of Atrial Systole
Remaining $\frac{1}{3}$ (30%) Ventricle
fill by blood (Active)
= 0.1 Sec

Ventricle Systole Start
 I^{st} Heart Sound = LUB
Due to closing of A.V. Valve

Ventricular Diastole
[II^{nd} Heart Sound = Dub]
Due to closing of Semilunar Valve.

Duration b/w -

- I & II - Sound = 0.3
- II & I - Sound = 0.5

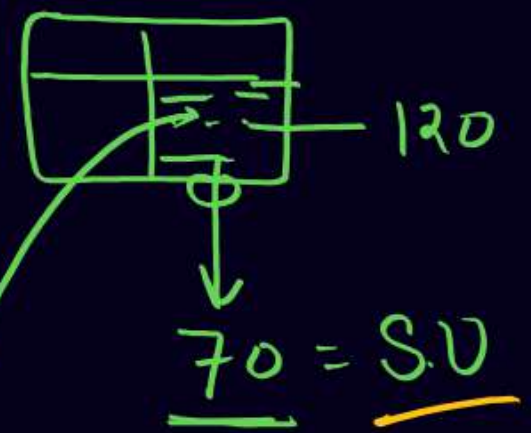
① **EDV** = (120 ml)
Each Ventricular

EDV $\propto$ Size of Ventricle
 (EDV $\propto$ SV)

② **ESV** = 50 ml (Fixed)

E.S.V. = E.D.V. - S.V.

120 - 70 = 50 ml



③ **SV** = 70 ml

S.V. = EDV - ESV

= 120 - 50 = 70 ml

④ **CO** = S.V. $\times$ HBR/min.

= 70 $\times$ 72

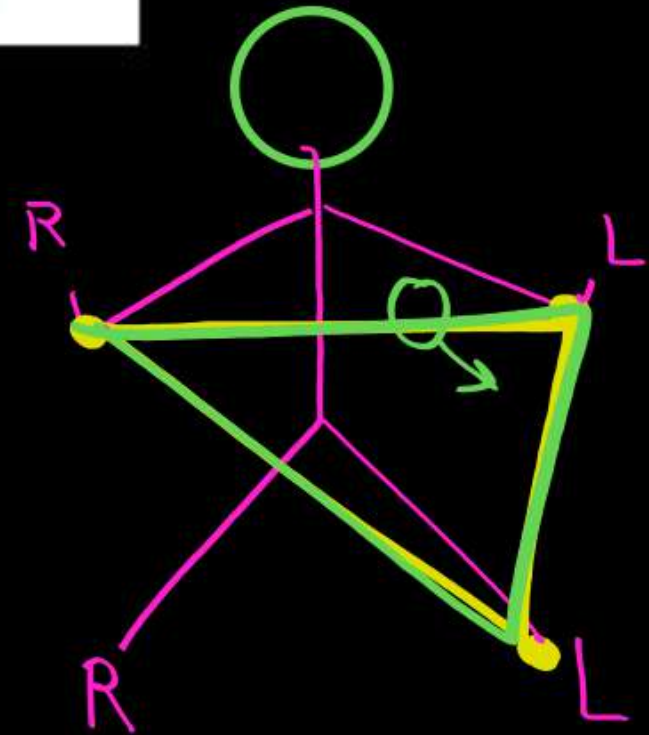
= 5040 ml/min.

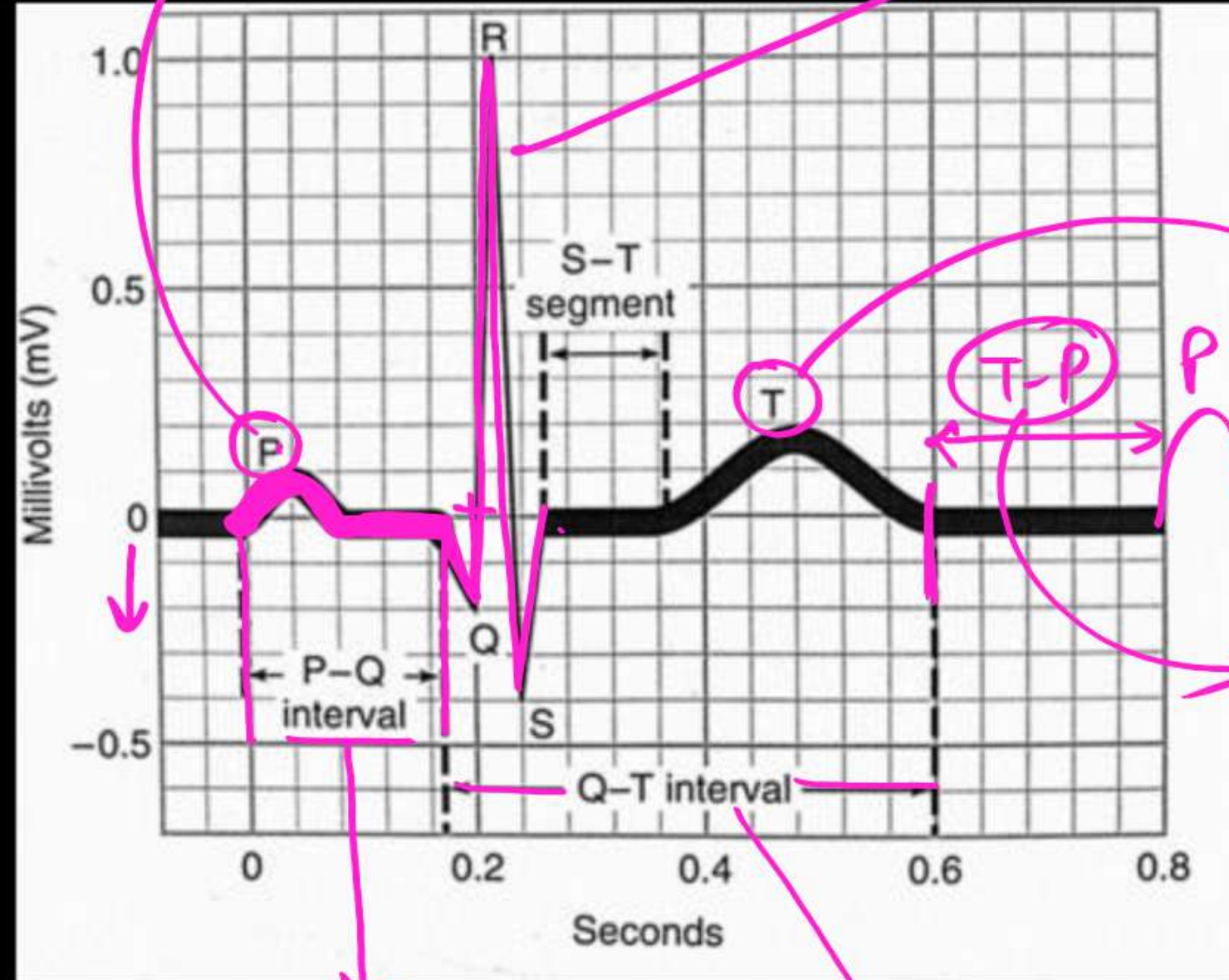
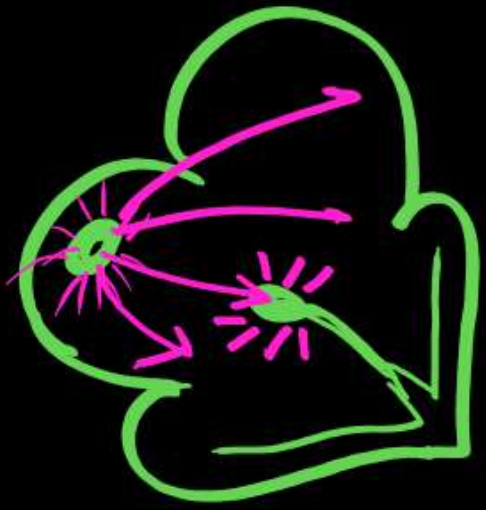
5 l/min.

15.3.3 Electrocardiogram (ECG)

You are probably familiar with this scene from a typical hospital television show: A patient is hooked up to a monitoring machine that shows voltage traces on a screen and makes the sound "... pip... pip... pip..... peeeeeeeeeeeeeeeeeeeeeee" as the patient goes into cardiac arrest. This type of machine (electro-cardiograph) is used to obtain an electrocardiogram (ECG). ECG is a graphical representation of the electrical activity of the heart during a cardiac cycle. To obtain a standard ECG (as shown in the

vvpmp





Depolarisation
in Atria / S.A. Node

QRS = Depolarisation in
Ventricular / AV Node
(Start after = Q)

Repolarisation / Relax
in Ventricular
(Relaxation of Ventricular)

Depolarisation &
Contraction in Atria.

Depolarisation &
Contraction in Ventricular } at the end
of T wave

DISORDERS OF CIRCULATORY SYSTEM

(1) High Blood Pressure (Hypertension):

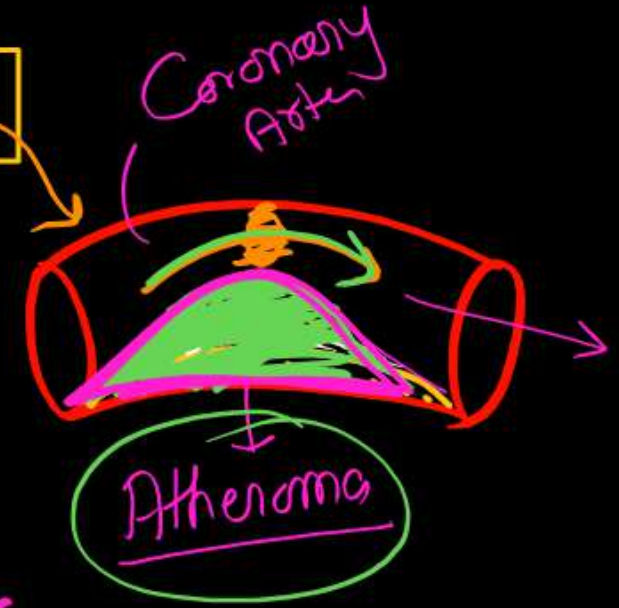
- Hypertension is the term for blood pressure that is higher than normal (120/80). If repeated checks of blood pressure of an individual is 140/90 (140 over 90) higher, it shows hypertension.
- High blood pressure leads to heart diseases and also affects vital organs like brain and kidney.

$$\left. \begin{array}{l} \text{Systolic} = \underline{120} \text{ mmHg} \\ \text{Diastolic} = \underline{80} \text{ mmHg} \end{array} \right\} 120 / 80 - 90$$

Handwritten notes: The number 140 is circled in orange with an arrow pointing to the 120. The number 90 is circled in orange. There are also some green and orange scribbles.

@ (2) Coronary Artery Disease (CAD):

- Coronary Artery Disease, often referred to as atherosclerosis, affects the vessels that supply blood to the heart muscle.
- It is caused by deposits of calcium, fat, cholesterol and fibrous tissues, which makes the lumen of arteries narrower. blood flow = ↓



(3) Angina:

- It is also called 'angina pectoris'. A symptom of acute chest pain appears when no enough oxygen is reaching the heart muscle.
- Angina can occur in men and women of any age but it is more common among the middle-aged and elderly.
- It occurs due to conditions that affect the blood flow.

(4) Heart Failure:

- Heart failure means the state of heart when it is not pumping blood effectively enough to meet the needs of the body.
- It is sometimes called congestive heart failure because congestion of the lungs is one of the main symptoms of this disease.

(5) Heart attack

Death of a part of Cardiac Muscles

(6) Cardiac arrest → Complete Death of Cardiac Mus.
CO = 0
Pip-pip-pip-peeEEEE ✓

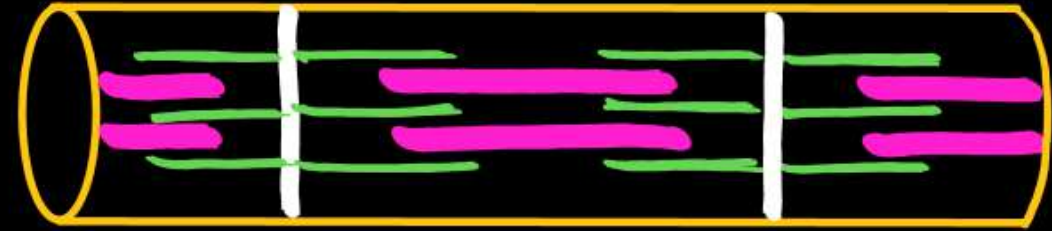


@DR_MANISH_DU
BEY_MD_SIR_BMS

Locomotion

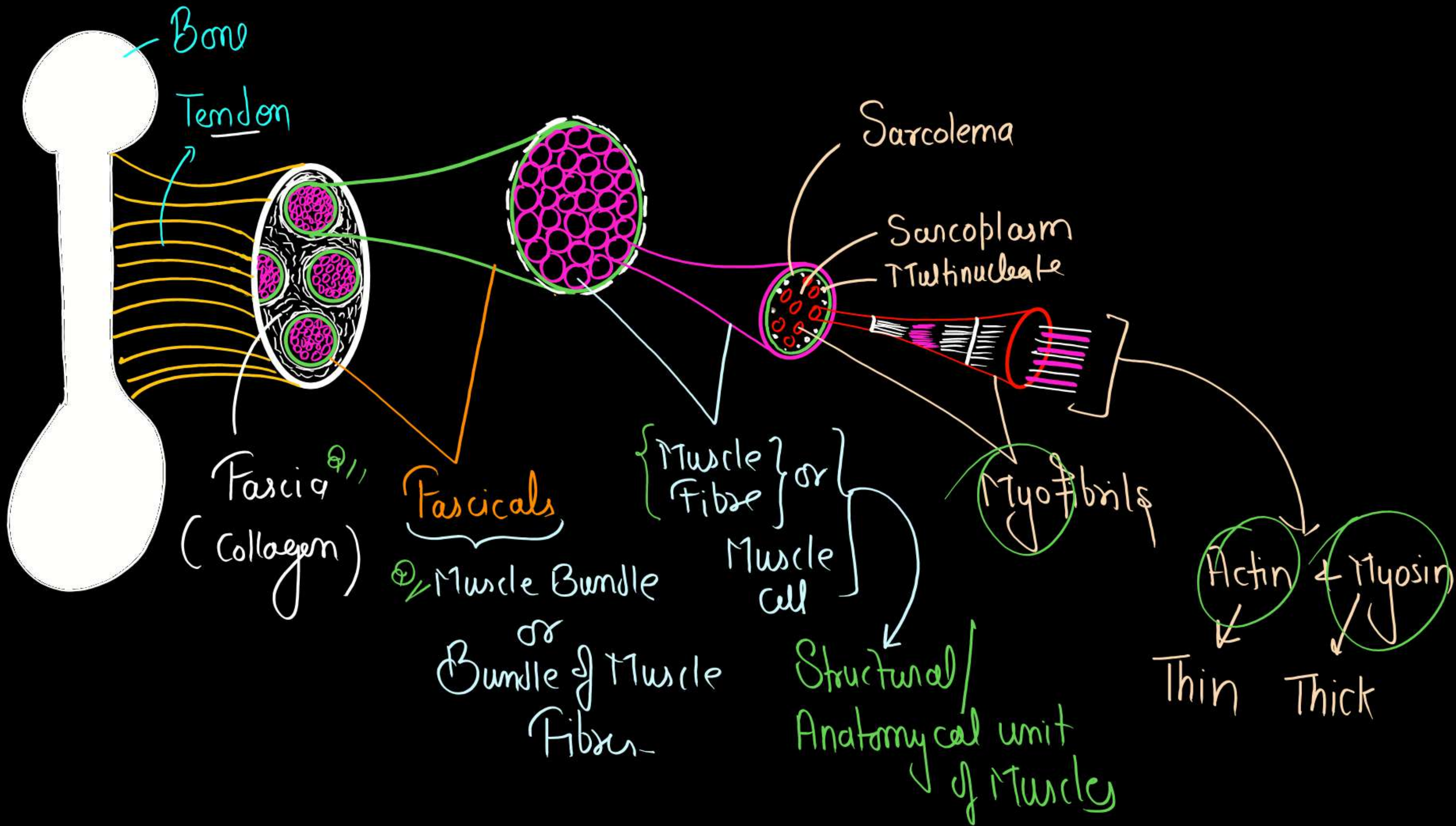
Skeletal Muscles

- Voluntary ✓
- Striated
- Striped
- Multinucleated

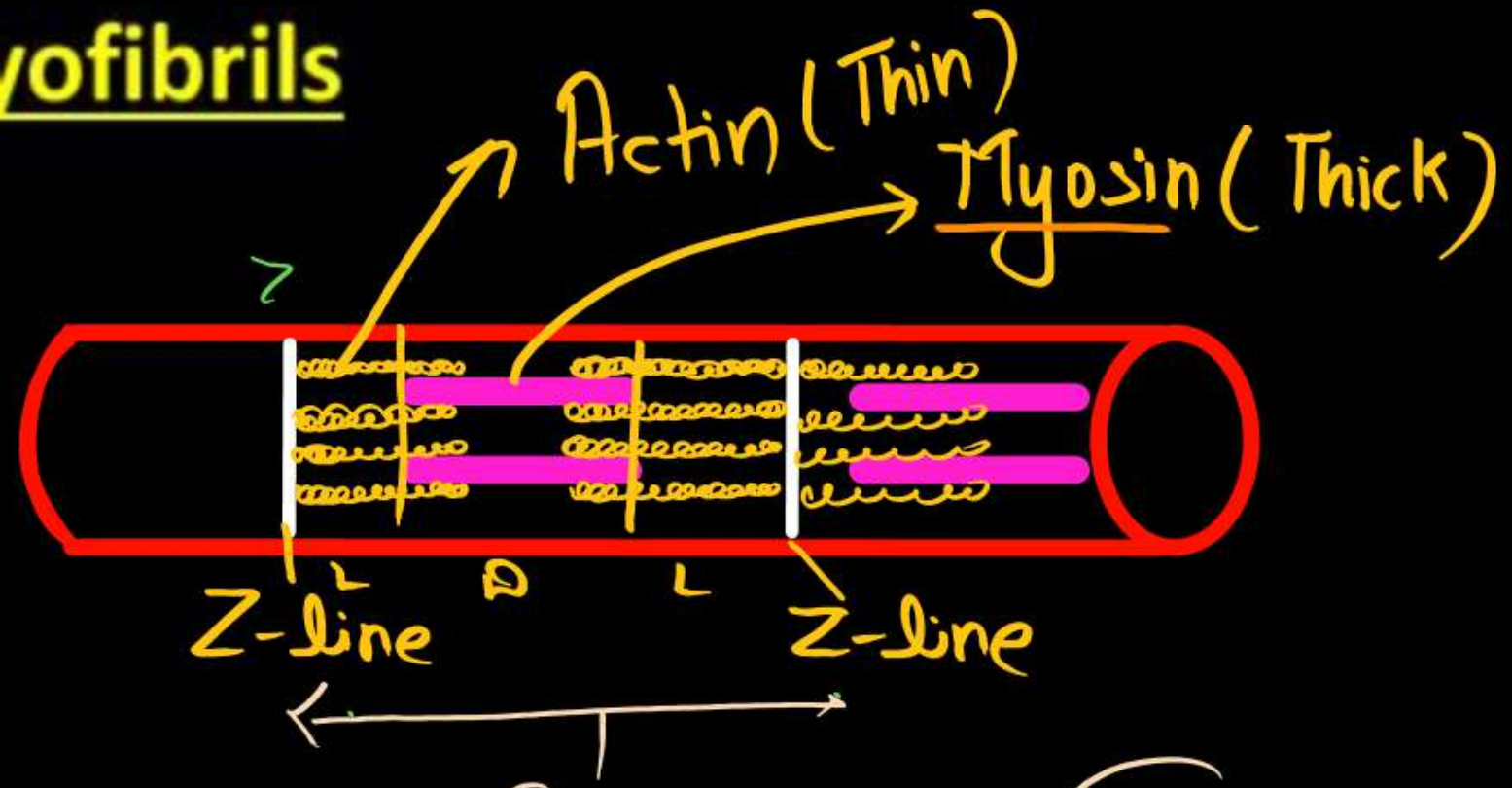


Light & Dark band

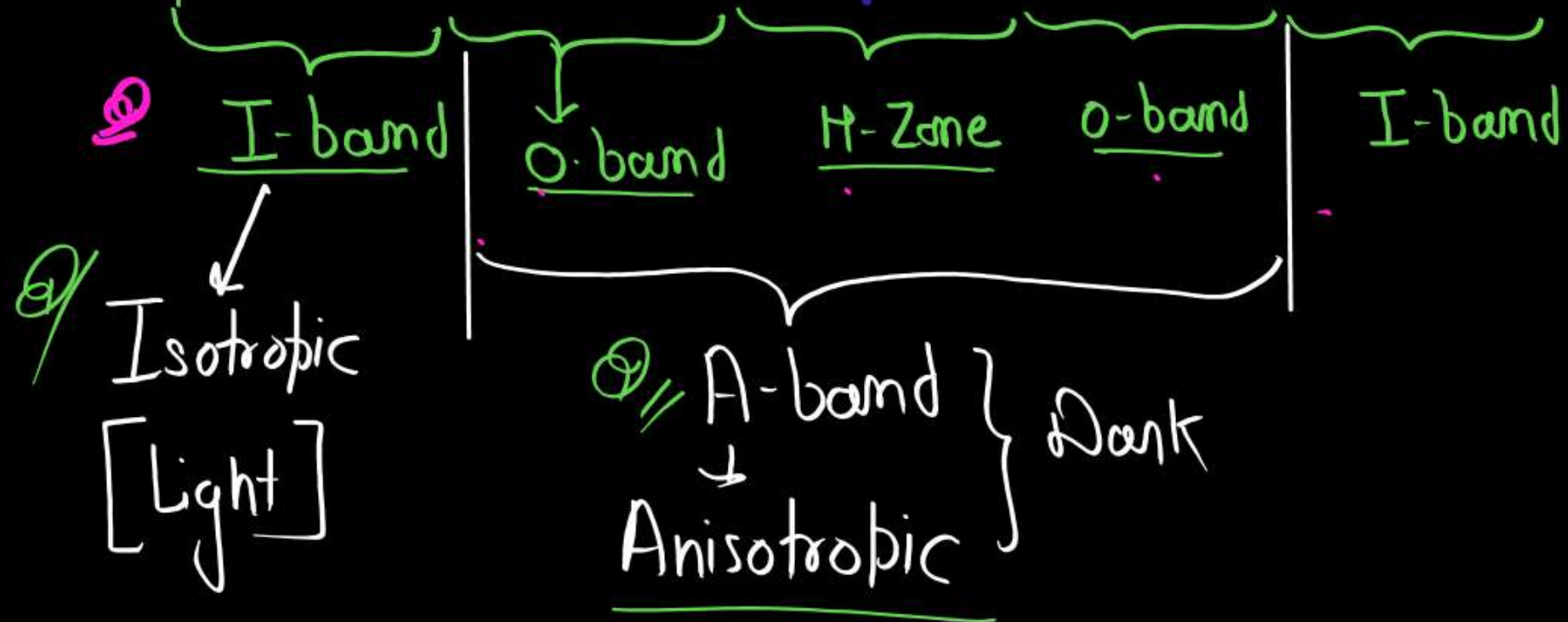
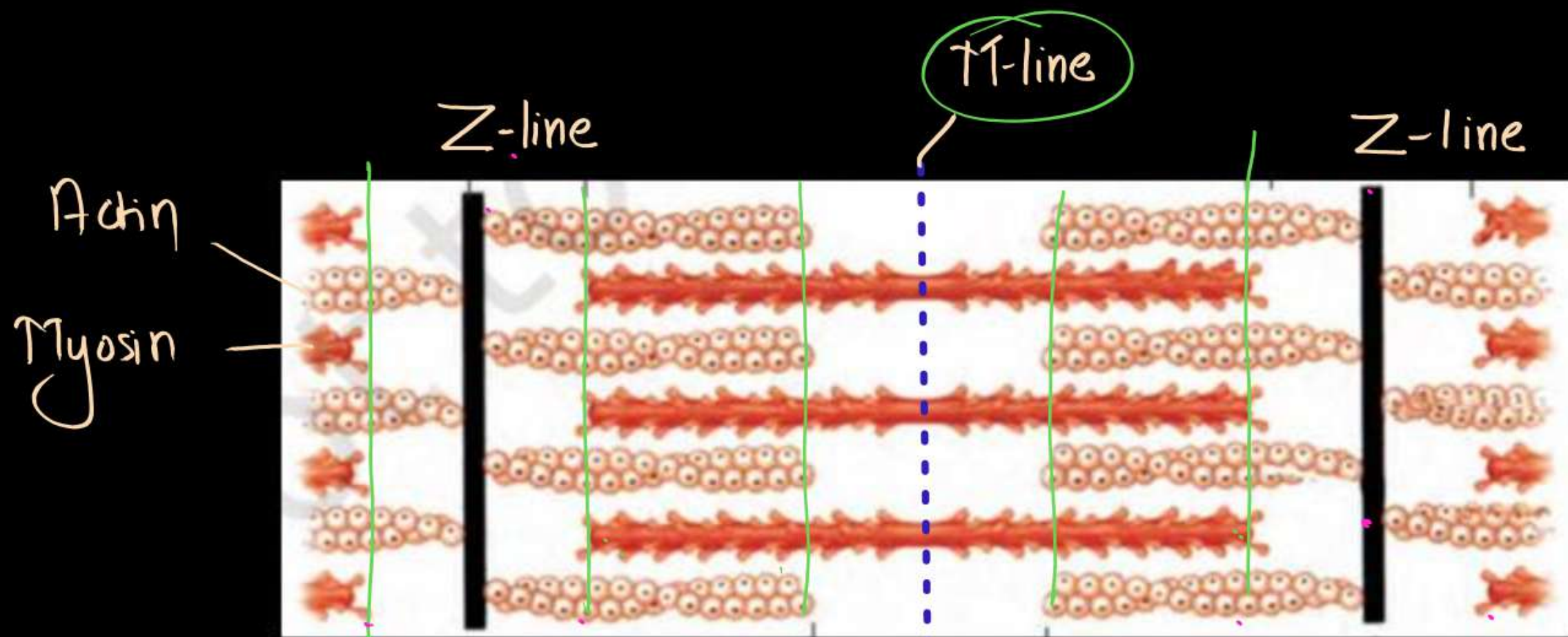
Transverse band //



Structure of Myofibrils



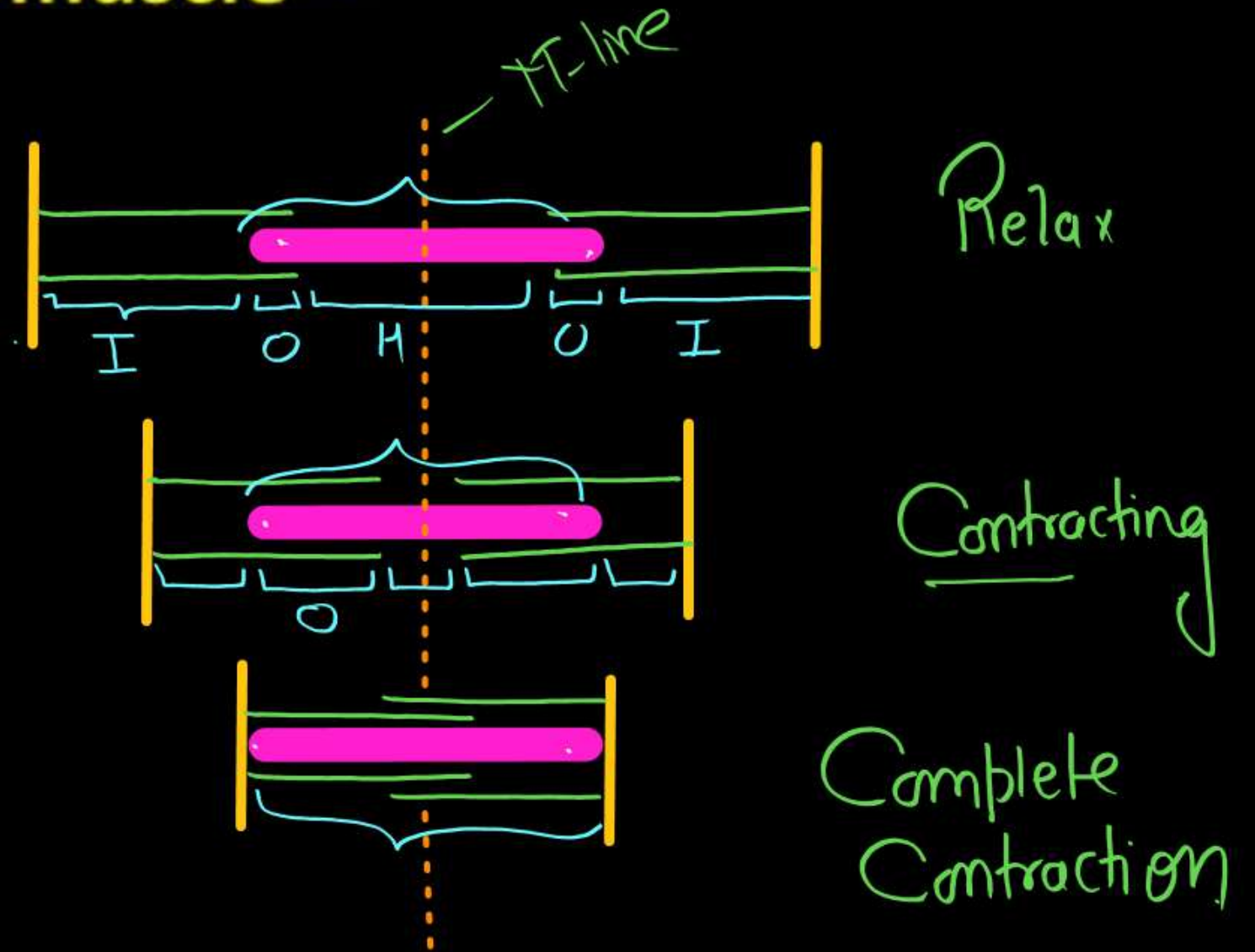
Sarcomere = Functional
unit of Muscle.



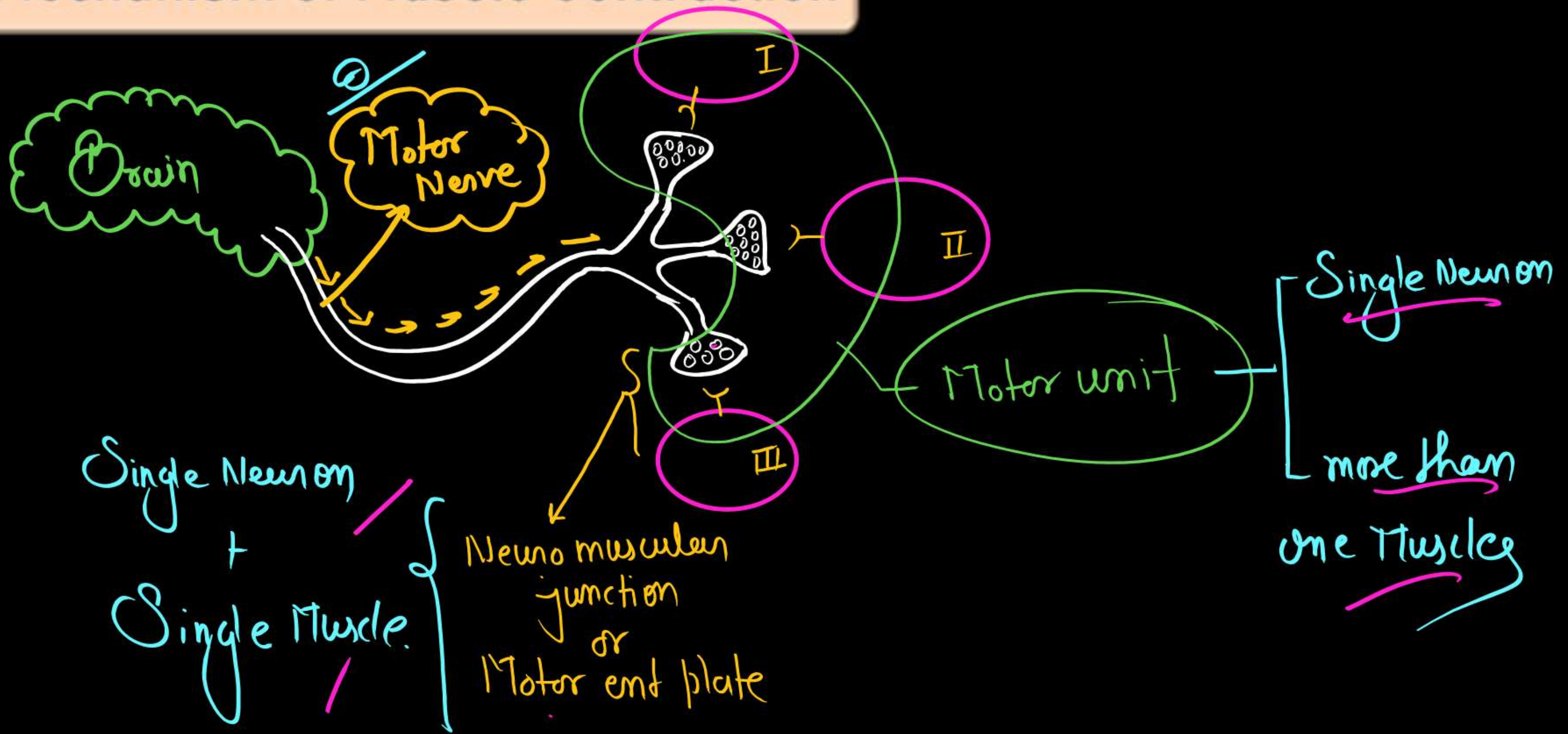
M-line + ht b/w =
 A-band / H-Zone
 Z-line + ht b/w
 I-band

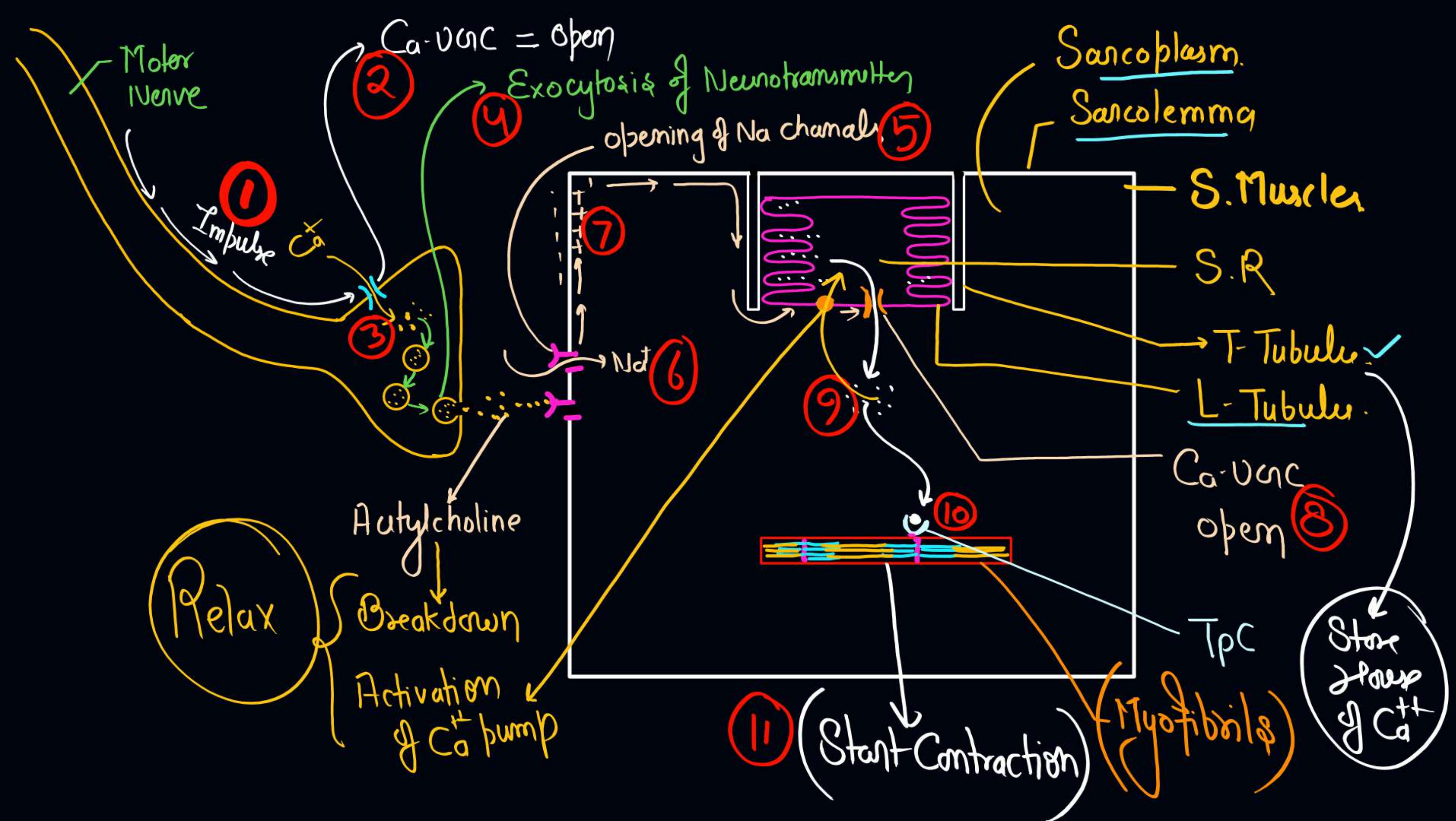
During Complete contraction of muscle –

- Length of Sarcomere = ↓
- Length of I Band = ↓
- Length of H Zone = ↓
- M-line = Disappear
- Length of O band = ↑
- Length of A band = Unchange

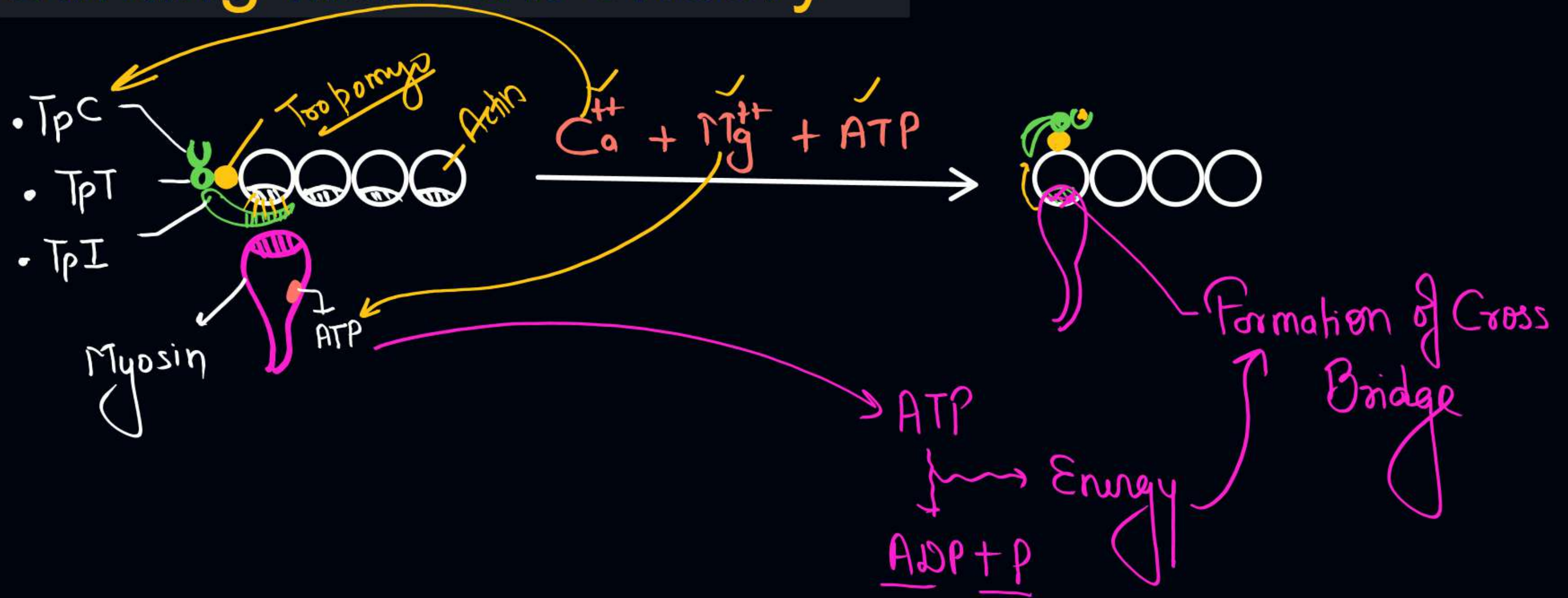


Mechanism of Muscle Contraction

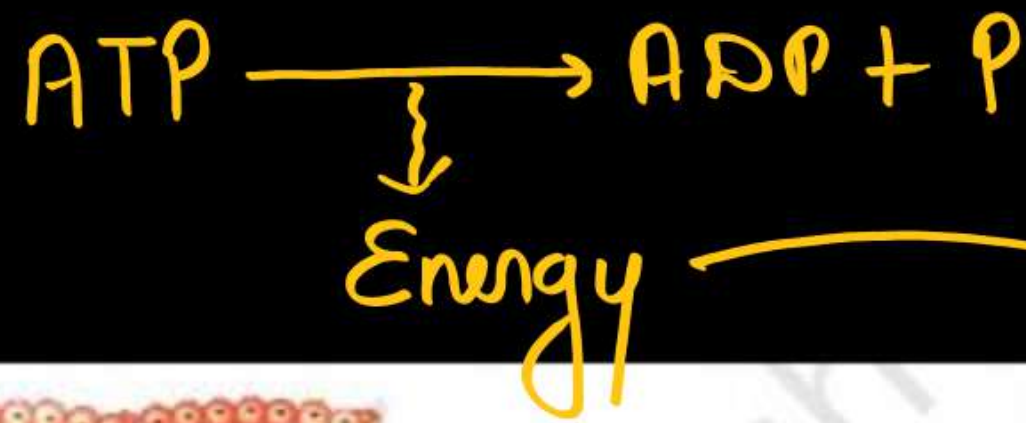




Sliding filament Theory

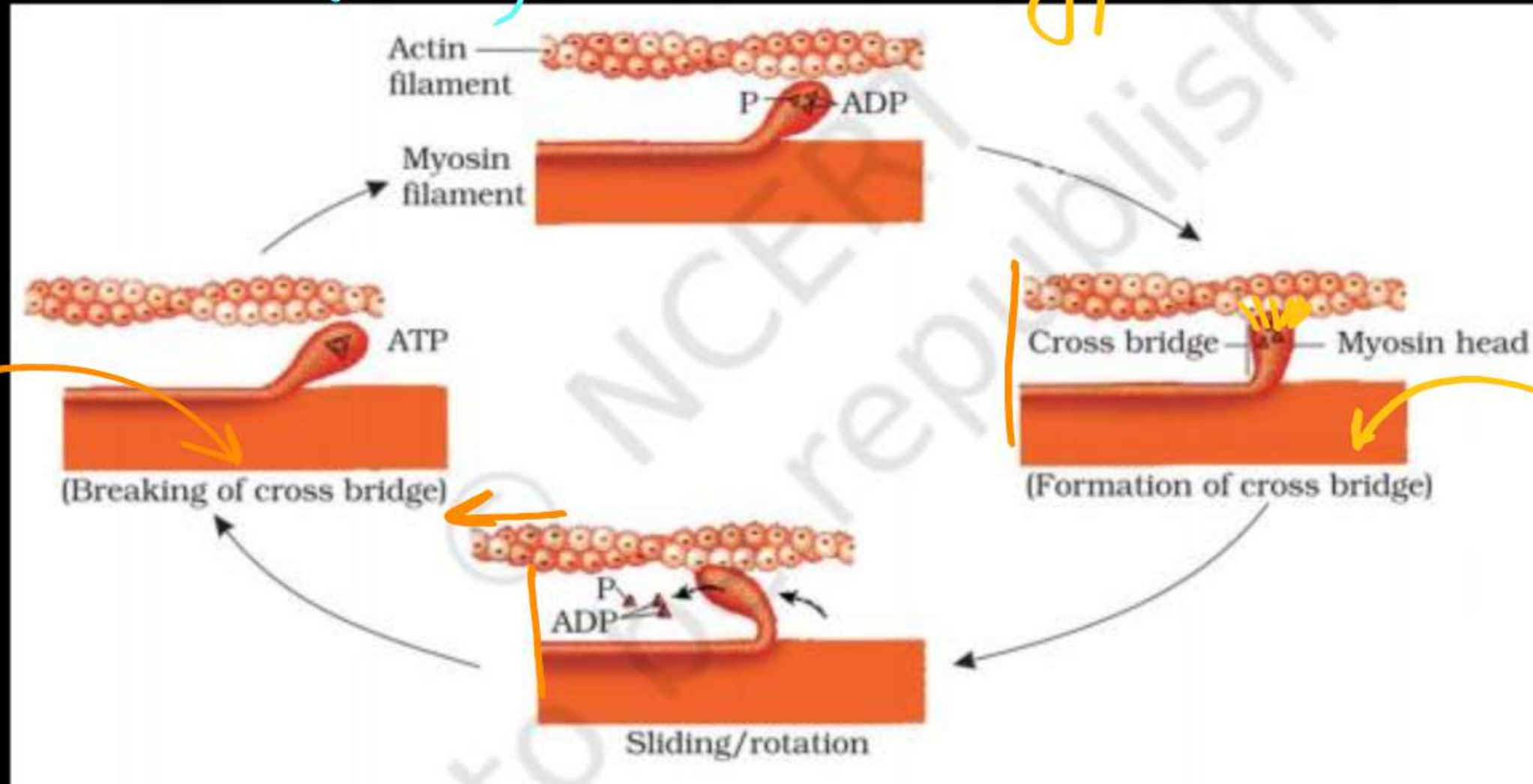


Crossbridge Formation = Breakdown of ATP
Crossbridge Breakdown = Formation of ATP



Breakdown of Cross bridge

Formation of Cross Bridge



ATP

ADP
+ High Energy Phosphate

Types of Muscles

skeletal

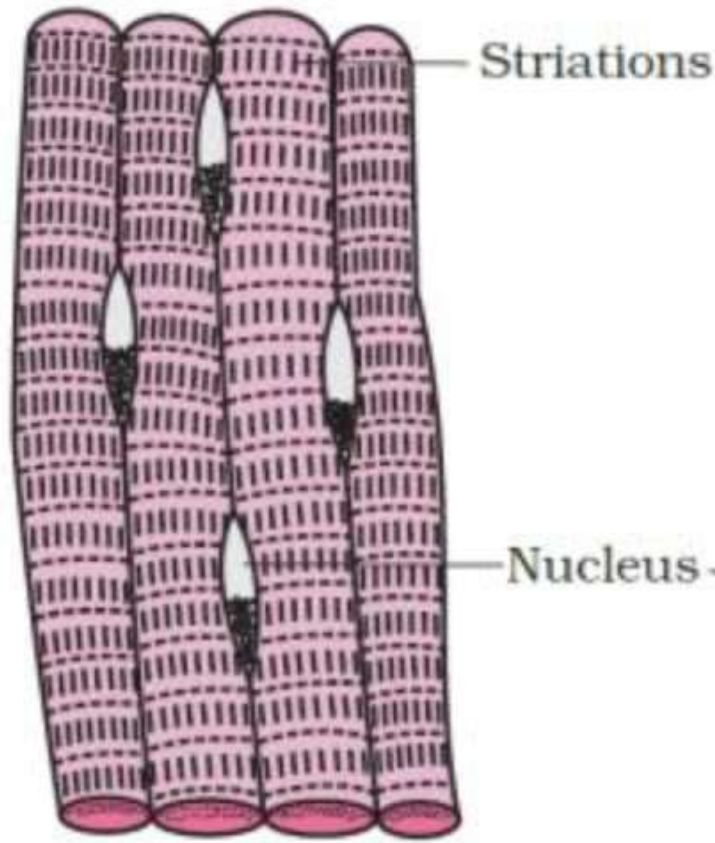
smooth

Cardiac

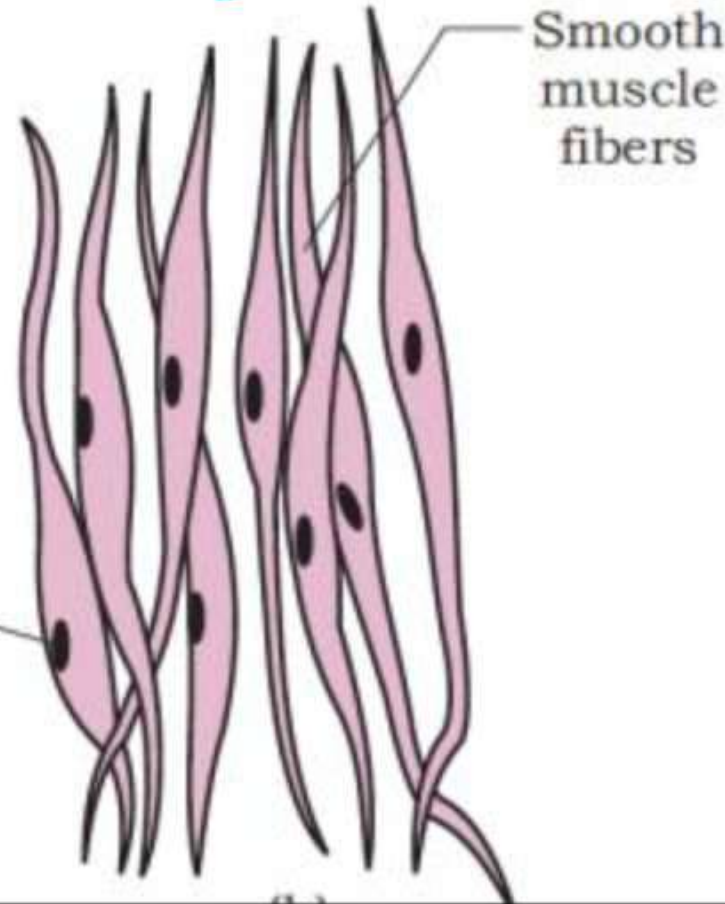
Limbs
Muscles

organs

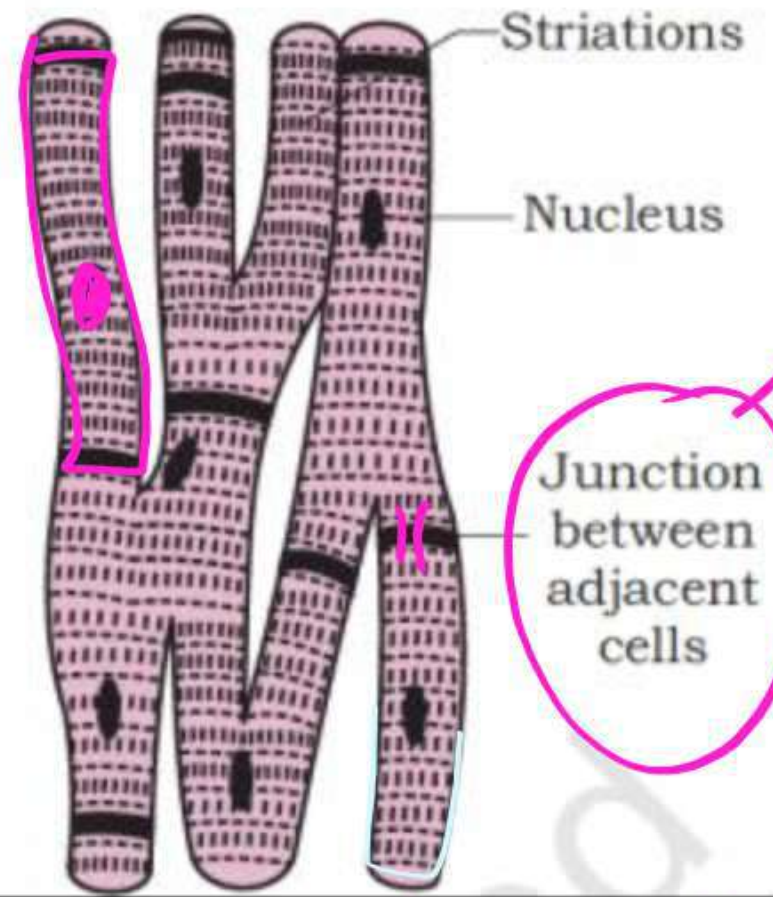
Heart



Cylindrical / unbranched
Multinucleated
Striated
Voluntary



Spindle Shape
Uninucleated
Unstriated
Involuntary



Cylindrical / Branched
Uninucleate
Striated
Involuntary

Human Skeleton (206)

Axial (80)

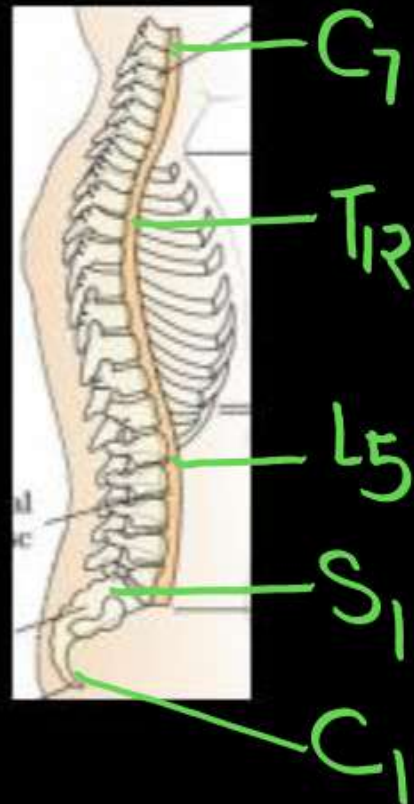
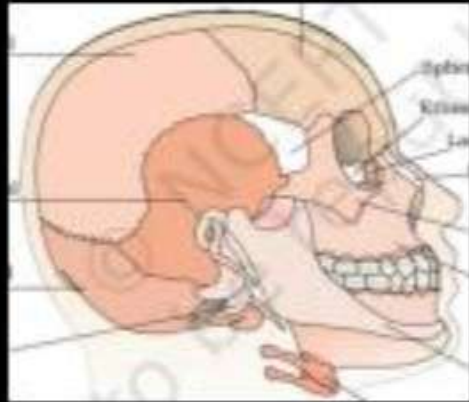
Appendicular (126)

Head (22)

V. Column (26)

Ribs + Sternum (24 + 1)

Skull (22)
+ Hyoid (1)
+ Ear Ossicles (3 + 3)
Cranium (8)
Facial (14)



Girdle (6)

Limb bones (120)

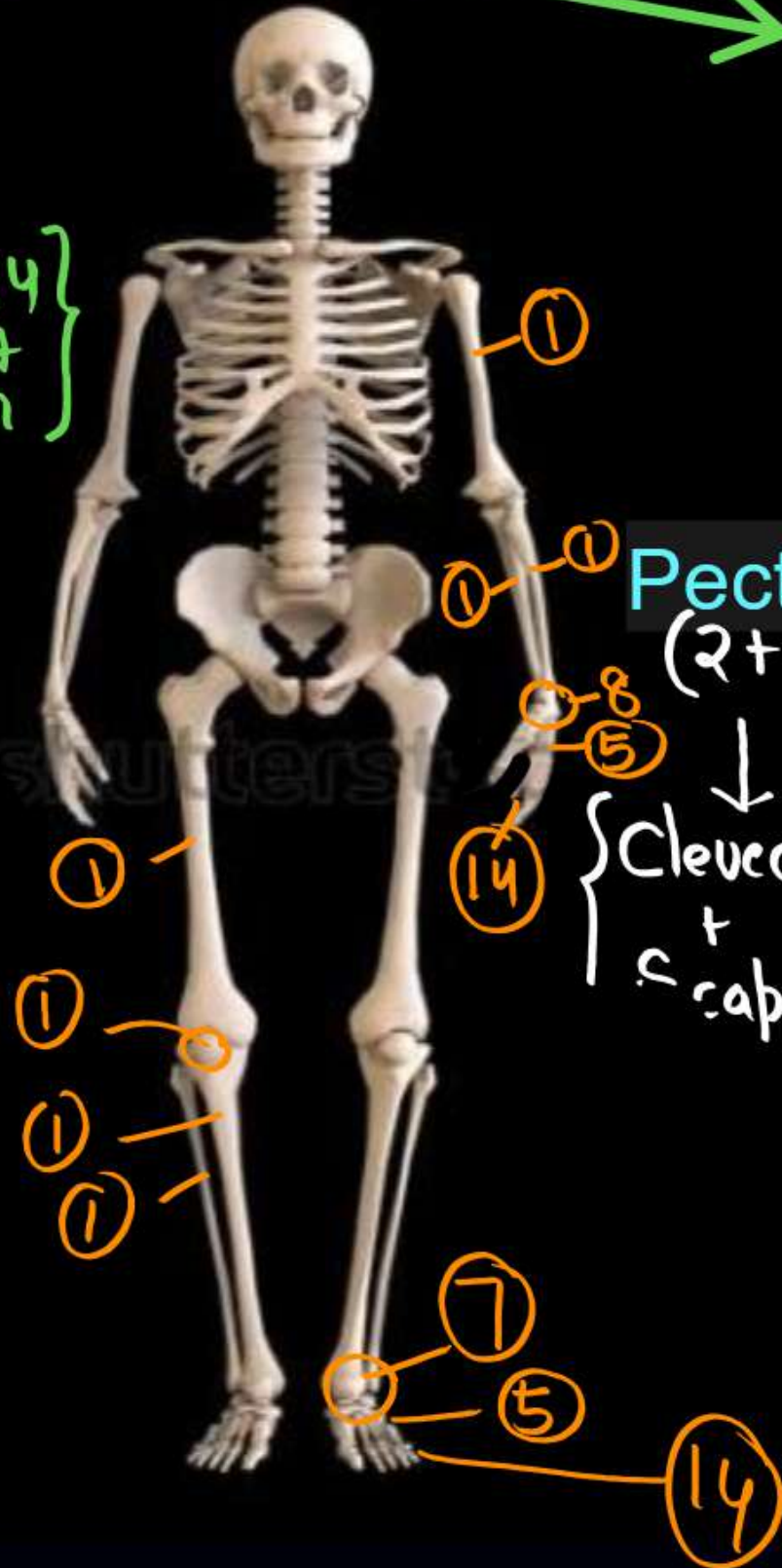
$30 \times 4 = 120$

Pectoral (2 + 2)

Pelvic (1 + 1)

Cleavage + scapula

Coxal

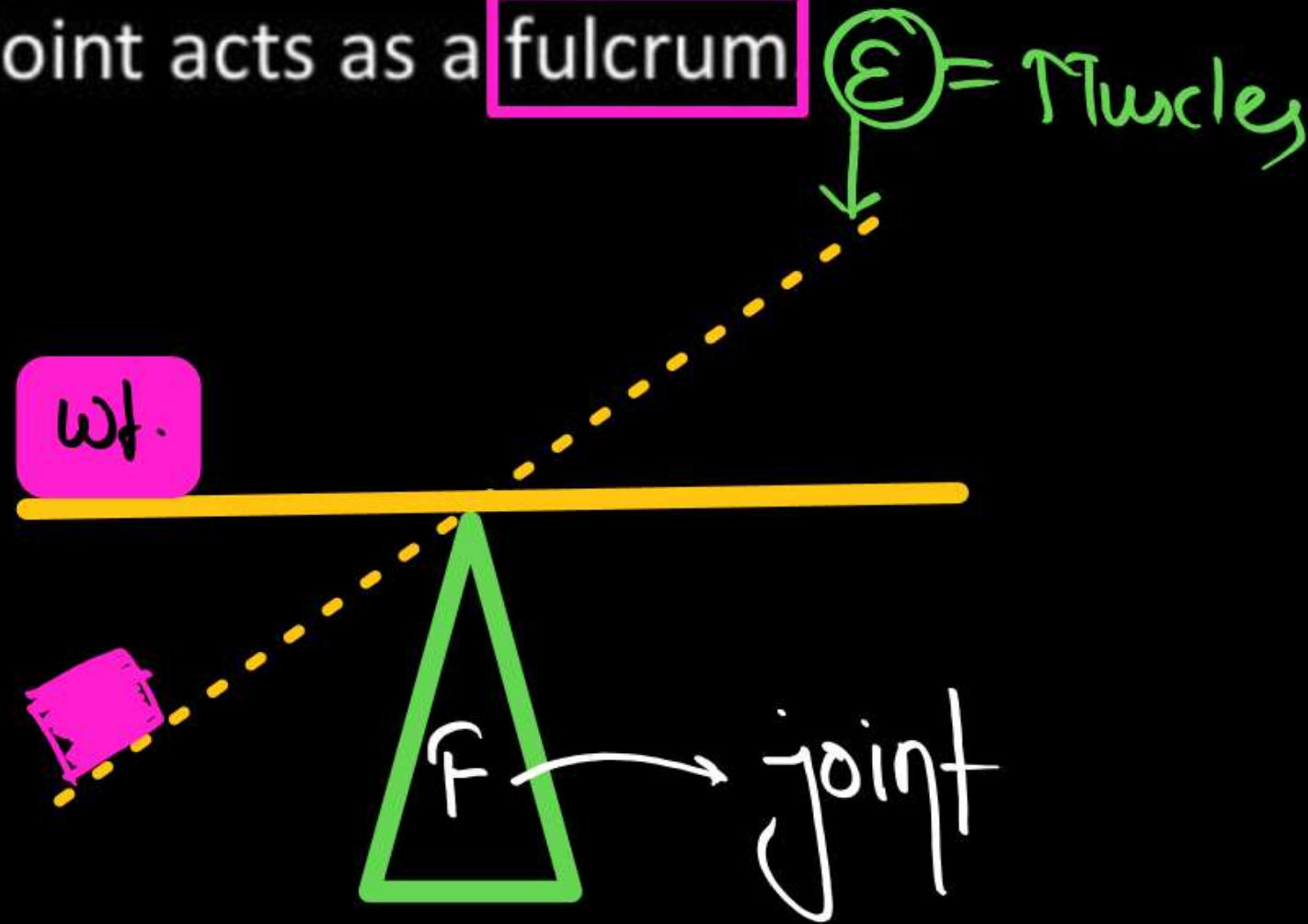


Joints

vvvvvmp $1Q = 100\%$

Joints are **points of contact** between bones, or between bones and cartilages.

Force generated by the **muscles** is used to carry out movement through joints, where the joint acts as a **fulcrum**



Joints

Fibrous joints

(Immovable)

eg → Sutures of Skull

Cartilaginous joints

(Semimovable)

eg → Pubic Symphysis
→ Intervertebral Disc

Synovial joints

(Movable)

Pivot b/w Atlas & Axis

Gliding joints • Wrist & Ankle

Saddle joint 
• Carpal & Metacarpal of Thumb

Hinge

• Knee & Elbow
• b/w Phalanges

Ball and Socket joints

• Shoulder
• Hip

Condyloid joints

• Skull & Atlas
• Metacarpal & Phalanges

Disorders

20.5 DISORDERS OF MUSCULAR AND SKELETAL SYSTEM

Myasthenia gravis: Auto immune disorder affecting neuromuscular junction leading to fatigue, weakening and paralysis of skeletal muscle.

Muscular dystrophy: Progressive degeneration of skeletal muscle mostly due to genetic disorder.

Tetany: Rapid spasms (wild contractions) in muscle due to low Ca^{++} in body fluid.

Arthritis: Inflammation of joints.

Osteoporosis: Age-related disorder characterised by decreased bone mass and increased chances of fractures. Decreased levels of estrogen is a common cause.

Gout: Inflammation of joints due to accumulation of uric acid crystals.

Hormones

I.R

Hypothalamus

Secretory Neurons (Nuclei)

R.H.

PRh
GnRH
CRh
TRh

Portal System

A. Pituitary

PP

Supra optic

Para Ventricular

GHRH
Somatostatin

① Somatotrophic (STH)
or
Growth Hormone

GH

② T.S.H

T₃/T₄ Thyroxin

③ ACTH

Cortisol

④ Gonadotrophin

MSH

⑤ PRL

L.H

F.S.H

Antrogen Sperma.

ovulation
Corpus luteum
Preg.

SMP

Development of Folli
Estrogen.

Vasopressin
[ADH]

[Oxytocin]

⑥ { Contraction of B.U
Absorption of water
from - [ACT
CD

• Contraction in
1) Mammary glands
[Milk Ejection]
2) - Uterus
[Birth]

(Both Synthe. by Hypothalamo.)

• Lactation
• D. Mammary

Disorders

~~vvvmp~~

Hyposecretion of G.H.

Hypertecretion of G.H.

Diabetes Insipidus
(Hyposecretion of ADH)

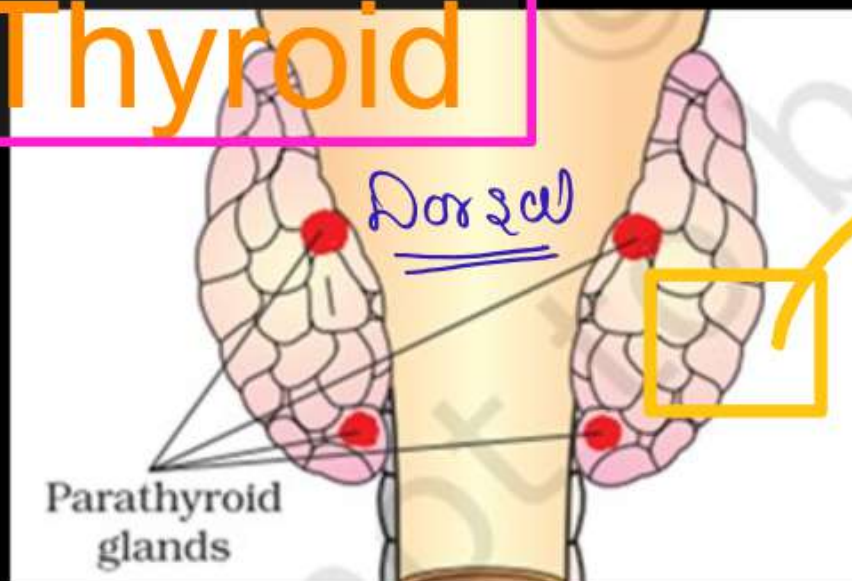
Pituitary Dwarfism
(Midget) (↓ of G.H.)

Acromegaly → Hypertecretion in late stage
↳ Premature Death if unchecked

Gigantism — Hypertecretion in Early stage

Diabetogenic effect → ↑ - G.H.
↳ ↑ - Blood Glucose level.

Thyroid



Thyroid

B.O.



Stromal Tissue

Follicles

[Simple Cuboidal Epi. tissue]

Thyroglobulin [Store form of Thyroxine]

Parafollicular Cells

Release - **T.C.T.**

[Hypocalcemic hormone]

↓ - Blood Ca^{++} level

by —

- Adsorption of blood Ca^{++} in bone matrix
- Elimination of Extra Ca^{++} through kidney

✓ Thyroxine = Iodinated form of Tyrosine Amino acid.

- ✓ • Tri-iodothyronine [T_3] → Most active form
- ✓ • Tetraiodothyronine [T_4] = Thyroxine ✓

Function ⇒

- 1) - Regulation of BMR ✓
- 2) - ↑ Erythropoiesis = ↑ Formation of R.B.C
- 3) - ↑ Blood Glucose level, Breakdown of
 - Fat
 - Protein
 - Carbo.
- 4) - Maintain water-Electrolyte Balance

Disorders

~~vvvvvvvvvv~~

Hyposecretion

of thyroxine

1) - Simple Goiter

2) - Cretinism
 (During Pregnancy)
 Deaf Mutism
 Abnormal Skin
 Low I.Q

3) - Irregular Menstrual
 In adult Female.

Hypersecretion

of Thyroxine.

✓ • Exophthalmic Goiter
 or

• Grave's Disease

• Diabetogenic Effect
 = ↑ - thyroxine

Parathyroid = (2. pairs)

Release = P.T.H
[Hypercalcemic]

↑ - Blood Ca^{++} & ↓ Blood Phospho.

1) - Stimulate = Osteoclast
(↑ - Breakdown of Bone)

2) - Stimulate = J.G. Cells of Kidney
Calcitriol = ↑ absorption
of Ca^{++} from gut

Disorders

Hyposecretion
of P.T.H

↓
[↓ - Blood Ca^{++} level]

↓
(Tetany)

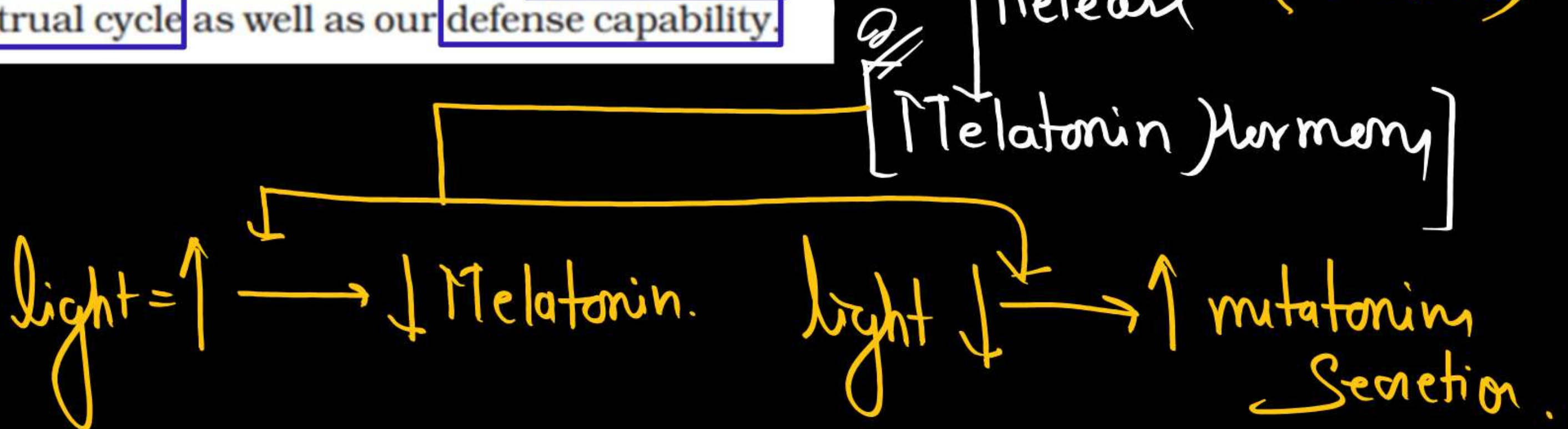
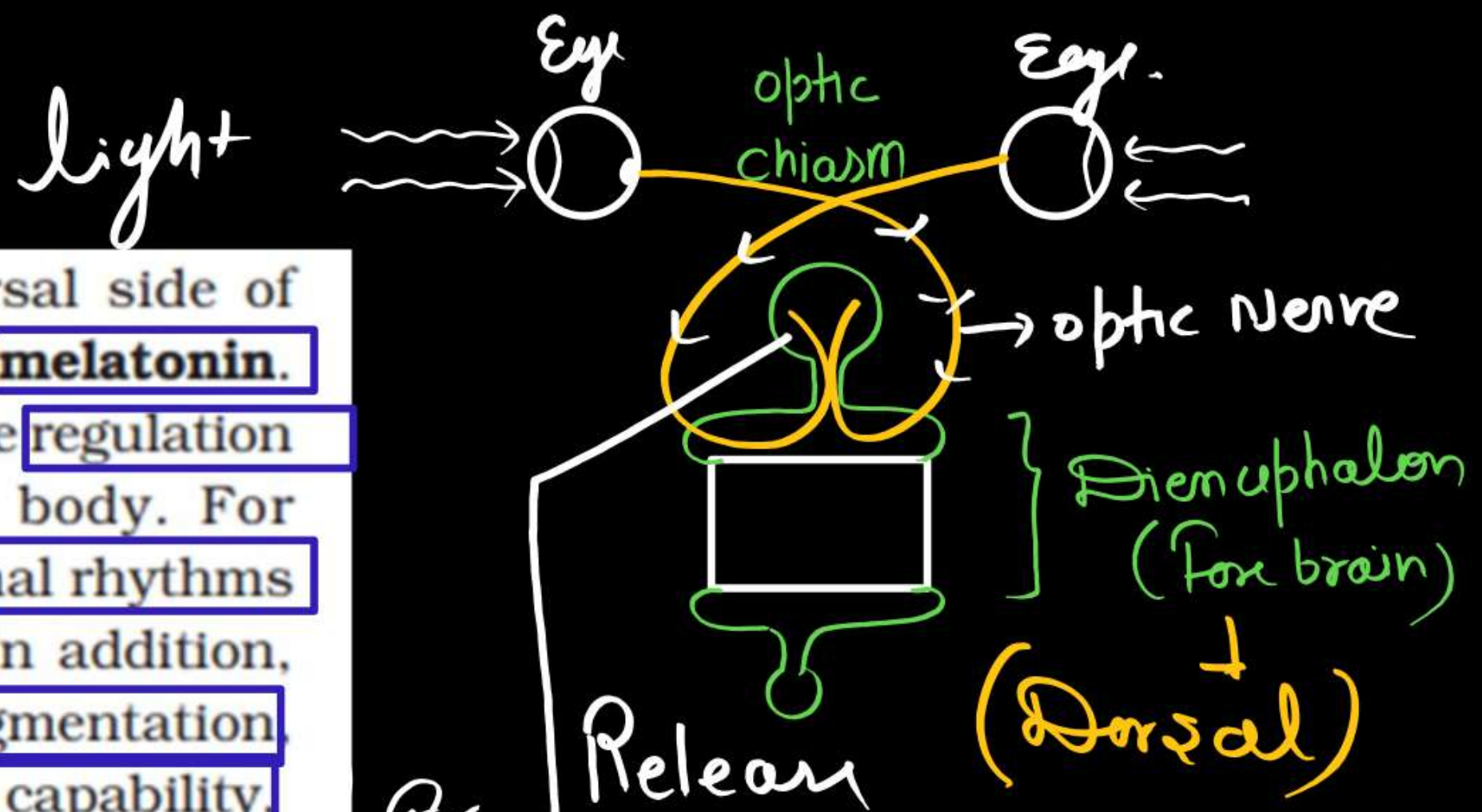
Hypersecretion
of P.T.H

↓
↑ - Breakdown of bone

↓
(Osteoporosis)

PINEAL GLAND

The pineal gland is located on the dorsal side of forebrain. Pineal secretes a hormone called **melatonin**. Melatonin plays a very important role in the regulation of a 24-hour (diurnal) rhythm of our body. For example, it helps in maintaining the normal rhythms of sleep-wake cycle, body temperature. In addition, melatonin also influences metabolism, pigmentation, the menstrual cycle as well as our defense capability.



[Thymus]

Release

[Thymosin Hormone]

Cell Mediated Immunity (CMI)

or

T-lym. Mediated Immunity

T_H-cell

T_K-cell

T_S-cell

T-Memory cell

Maturation

&
Differentiation
of T-lym

Humoral Immunity

or

B-lym Mediated Immunity (B.M.I)

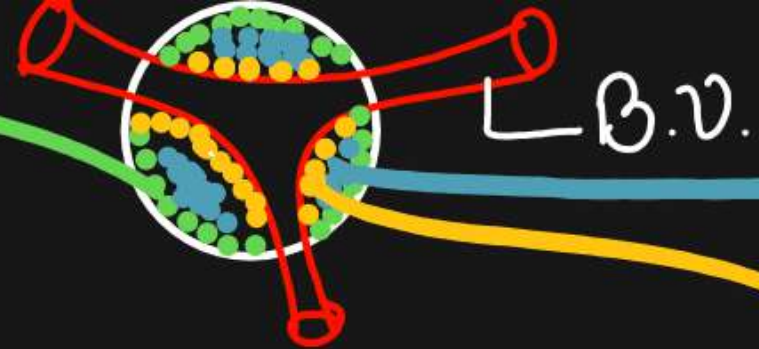
Plasma cells

Memory

Production

(Antibody)

Pancreas



[α -Cells]

Release - Glucagon [Hyperglycemic]

↑ - Blood Glucose by -

1) - Gluconeogenesis = Fat/Protein > Break

2) - Glycogenolysis = Glycogen

3) - ↓ Cellular uptake of Glucose

[β -Cells]

Release - Insulin [Hypoglycemic]

↓ - Blood Glucose level by -

↑ - Cellular uptake of Glucose

[δ -Cells]

Somatostatin

Inhibit ↓ α -cell, β -cell

Disorders

Diabetes
Mellitus

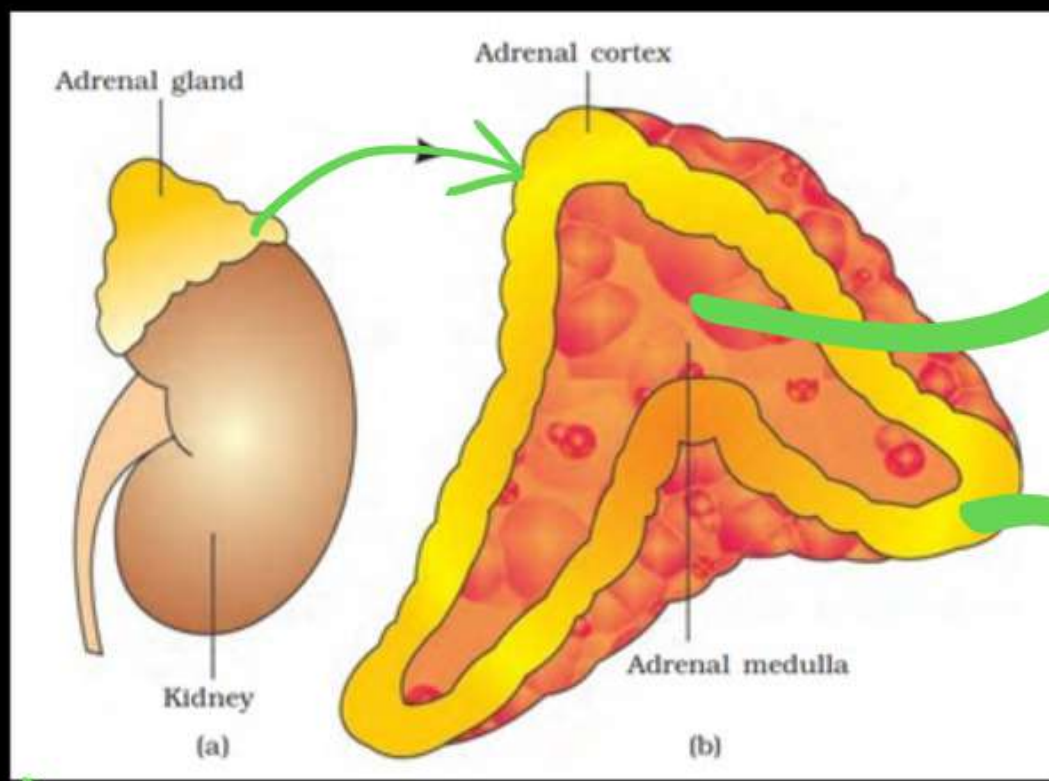
↑ Blood Glucose

↑ - Ketone bodies

↑ - Glucosuria

} Hyposecretion of Insulin

~~vulgar~~



Medulla [Inner]

- Origin = Ectodermal
- Nature of Hormone = Protein / Emergency Hormones or F.F. Hormones or Catecholamines

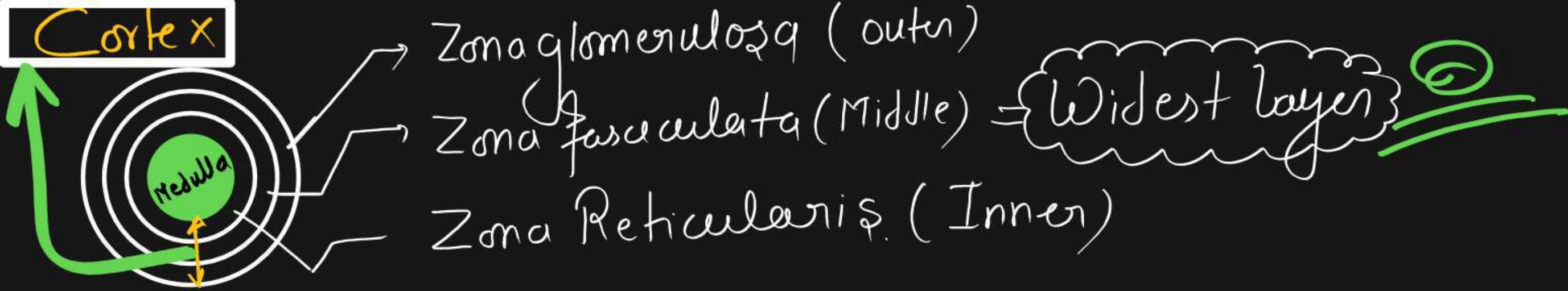
① Adrenalin [80%]

② Noradrenalin [20%]

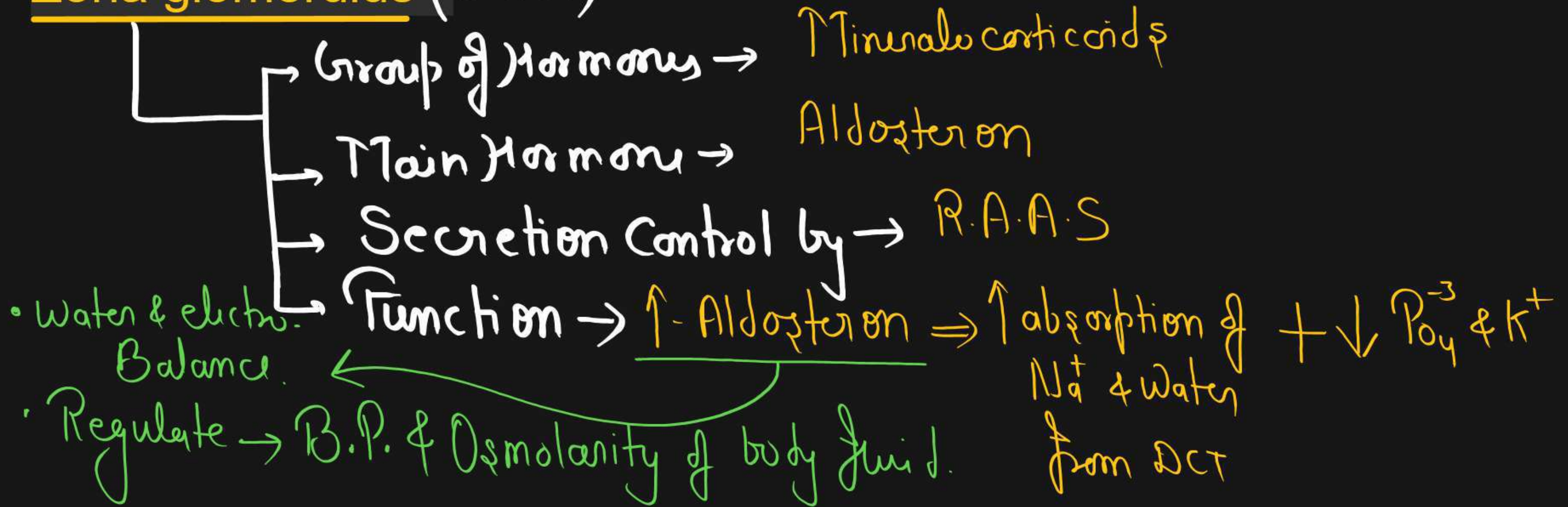
- 1) - ↑ alertness
- 2) - Pupillary dilation, ③ piloerection
- 4) - Sweating etc
- 5) - ↑ heart beat, Rate of Respiration
- 6) - Break down of Glycogen (↑ Blood Sugar)
- 7) - " " of Lipid & Protein.

Cortex [Outer]

- Origin = Mesodermal
- Nature of Hormones = Steroids
- Vitally Imp. for Life



Zona glomerulus (outer)



Zona fasciculata (Middle)

- 1) - Group of Hormone → Glucocorticoids
- 2) - Main Hormone → Cortisol (Hydroxycortisone)
- 3) - Secretion Control by → A.C.T.H A. Pituitary (Life Saving Hor.)
- 4) - ~~Spoke~~ Function
 - Stimulate - Formation of R.B.C
 - Antistress Hormone.
 - ↑ - Blood glucose level.
 - Act as Immuno suppressor
 - Used in ⇒ Autoimmune disorders ✓
 - ⇒ Organ Transplantation ✓
 - ⇒ Allergy ✓
 - ⇒ Inflammation ✓

Zona reticulosis (Inner) → Sex Corticoids

Female Sex Hormone } ⇒ Estradiol

Male Sex Hormone } ⇒ [D.H.E.A]
(Androgen)

- Responsible
- Body Hair
 - Pubic Hair
 - Facial Hair at Puberty

Disorders →

Hyposecretion

⇒ Addison's (↓ Gluco. Miniralo.)
Re-NEET

Hypersecretion

⇒ Cushing (↑ Glucocorti.)
Conn's (↑ Mineralocor.)



Hormone of - vvv

Imorga.

1) - Heart = ANF

↳ Vasodilator -
inhibit RAAS -
Diuretic

2) - Kidney (J.G. cells) = Erythropoietin = ↑ Erythropoiesis

3) - GIT = Gastrin ✓ = ↑ Gastric juice

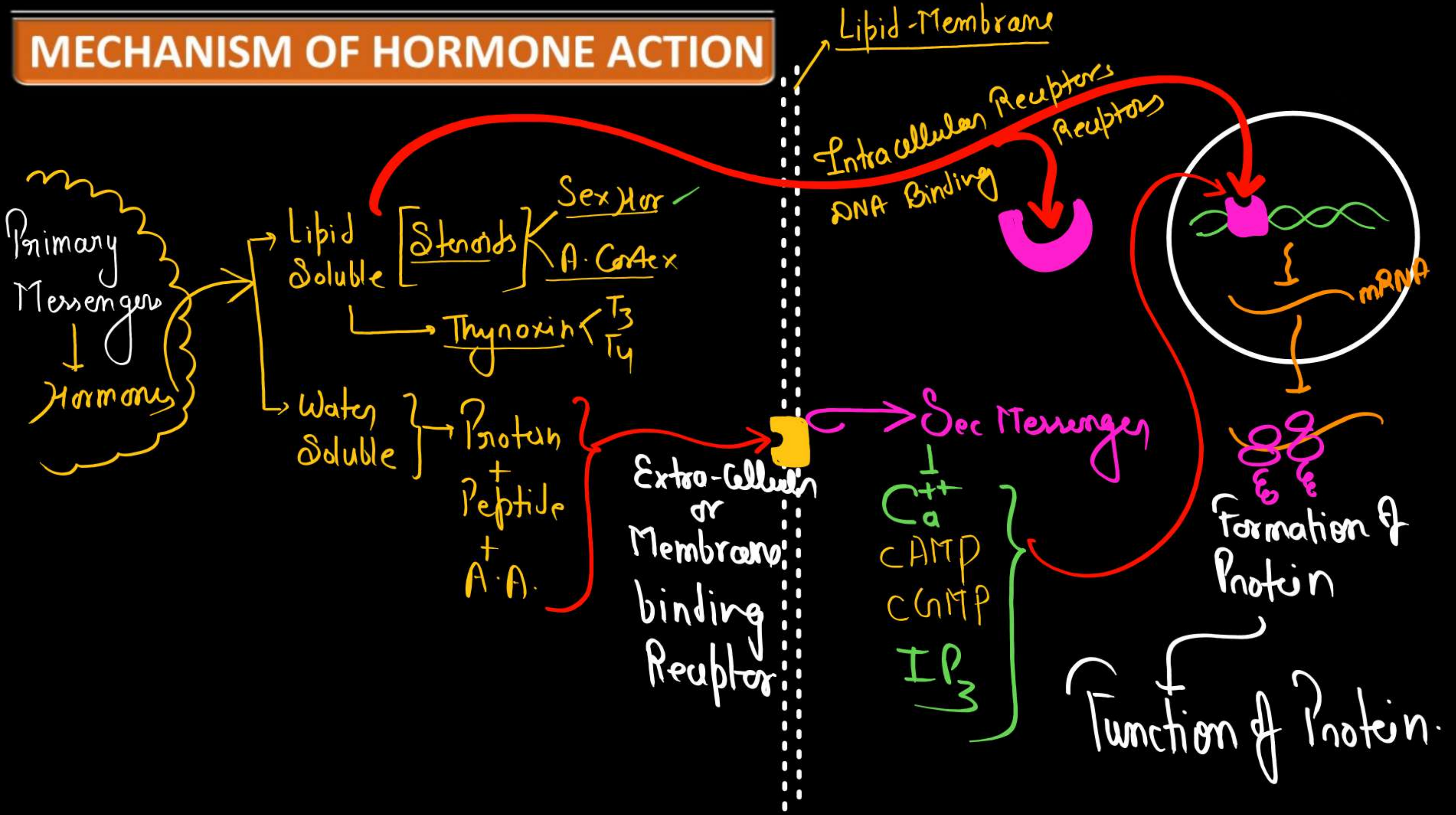
GIP ✓ = Inhibit Gastric Motility & HCl

Secretin = ↗ Stimulate Secretion of Alkaline fluid from - [Liver
Pancreas

CCK = Stimulate ↗ Pancreatic juice
Bile juice.

vvv

MECHANISM OF HORMONE ACTION





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Excretory

Types of nitrogenous waste

Ammonia

- Pri. Nitrogenous Waste
- Highly Toxic → Not Storable
- Readily Soluble → For removal require lot of water
- Ammonia diffuse through Skin & Gills = NH_4^+
- Kidney donot play any Significant Role in its Removal

Ammoniotelic

eg -
Aq. Arthropods ✓
Aq. Amphibia (larva) ✓
Bony fish ✓

Types of nitrogenous waste

Urea

- Produced in Live from Ammonia (Ornithine Cycle)

- Less Toxic = Small amount Storable

Blood = 30 mg/100ml Kidney = Maintain

Osmolarity Gradient (Counter Current)

- Less Soluble → For removal require less water

Ureotelic

- eg- Terrestrial Amphibia (Adult)
- Marine fish (Carti.)
 - Mammals

Types of nitrogenous waste

Uric acid

- In Birds & Reptile → Major End Product of Protein Metabolism
- In Mammals uric acid also produced

by breakdown of Purines (A & G)

A + G → Xanthine → Uric acid Crystals

- Uric Acid — Least Toxic ✓
— Least Soluble → for removal
Require least water

Uricotelic

eg - Reptile ✓
Birds ✓
❌ Land Snail ✓
Cockroach (Arthropods) ✓

Gouty
Arthritis

Types of Nephron

Cortical Nephron

85%

Small

L.H = Short

U.R = -nt

Max^m part = Cortex

Work during
Normal Condition

Juxtamedullary Nephron

15%

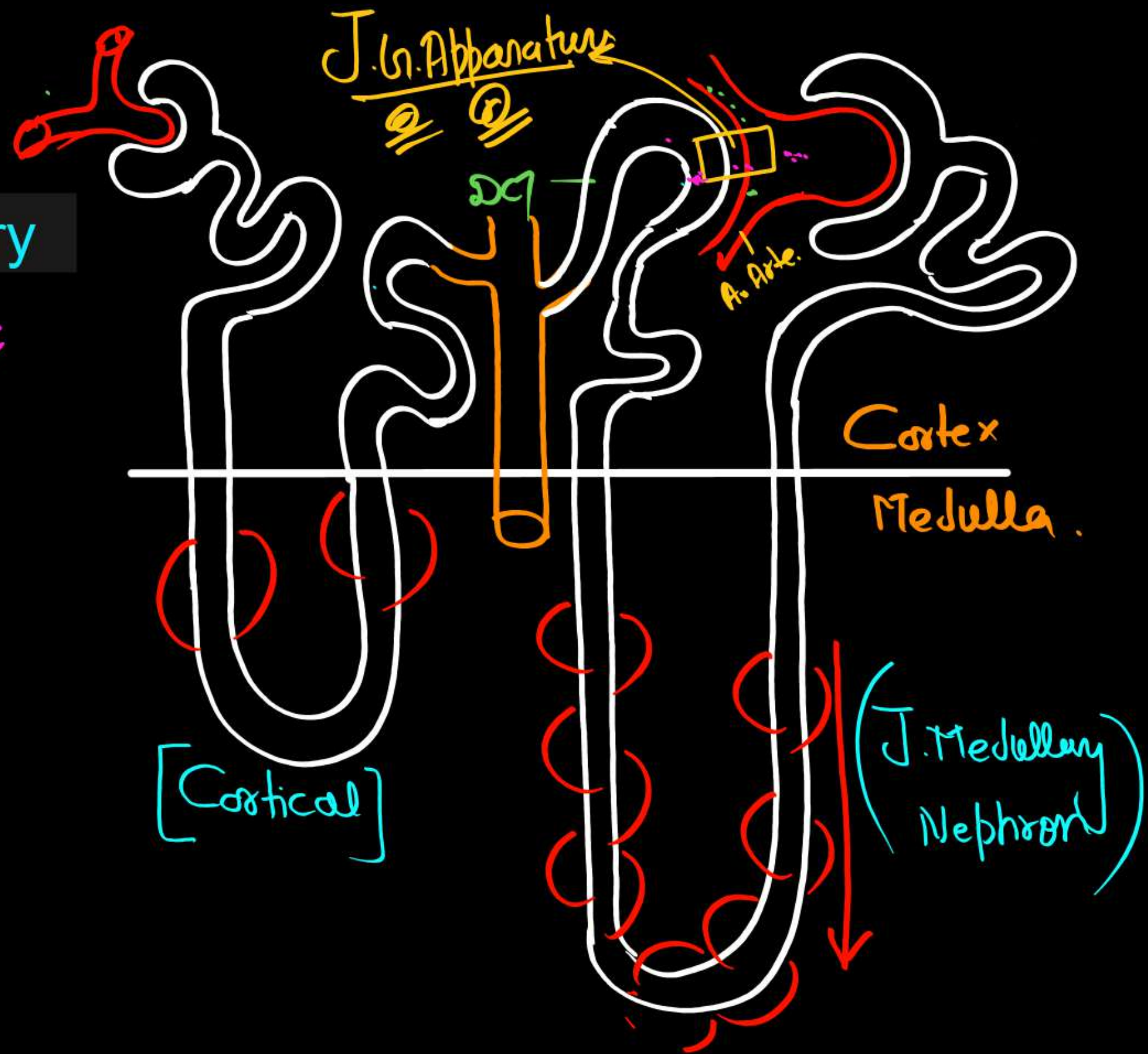
Large

Long

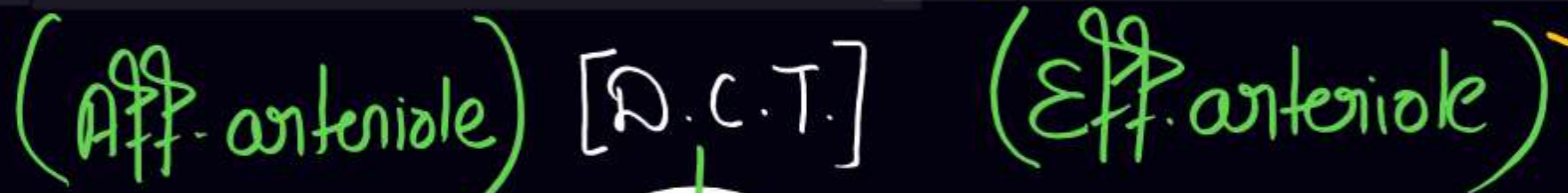
+nt

= Medulla

Emergency



Juxta Glomerular apparatus



② J. Cr. Cells →

1) Renin = $\uparrow$ G.F.R / $\uparrow$ B.P.

2) - Erythropoietin = $\uparrow$ Erythropoiesis $\rightarrow$ Formation of R.B.C

3) - Calcitriol (Vit D₃) = ↑ absorption of Ca⁺⁺ from gut food.

Macula
cells of DCCT
+
J.G. Cells
↓
Modified Smooth
Muscles of Aff. arterial

(Aff. antennale)

[D.C.T.]

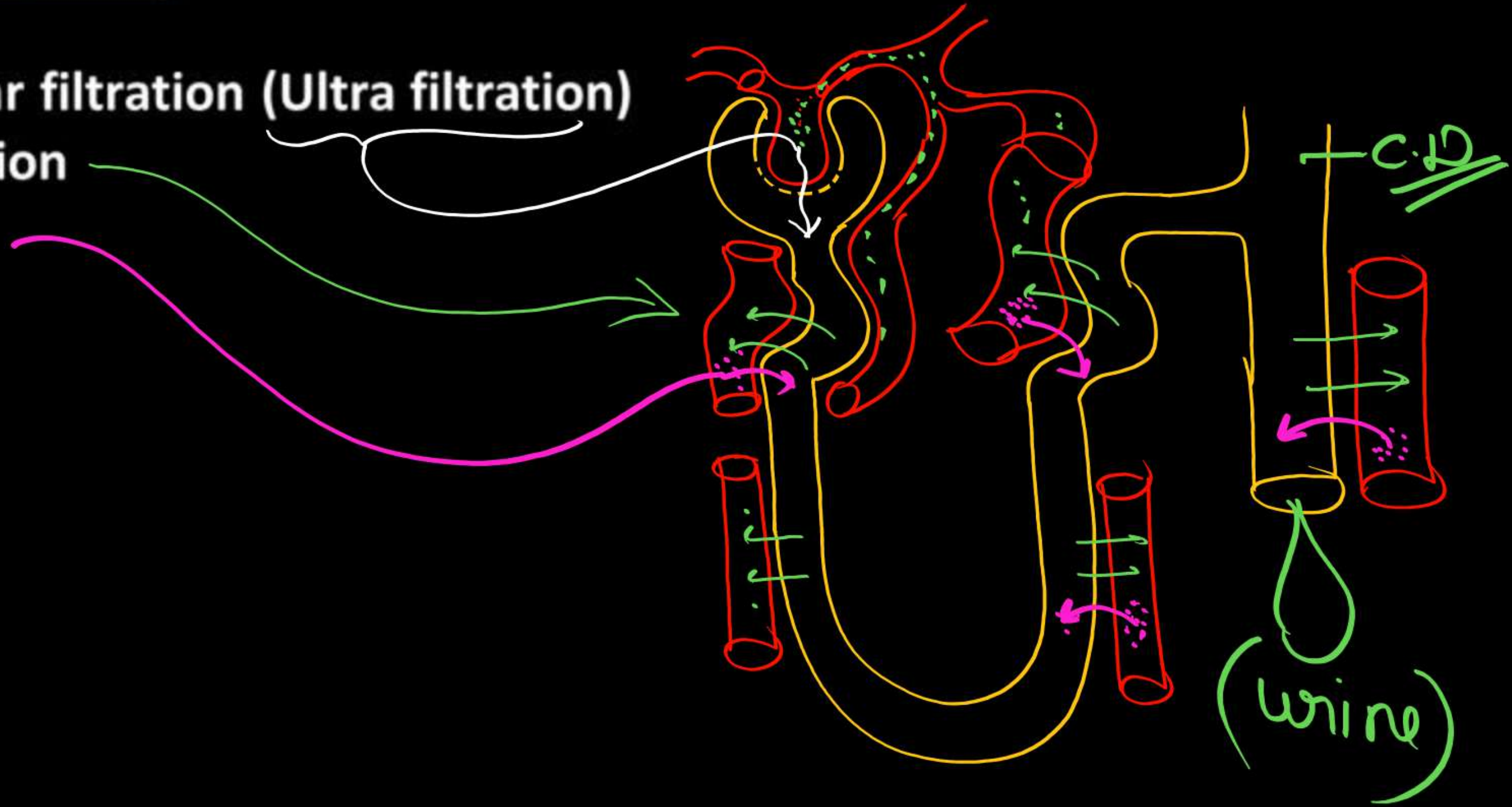
(Eff. arteriole)

Macula
cells of RCT
+

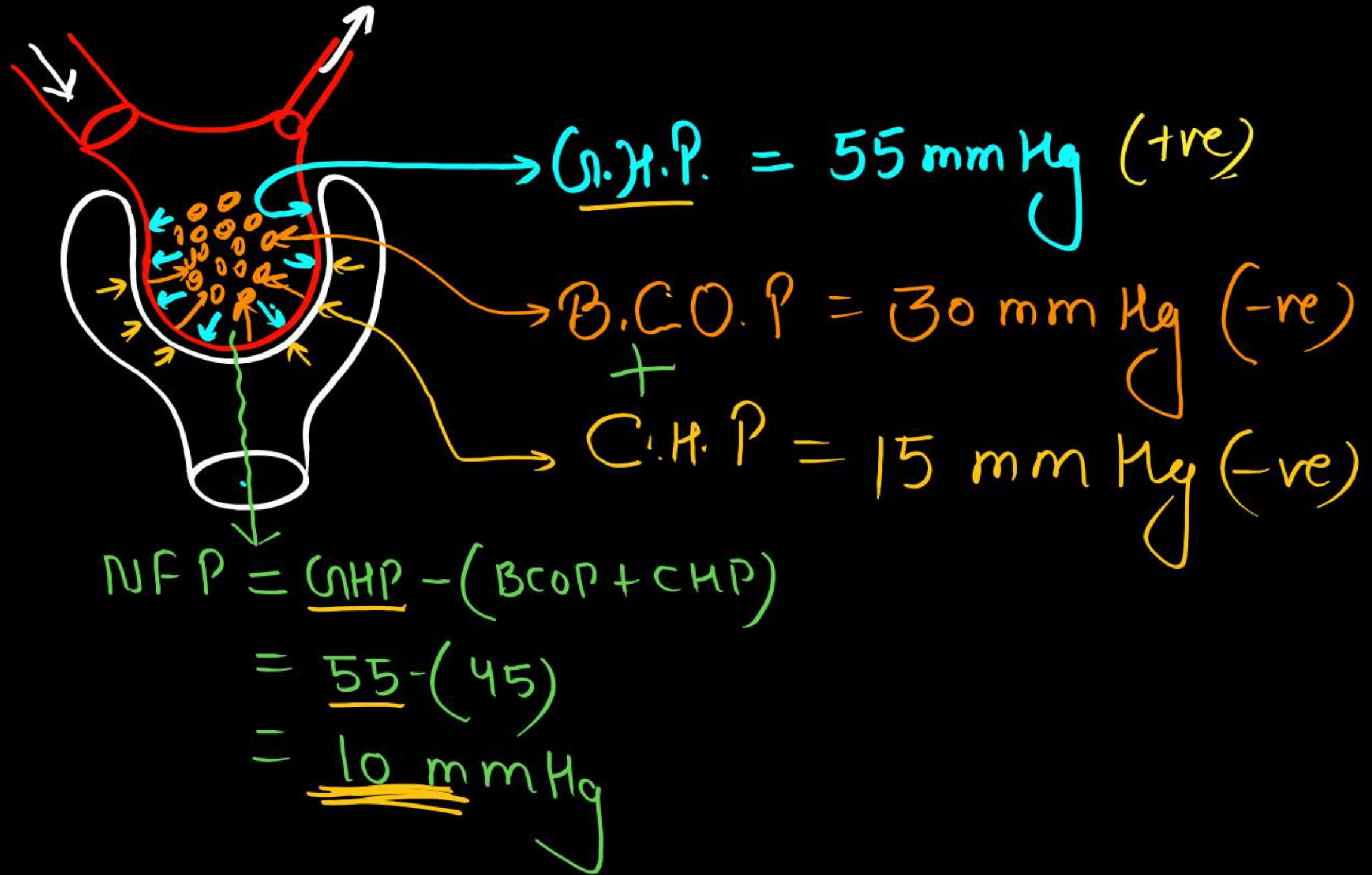
J.G. Cells ←

Urine Formation

- (1) Glomerular filtration (Ultra filtration)
- (2) Reabsorption
- (3) Secretion



(1) Glomerular filtration (Ultra filtration)



2) GFR

vvv fmp
NEET

1100-1200 ml blood
flow by both kidney
Per minute

or
 $\left\{ \frac{1}{5} \text{ of Cardiac out put} \right\}$

99% Reabsorbed
in Blood



Blood Corpuscles - not
Plasma protein - not

vvv fmp

Filtrate

125 ml/minute [180 l/Day]

{ G.F.R }

Only 1% Convert into urine

= 1.8 l/Day (1.5 l/Day)

Tubular reabsorption

PCT → Brush Bordered Cuboidal

Max^m = 70-80% → Ions & water

Glucose
Amino acids
Vit-C
+ve ions (Na^+)
-ve ions
Water

100% } Active Transport

Passive Transport

Osmosis

LH ← = Squamous
↑ = Cuboidal

Min^m = 5%

↓ = H_2O Passive
↑ = Thin = Na^+
Thick = Na^+ + Active

DCT = Cuboidal

Conditional (15%)
- Na^+
- Water

Hypo.

ADH
↓
DCT/CD
+ Water

Adrenal glands

Aldosterone
↓
DCT

+ Na^+ } Active

CD = Cuboidal

Water = ADH
Ureter.

Absorb by Diffusion.

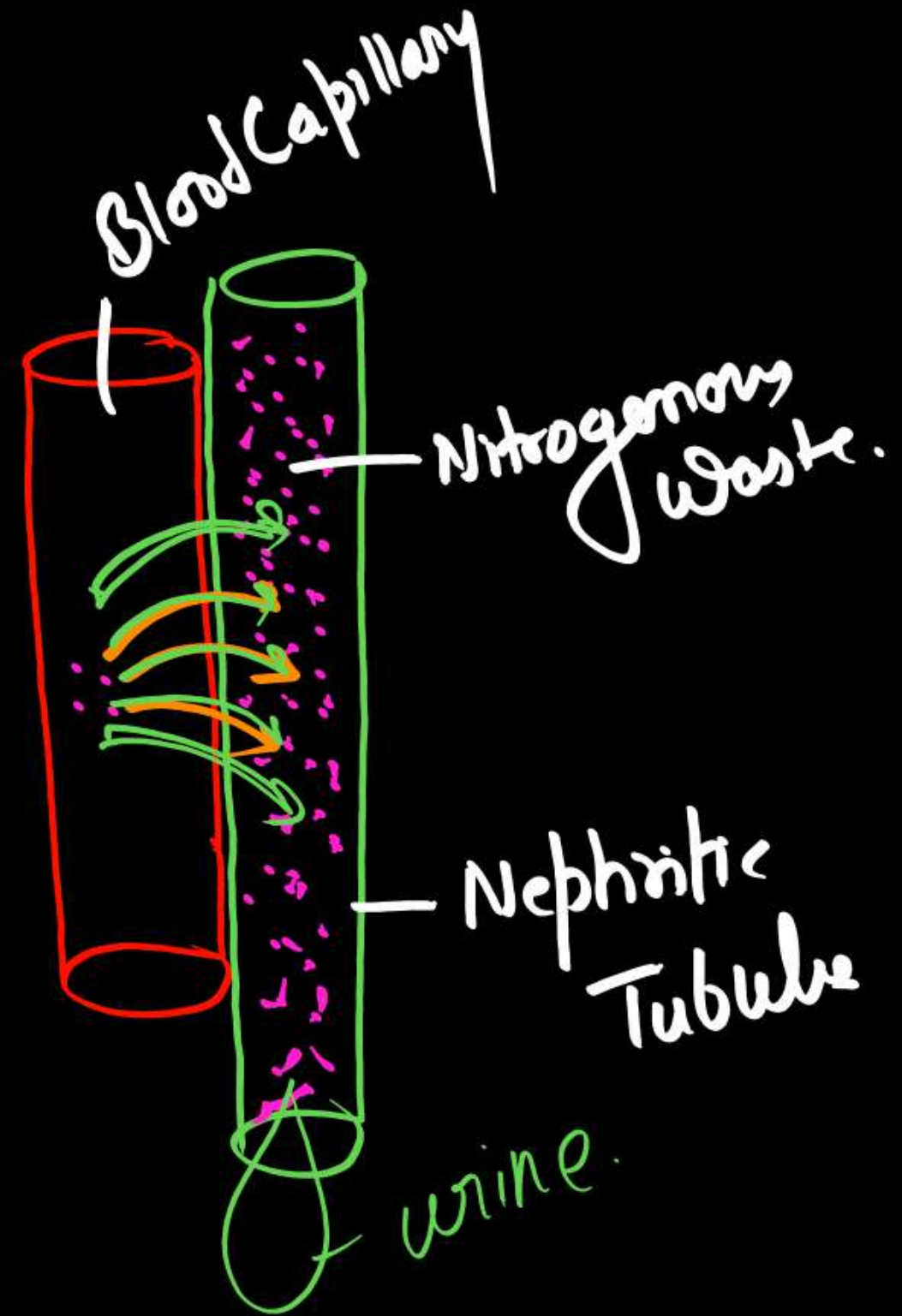
(3) Secretion - (Active)

a) - P.C.T. = H^+ , NH_3 , Drugs, Toxic Sub. etc.

b) - L.H. $\left\{ \begin{array}{l} \downarrow = X \\ \uparrow = \text{urea} \end{array} \right.$

c) - D.C.T = H^+ , NH_3 , K^+

d) - C.D = H^+ & K^+

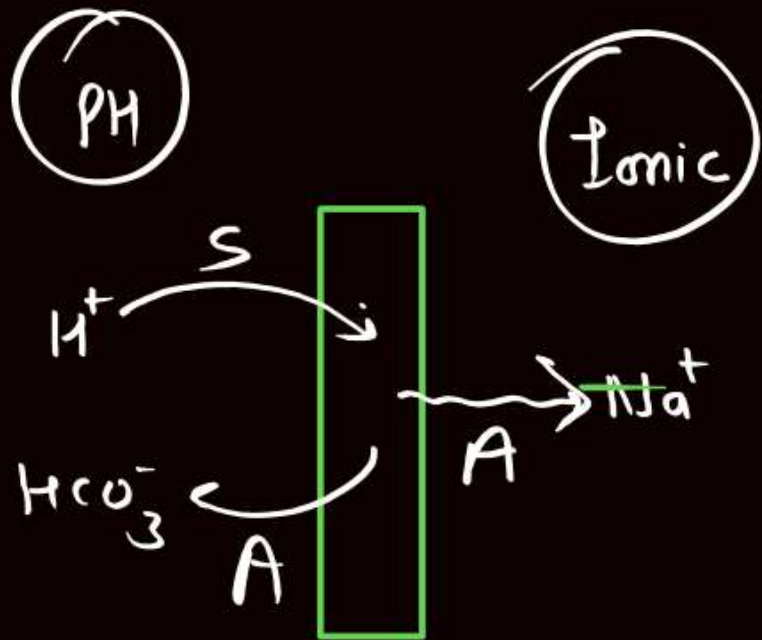


Function of Tubules

vvvvvfmjz

PCT

Maintain $\rightarrow$ pH & Ionic Bala.



LH

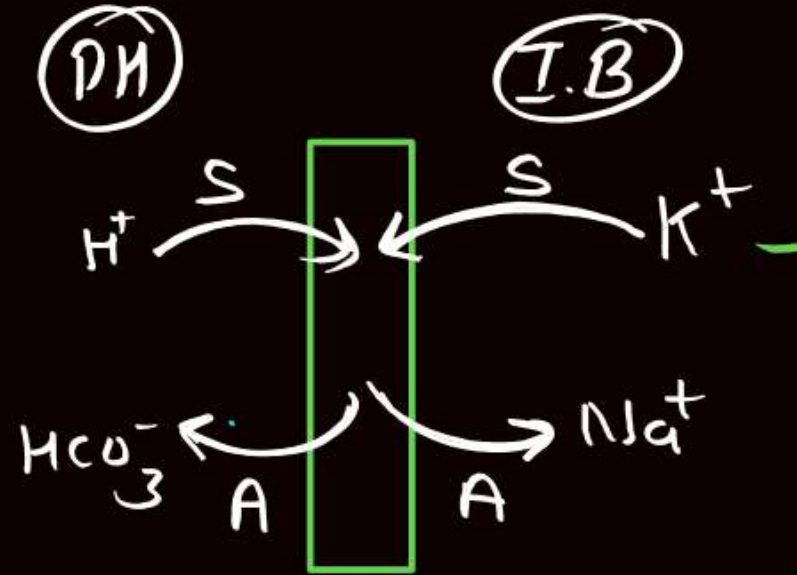
~~X~~

Absorption $\leftarrow H_2O$
 $\leftarrow NaCl$
 +
 Secætion = urea
 $\downarrow$
 Maintain
 Osmolarity Gradient



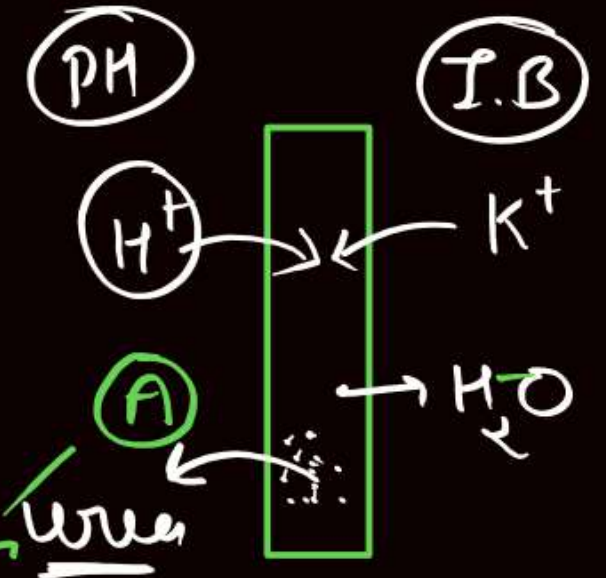
DCT

Maintain = pH & Ionic Bala.

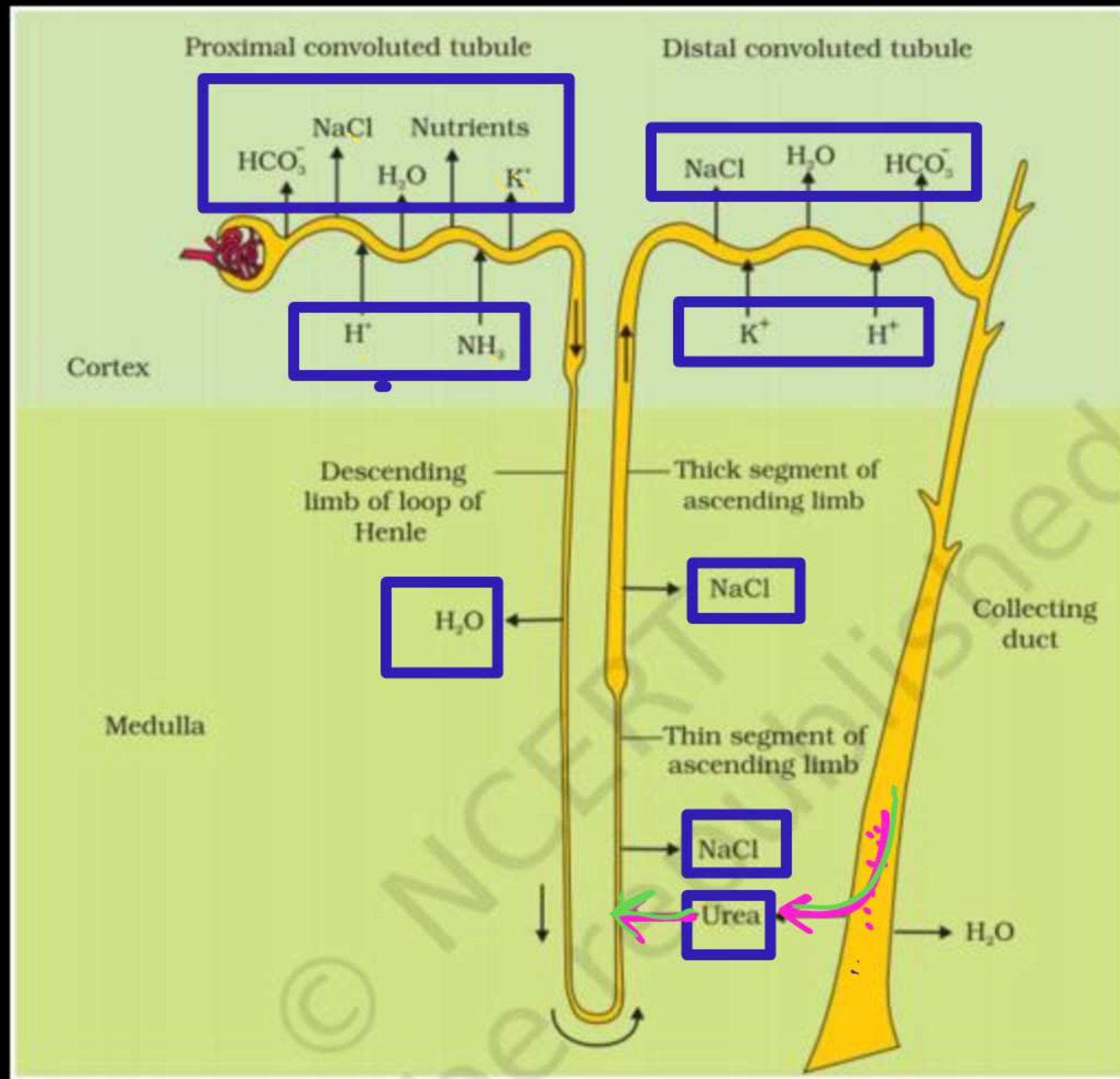


CD

Main = pH & Ionic



Absorption of -
 water = Cons. Urine
 Urea = Maintain
 Osmolarity



Counter Current Mechanism

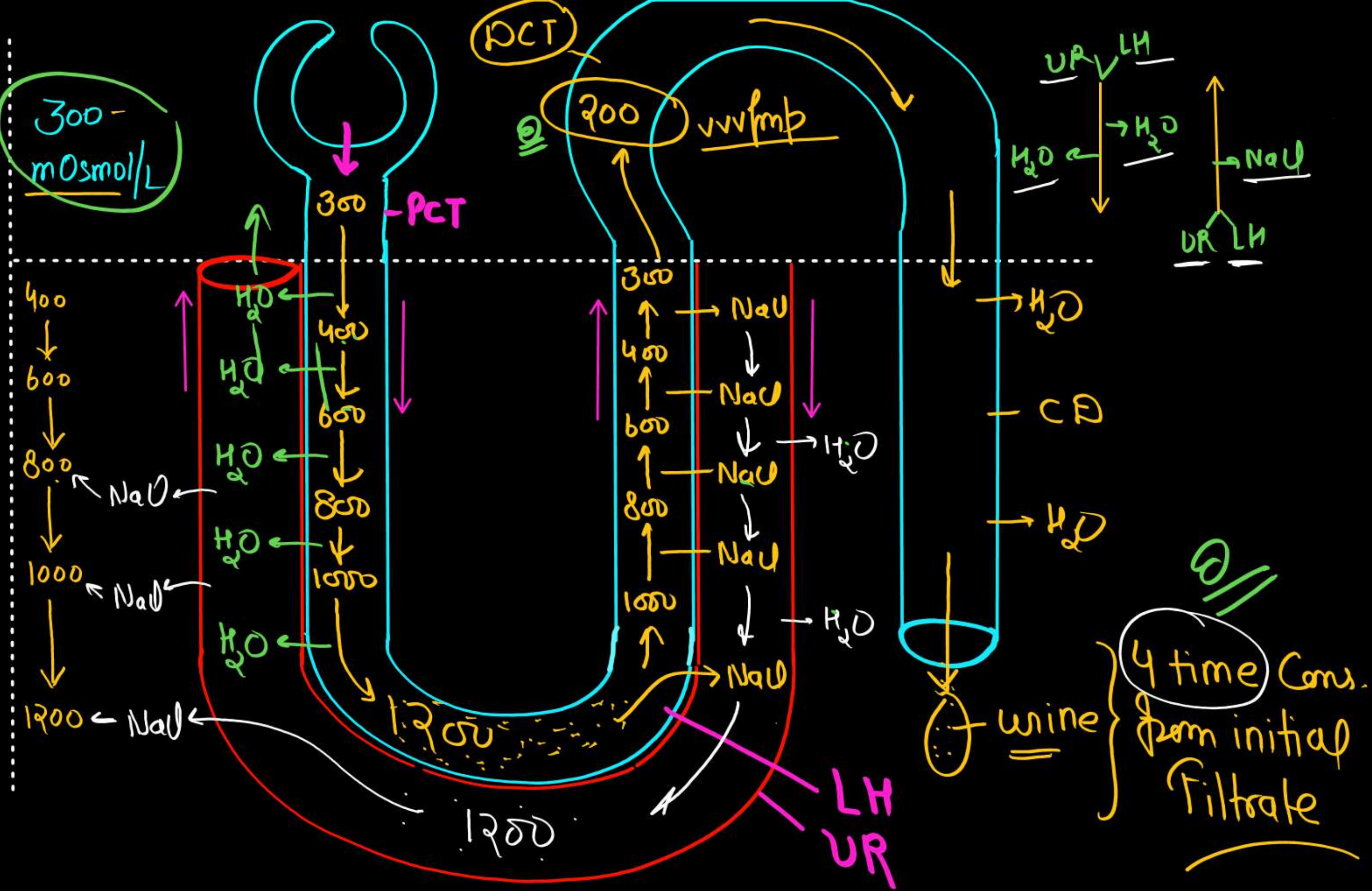
- Mammals have the ability to produce a concentrated urine. The Henle's loop and vasa recta play a significant role in this.
- Osmolarity gradient towards the inner medullary interstitium mainly caused by NaCl and urea.
- Human kidneys can produce urine nearly four times concentrated than the initial filtrate formed.

Cortex

300 -
mOsmol/L

Medulla

NaCl
+
Urea



Regulation of Kidney Function

(1) Regulation By Hypothalamus

if G.F.R. = ↓
or
B.P. = ↓
or
Blood Water = ↓
or
Blood Osmolarity = ↑
or
Dilution of urine = ↑

Stimulate

Hypothalamus
Osmolarity Receptors

A/P Pituitary

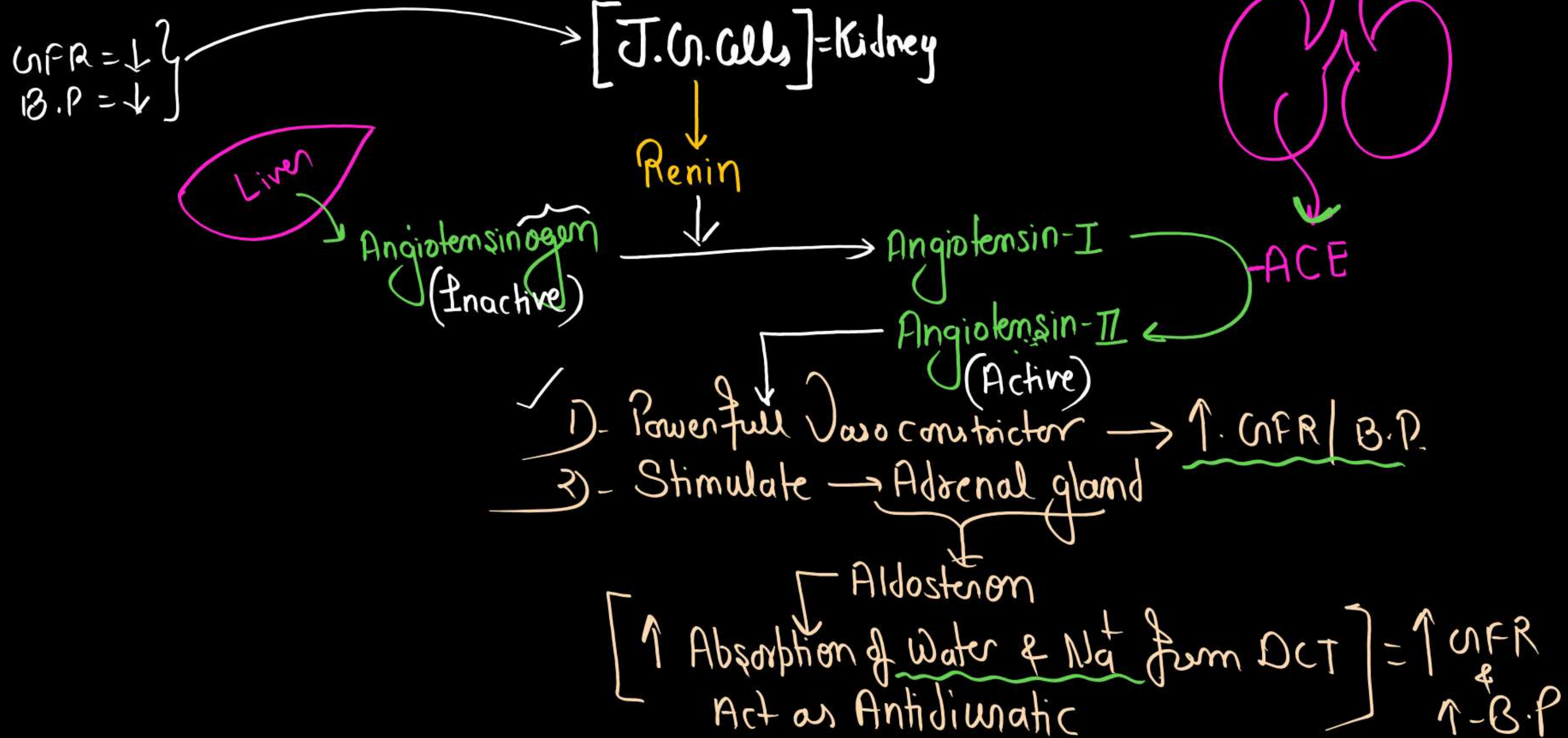
ADH (Vasopressin)

✓ ↑ absorption of
Water from DCT & CD

Constriction in
Blood Vess. = ↑ B.P
&
↑ GFR

↑ Cons. of urine &
↑ GFR / B.P etc
→ Feed back inhibition of ADH ✓

(2) Regulation By J.G. Cells of kidney (R.A.A.S.)



(3) Atrial Natriuretic Factor (ANF)

[Antagonistic of R.A.A.S.] → ↑ Pressure on wall of Heart

↓
↑ B.P.

↓
= [Wall of Atria] //

} Release

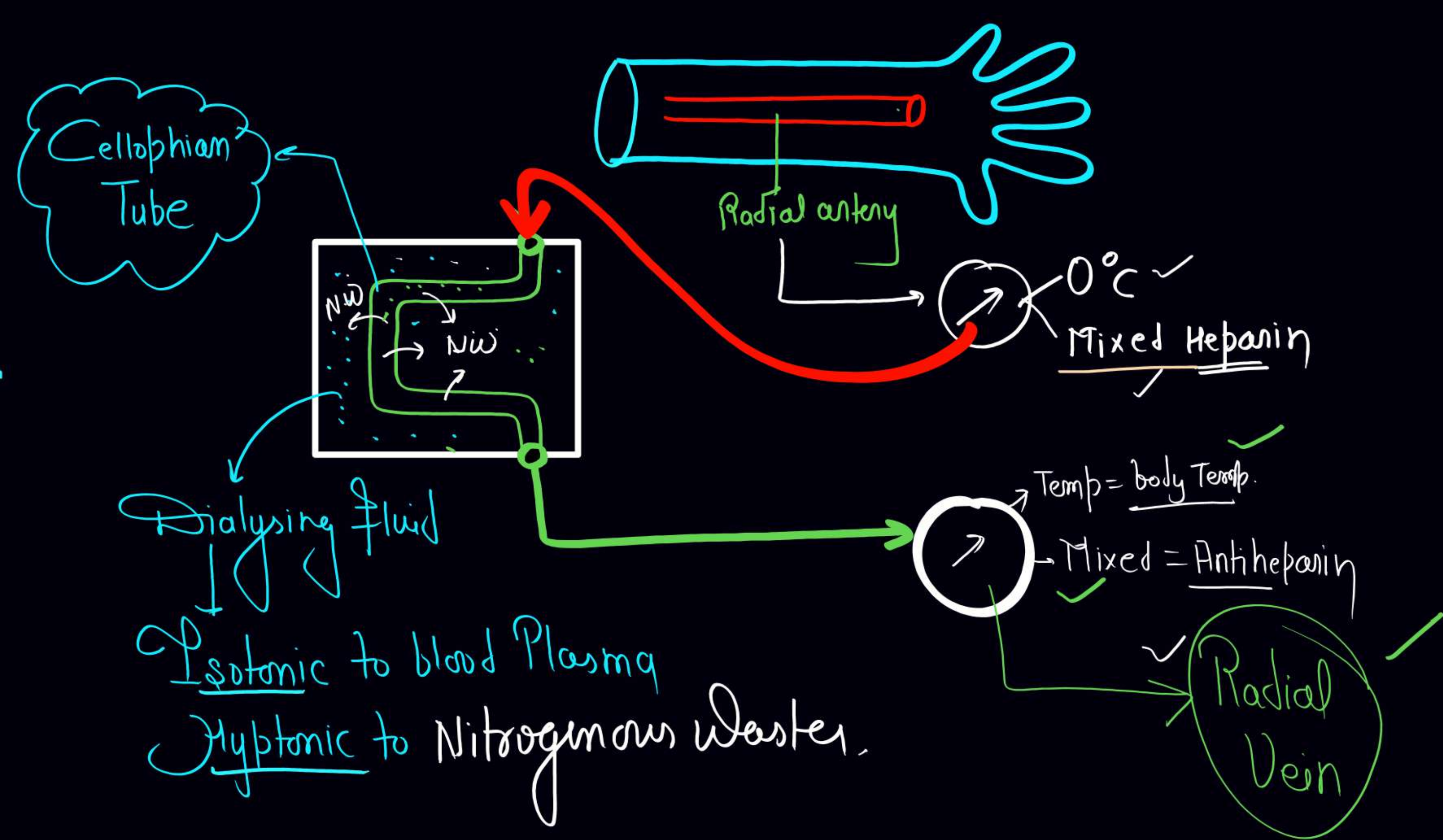
= [ANF] //

- 1) Powerful = Vasodilator
- 2) Inhibit = R.A.A.S & Aldosterone Secretion
- ✓ 3) Diuretic = ↑ Dilution of urine.

DISORDERS OF THE EXCRETORY SYSTEM

- Malfunctioning of kidneys can lead to accumulation of urea in blood, a condition called **uremia**, which is highly harmful and may lead to **kidney failure**. In such patients, urea can be removed by a process called **hemodialysis**. During the process of haemodialysis, the blood drained from a convenient artery is pumped into a dialysing unit called **artificial kidney**.





} Diabetes

- Glycosuria : - Presence of glucose in urine.
- Ketonuria : - Presence of ketone bodies in urine
- Renal calculi: - Stone or insoluble mass of crystallized salts (oxalates, etc.) formed within the kidney.
- Glomerulonephritis: - Inflammation of glomeruli of kidney



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Neural Control.....

Human Brain - Central information Processing Organ

(1) Fore Brain

→ Telencephalon

→ Diencephalon

Cerebrum

olfactory lobes

Amygdala

Hippocampus

Thalamus

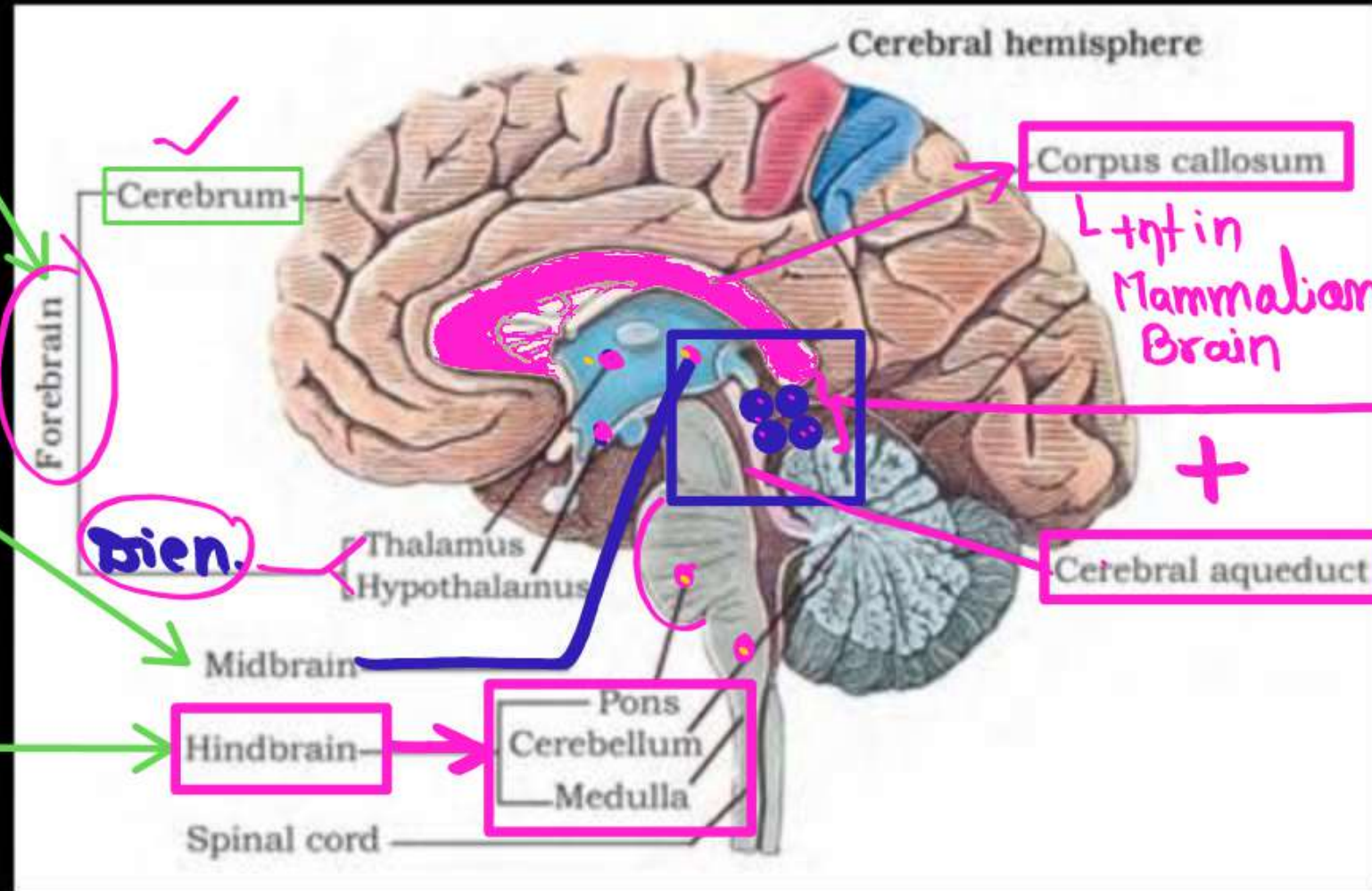
Hypothalamus

Epithalamus

Limbic System

(2) Mid Brain

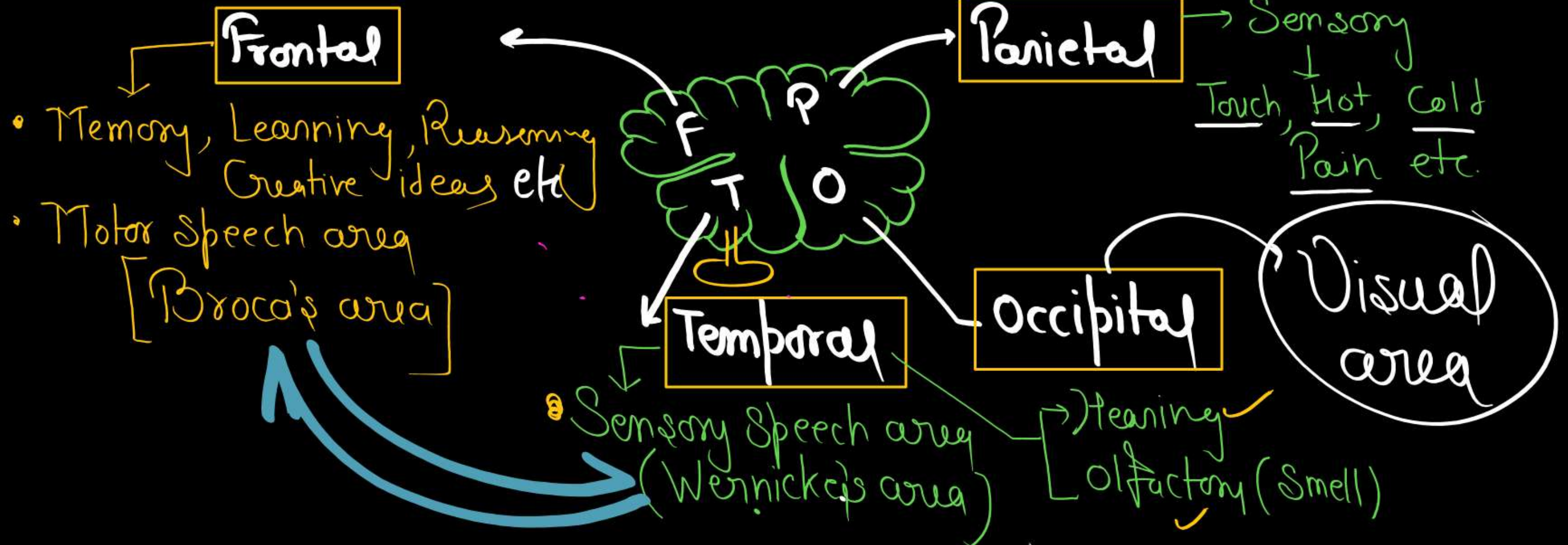
(3) Hind Brain



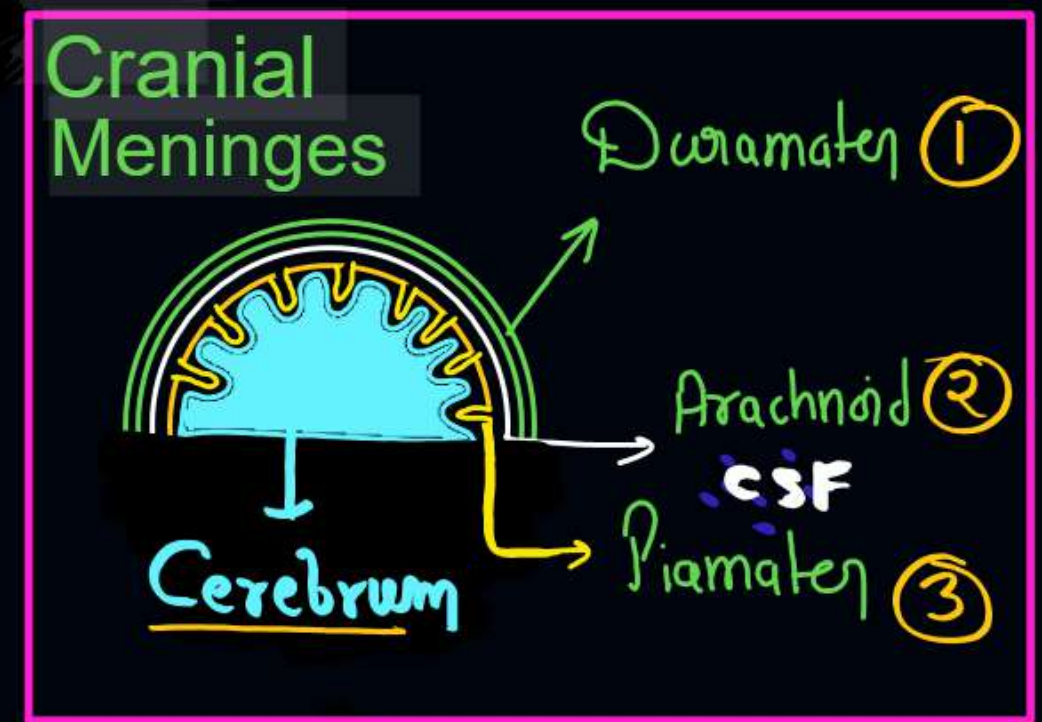
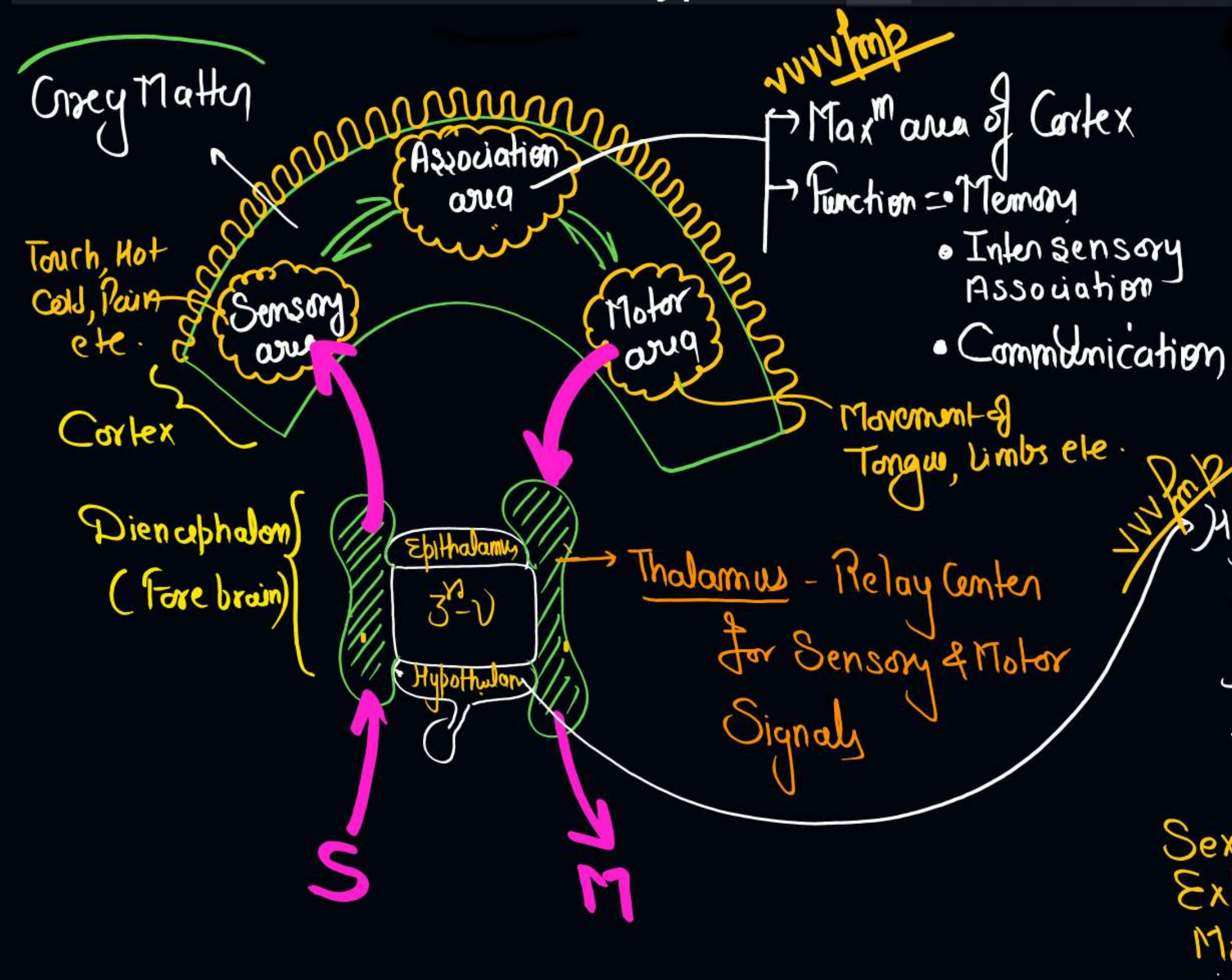
Corpus callosum

→ Thick bundle of Neurons, which Co-ordinate b/w both hemispheres
→ + not only in Mammals. Brain

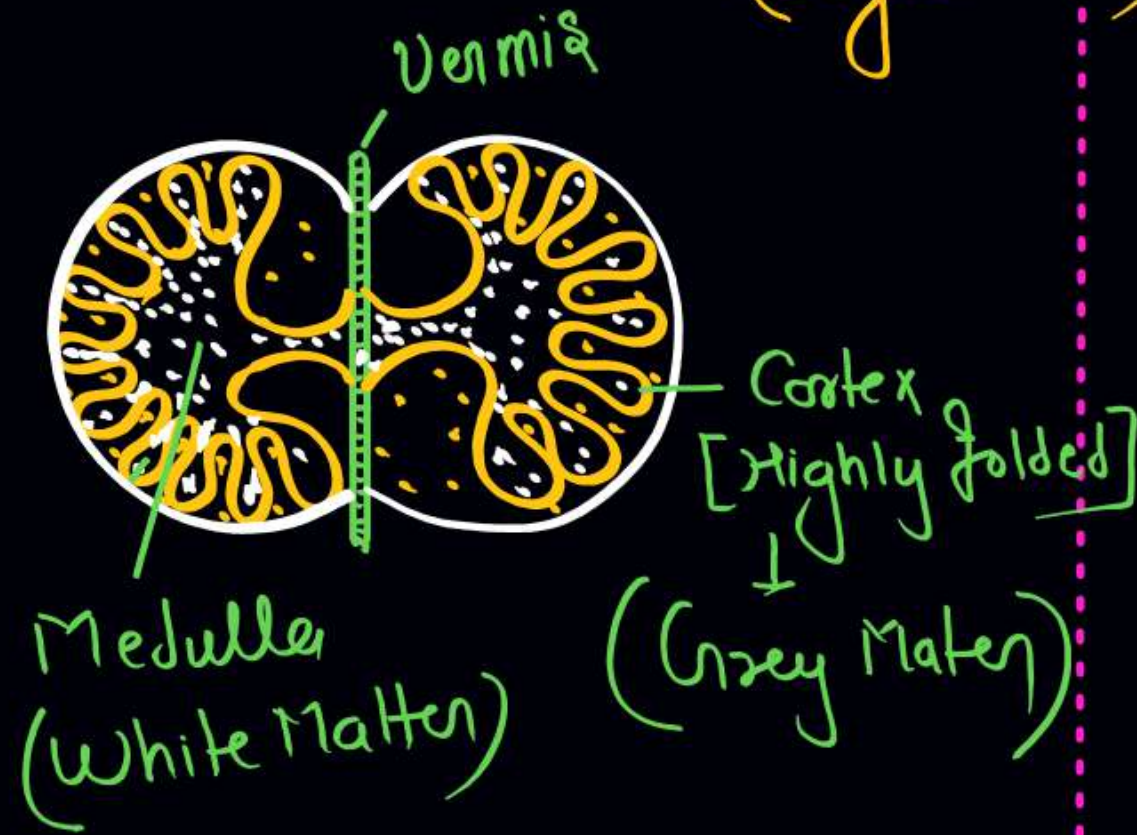
Lobes of Each Hemisphere →



Cerebral Cortex Contain 3 types of Functional area (Nucle)



Cerebellum = 2nd - (Largest Part)



Function = Balancing & Coordination of Voluntary Movement

Medulla oblongata

Control Involuntary action of body System

Control - Respiration

- Cardiovascular Reflex

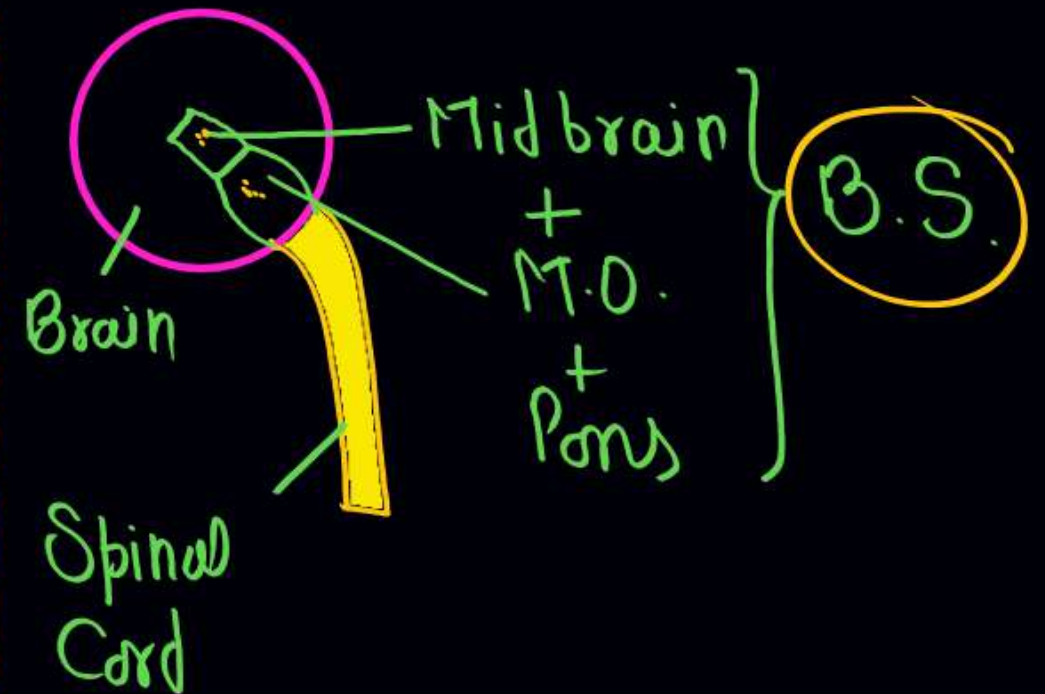
- Gastric - Secretion

Pons

• Bundle of Nerve fibres tracts → Interconnect different Regions of Brain.

• Pneumotaxic Center →
↓ Duration of Inspiration →
↑ Rate of Respiration.

Brainstem



There is a wide diversity in living organisms in our biosphere. Now a question that arises in our minds is: Are all living organisms made of the same chemicals, i.e., elements and compounds?

Living Organism { Animals
Plants.
Microbes.
etc.

- Q. {
- * C, H & O etc. +nt
 - * C & H most abundant
 - * Si Negligible

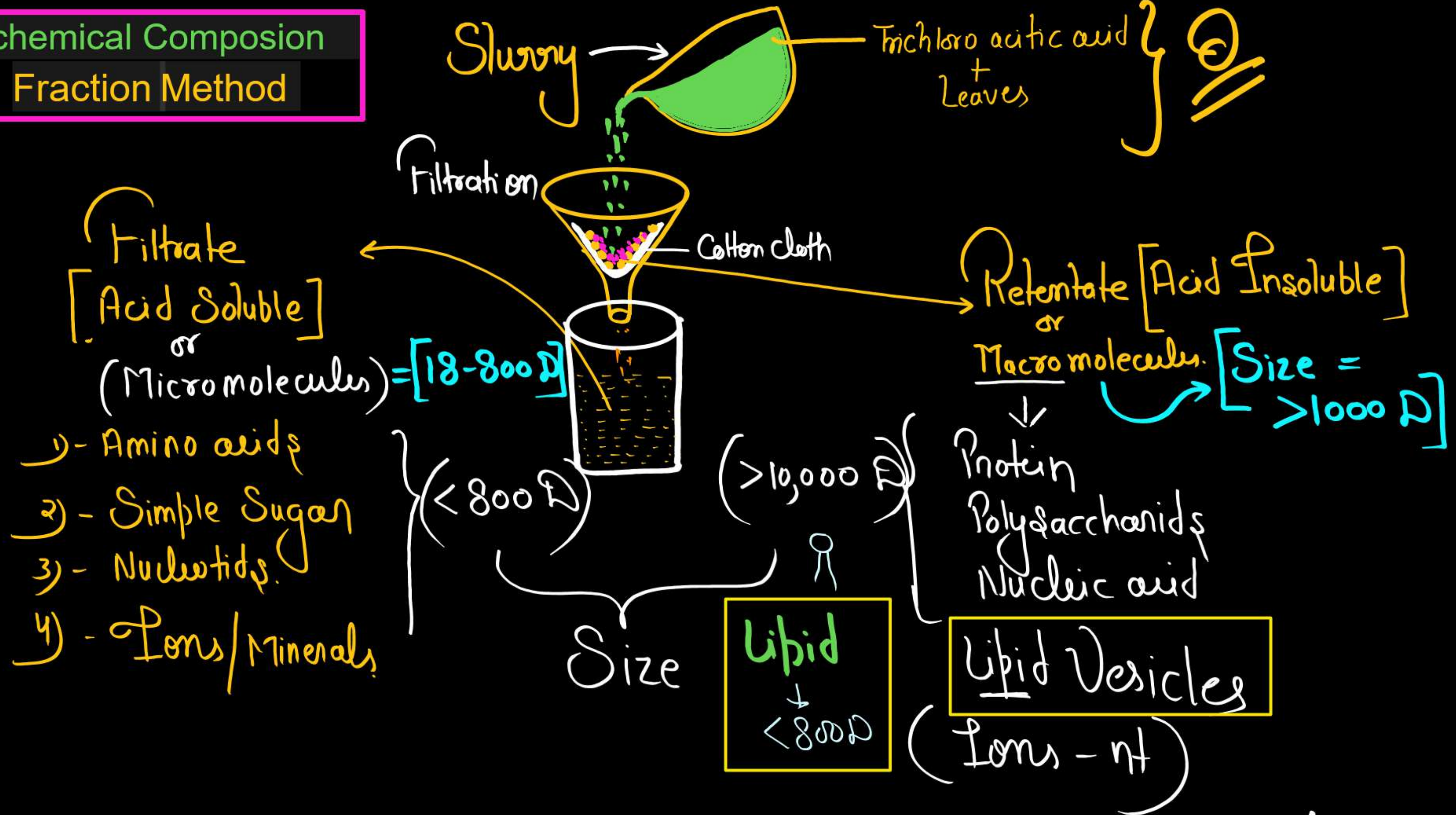
Earth Crust (Non living)

C, H & O etc +nt

Less.

most abundant

chemical Composition Fraction Method



Primary Metabolite.

→ Produce in all living Organism →
Animals, Plant, Bacteria & Fungus

- Involve in Metabolic Reaction
- Identifiable Role = Role in Growth, Development, Reproduction etc.
- Produce in large Quantity during Growth Phase.

eg - Amino acid
Lipid
Carbohydrate
Nucleic acid etc

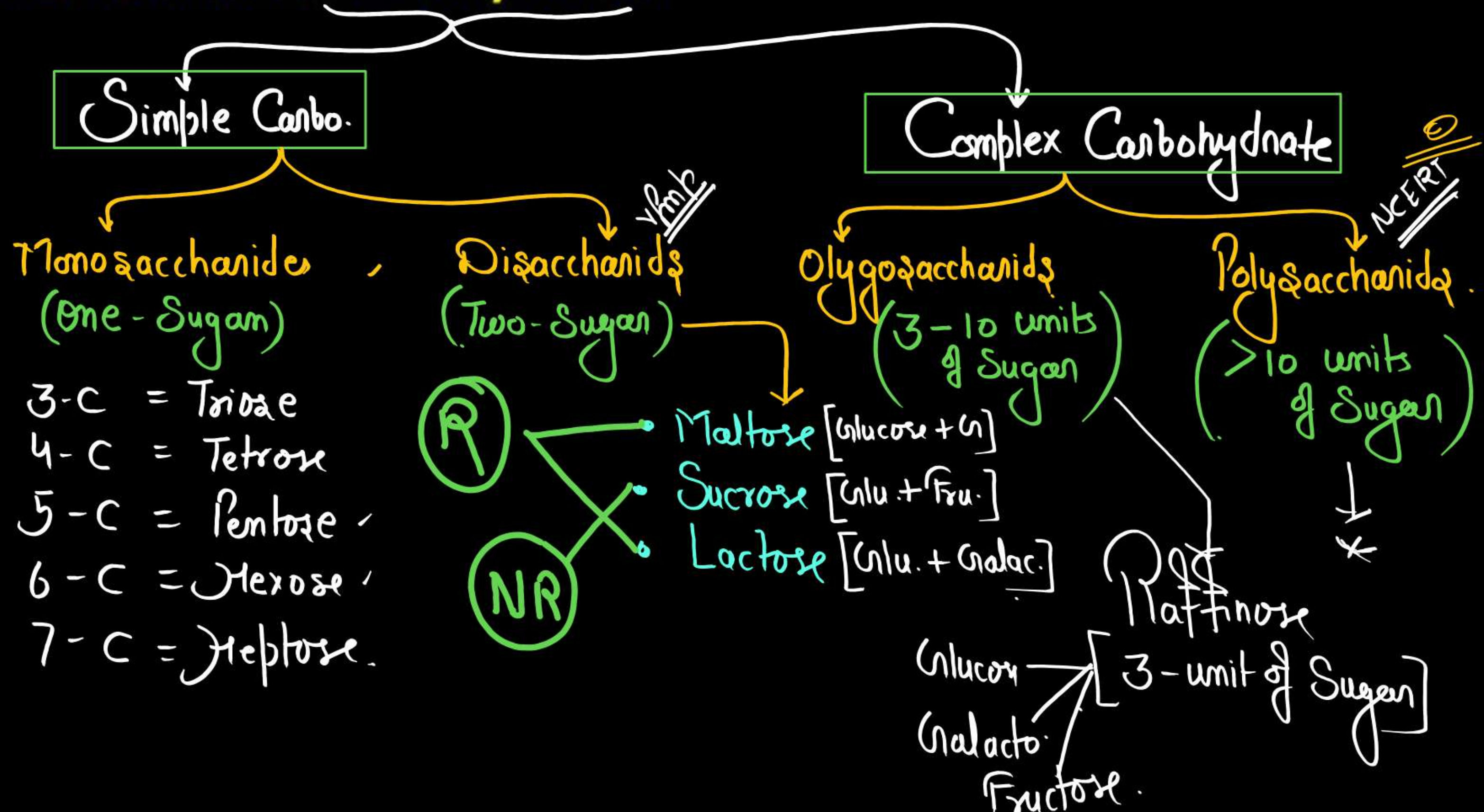
Sec. Metabolite

Produce in only - Plant, Bacteria & Fungus
(not in animals)

TABLE 9.3 Some Secondary Metabolites

Pigments	Carotenoids, Anthocyanins, etc.
Alkaloids	Morphine, Codeine, etc.
Terpenoides	Monoterpenes, Diterpenes etc.
Essential oils	Lemon grass oil, etc.
Toxins	Abrin, Ricin
Lectins	Concanavalin A
Drugs @ //	Vinblastin, curcumin, etc.
Polymeric substances	Rubber, gums, cellulose

Classification of Carbohydrates -



Poly saccharides $\rightarrow$ (Tastless, Water Insoluble)

(A) Homopolysaccharites

Starch $\rightarrow$ Plant Store Food
Monomer unit = α -Glucose.

Branches
+ Helical. Formed by =

Amylose (20%) + Amylopectin (80%)



Helix

Water Soluble
 α (1-4)

Blue colour
with I_2



Branched
Water insoluble
 α (1-4) & α (1-6)

(2) Glycogen -

Monomer = α Glucose

Animal Store Food $\rightarrow$ Liver
Skeletal Muscle

Branched $\rightarrow$ α (1-4)
 α (1-6)

Amylose - nt

(3) Cellulose

Monomer = β -Glucose

→ in cell wall of Plant & Algae

Digested by - Cellulase Enzyme

→ Bacteria & Protozoa

Wood Pulp - Cotton

- Paper formation

unbranched → $\beta(1-4)$

Chitin

Monomer = N-Acetylglucosamine (NAG)
↓
Derivative of Glucose

→ in - Fungal cell wall
- Exoskeleton of Cock.

Unbranched = $\beta(1-4)$

Inulin -

→ Monomer = β -Fructose

→ Store food → in root of Dahlia Plant

B) Hetero-polysaccharide

② Pepidoglycan

Hyaluronic acid

→ +nt in Connective Tissue

• +nt in = Bacterial Wall

• Polymer of → NAG + NAM

N-Acetyl Glucosamine

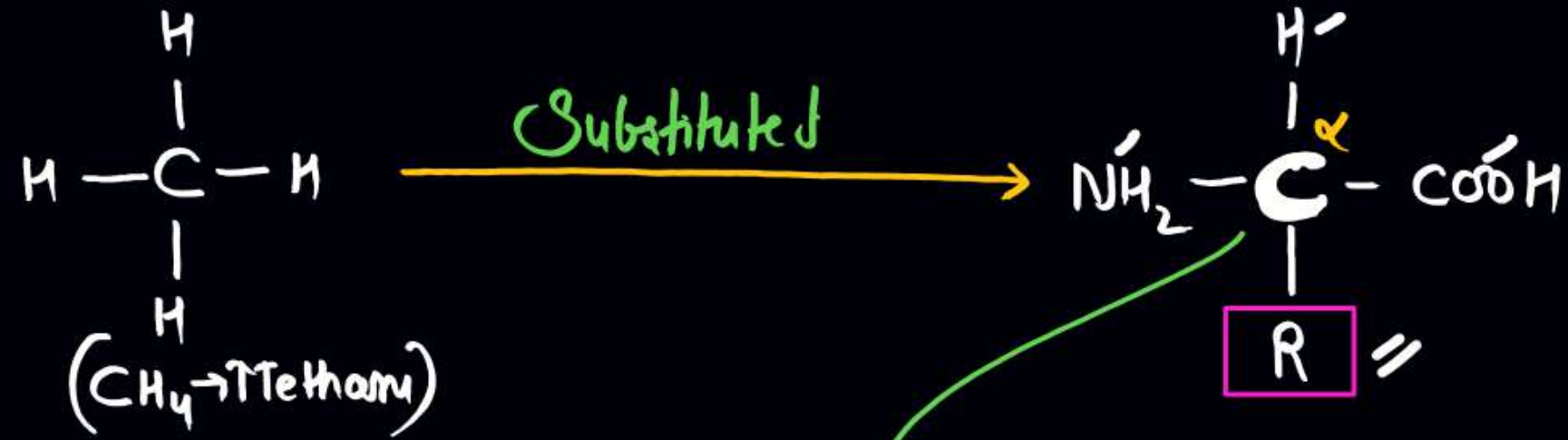
N-Acetyl Muramic acid

• linkage → $\beta(1-4)$ ②

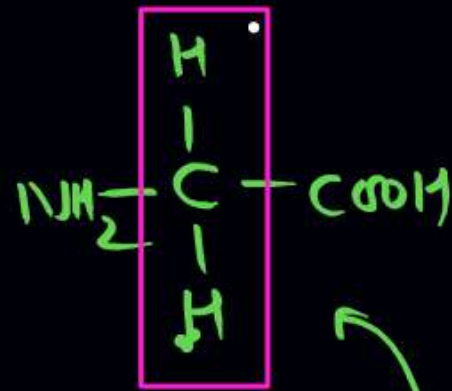
✓ prob

Amino Acids

- Mole.wt < 800 D
- +ve in acid soluble pool
- Synthesized by cells



[Chiral Carbon]



[Optically active]

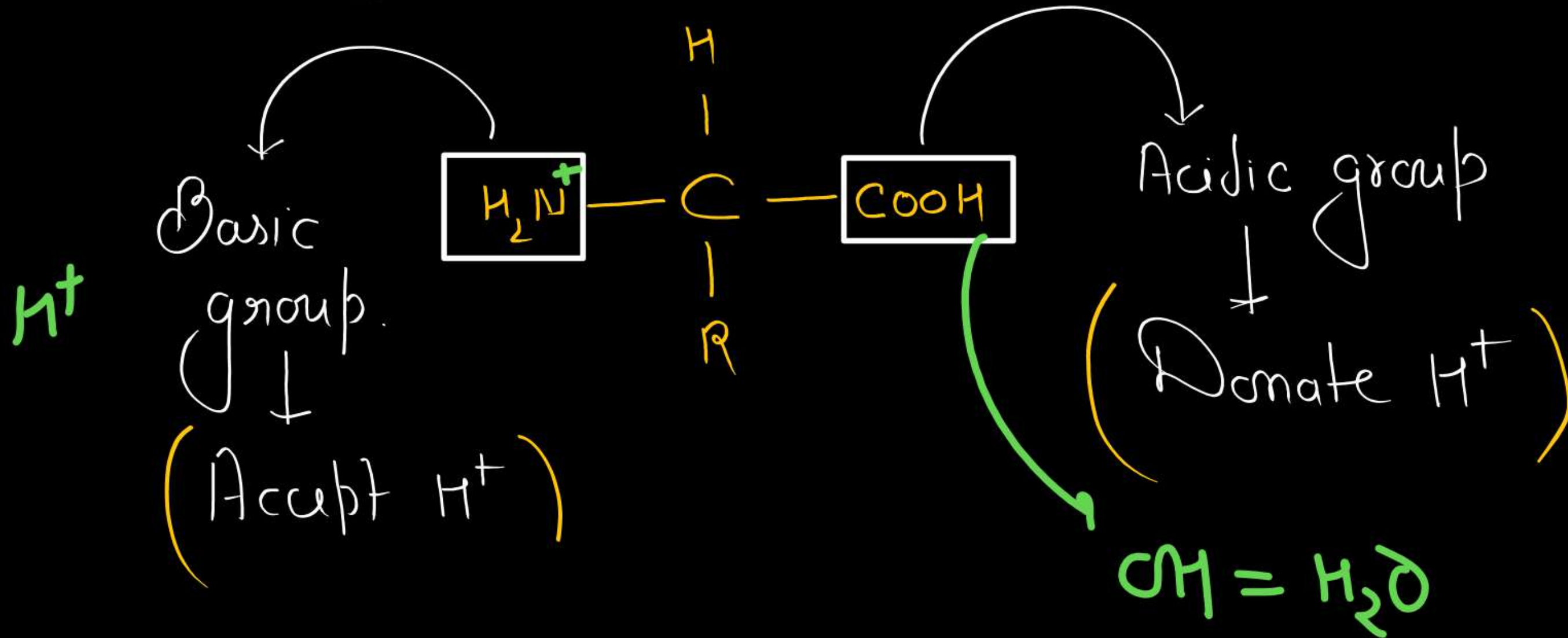
Exapt = Glycine

(Chiral Carbon
-H)

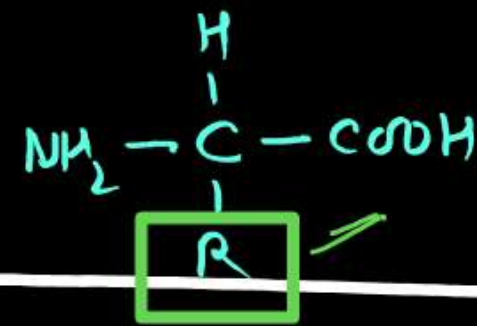
optically inactive

Zwitterionic form of amino acid

A particular property of amino acids is the ionizable nature of -NH_2 and -COOH groups. Hence in solutions of different pH, the structure of amino acids changes



Classification of amino acids



✓mp

Acidic A.A.



-ve charge

(Polar)

Water Soluble
Hydrophilic

eg → Glutamic acid - 5-C
Aspartic acid - 4-C

Basic A.A.



+ve charge

(Polar)

Water Soluble
Hydrophilic

eg → lysine - 2N
Histidine - 3N
Arginine - 4N

Neutral A.A.



No - charge

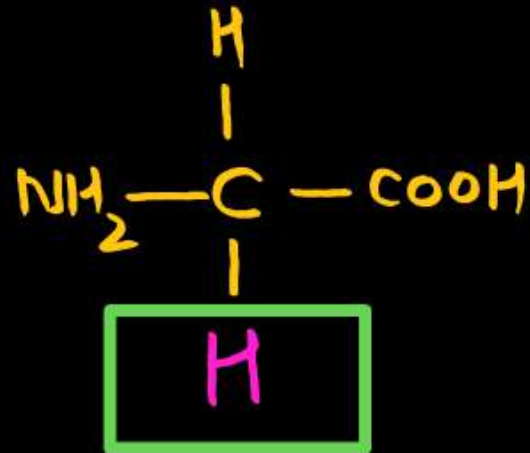
(Non polar)

Water insoluble
Hydrophobic

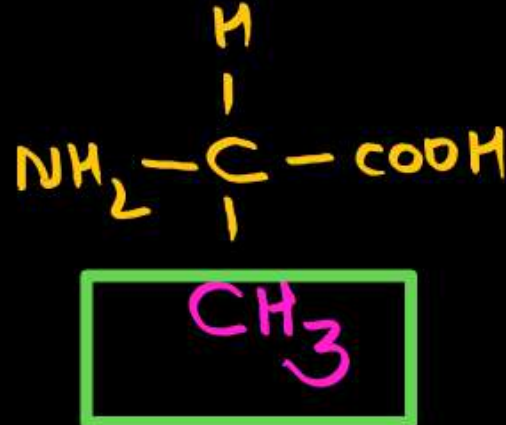
eg - Glycine
- Alanine
- Valine
- Leucine,
Isoleucine

Structure of Amino Acids

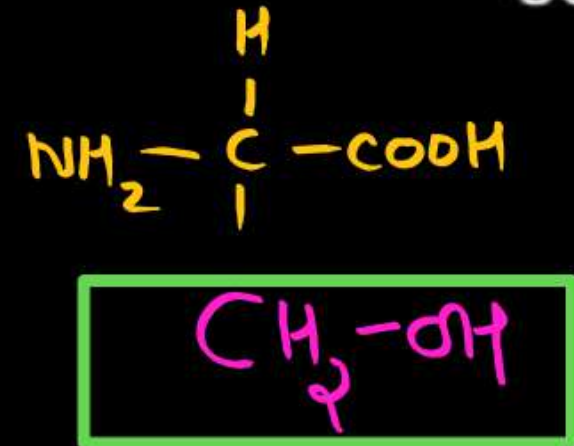
Glycine (Simple Str.)



Alanine

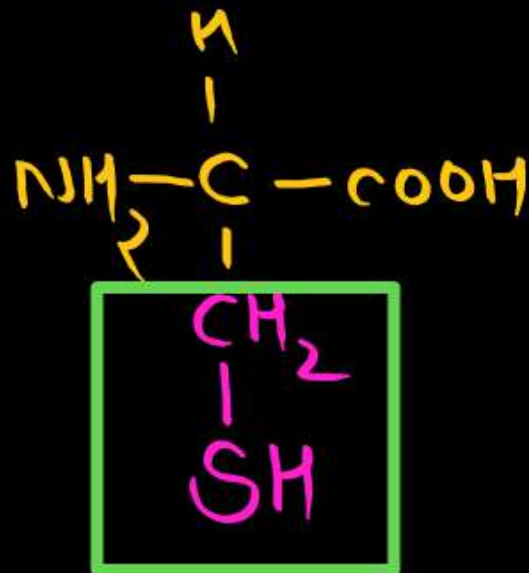


Serine

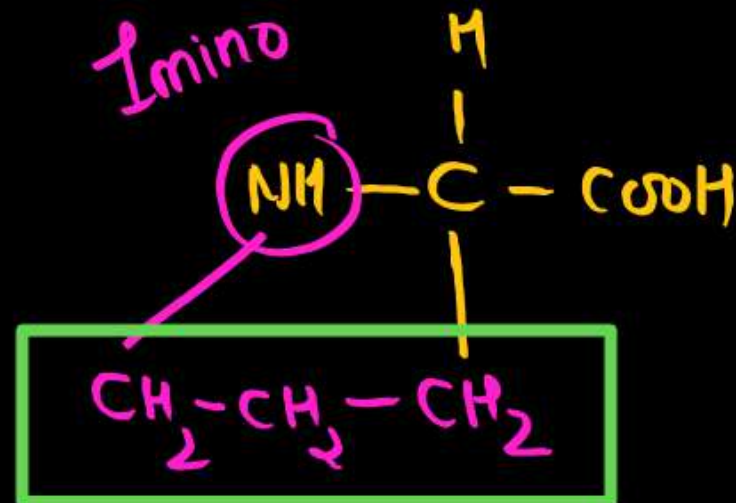


Complex Str.

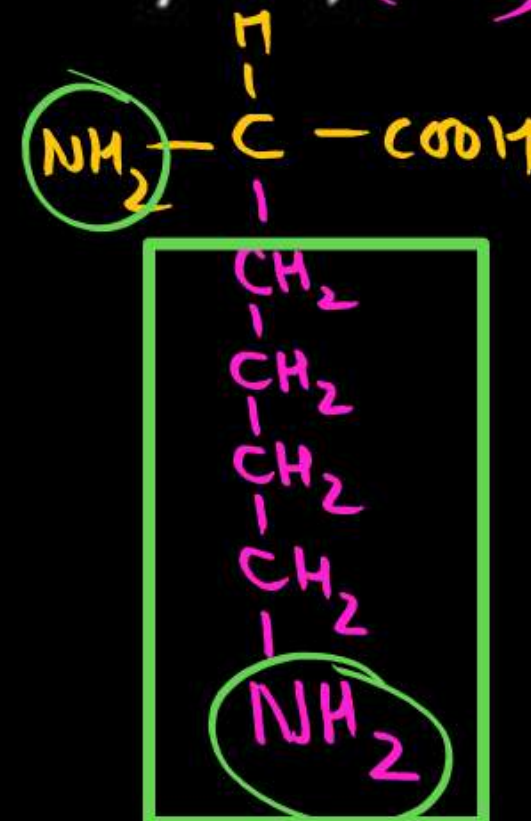
Cysteine, (S)



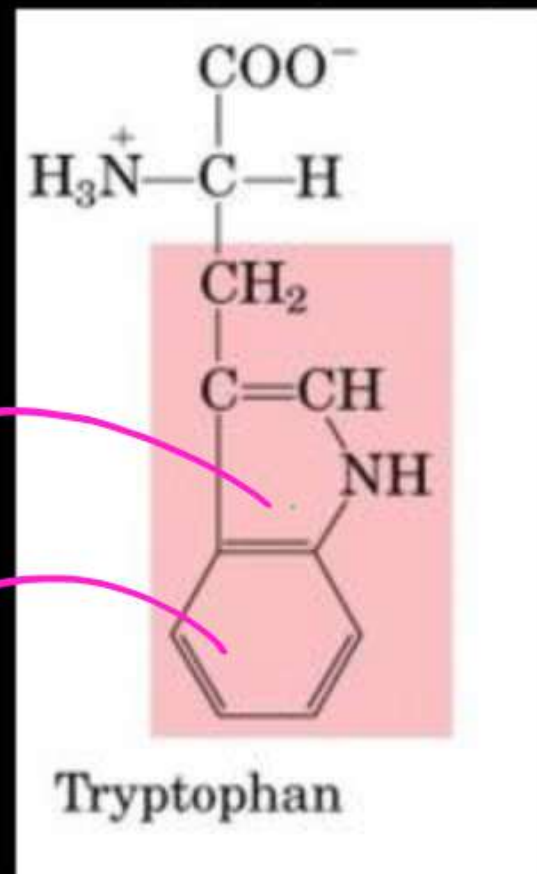
Proline,



Lysine, (+ve)



Tryptophan,



Protein

Structure

Bonding

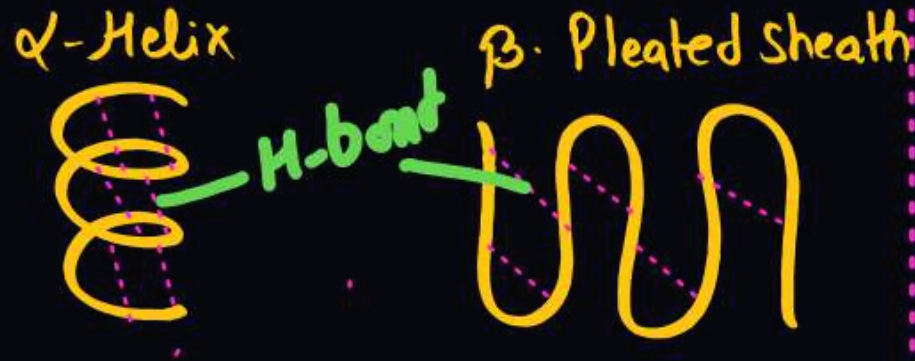
Example

Pri. Structure



Peptide bonds

Sec. Structure



Peptides + H-bond

α -Helix = Keratin
 β -Pleated Sheath = Silk

Ter. Structure



Peptide, H-bond, Disulphide
Ionic, Hydrophobic,
Vanderwaals etc.

Enzymes + Myoglobin

Qua. Structure



Non Co-Valeant
+
Disulphide

Insulin, Antibody (4)
Haemoglobin (4)

Function of Protein-

TABLE 9.5 Some Proteins and their Functions

Protein	Functions
Collagen	Intercellular ground substance
Trypsin	Enzyme
Insulin	Hormone
Antibody	Fights infectious agents
Receptor	Sensory reception (smell, taste, hormone, etc.)
GLUT-4	Enables glucose transport into cells

Collagen → Most abundant animal protein
→ +nt in Connective tissue. → Tendon, Ligament, Acellar ek Bones etc.

Tissue.

→ Triple Helix } Structure

→ Vit. C play imp. role in Formation



Ru.BisCO → Most abundant Plant Protein. in whole of the biosphere

ENZYMES (Bio catalyst) = Stru. = 3-D

vvvimp

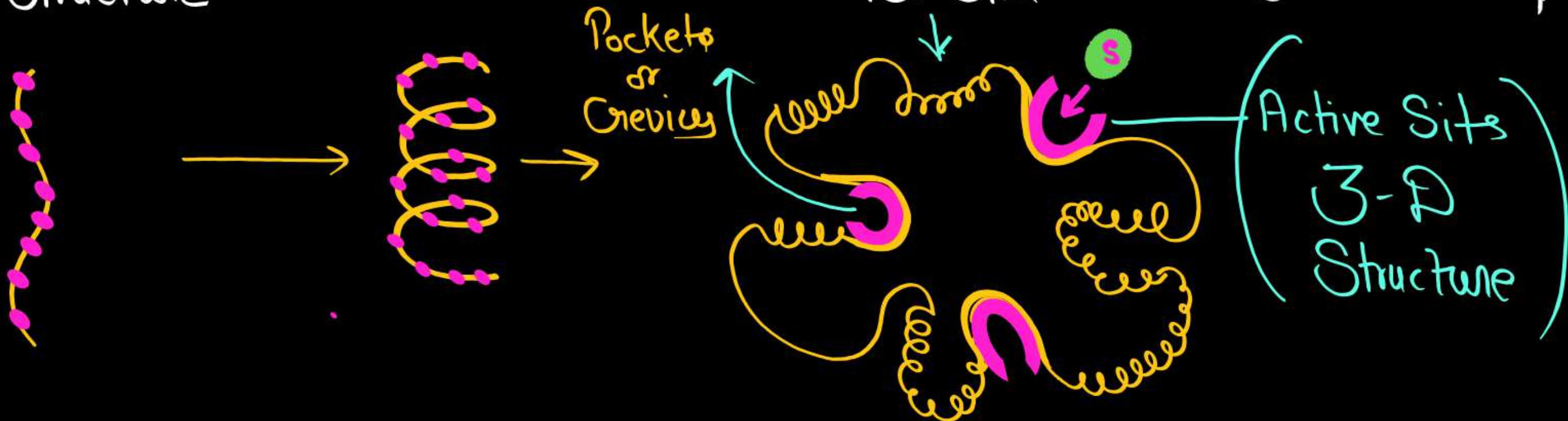
All Enzymes are protein.

Except $\left\{ \begin{array}{l} \text{Ribozyme} \\ \text{Ribonuclease} \end{array} \right\}$

Small Segments of RNA, which have Catalytic Activity

Structure of Enzyme.

Pri. Structure $\rightsquigarrow$ Sec. Stru. $\rightsquigarrow$ Ter. Stru.



- The reaction that occur in presence of catalyst proceed at very fast -



In absence of Enzyme
{ 200 mole. of H_2CO_3 }
formed / Hour

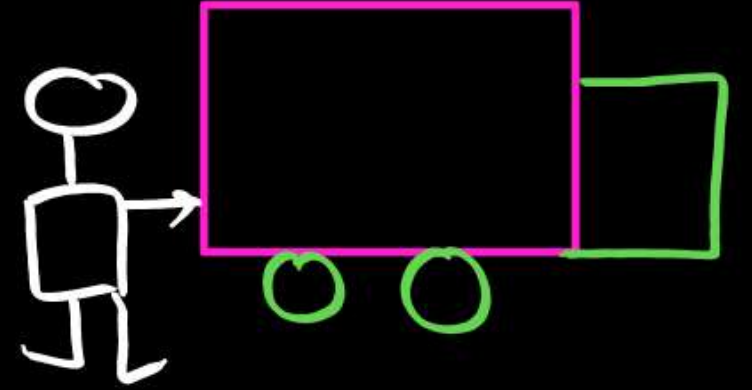
In presence of Enzyme.
eg - Carbonic anhydrase.

6,00,000 mole. of H_2CO_3
formed / Sec.

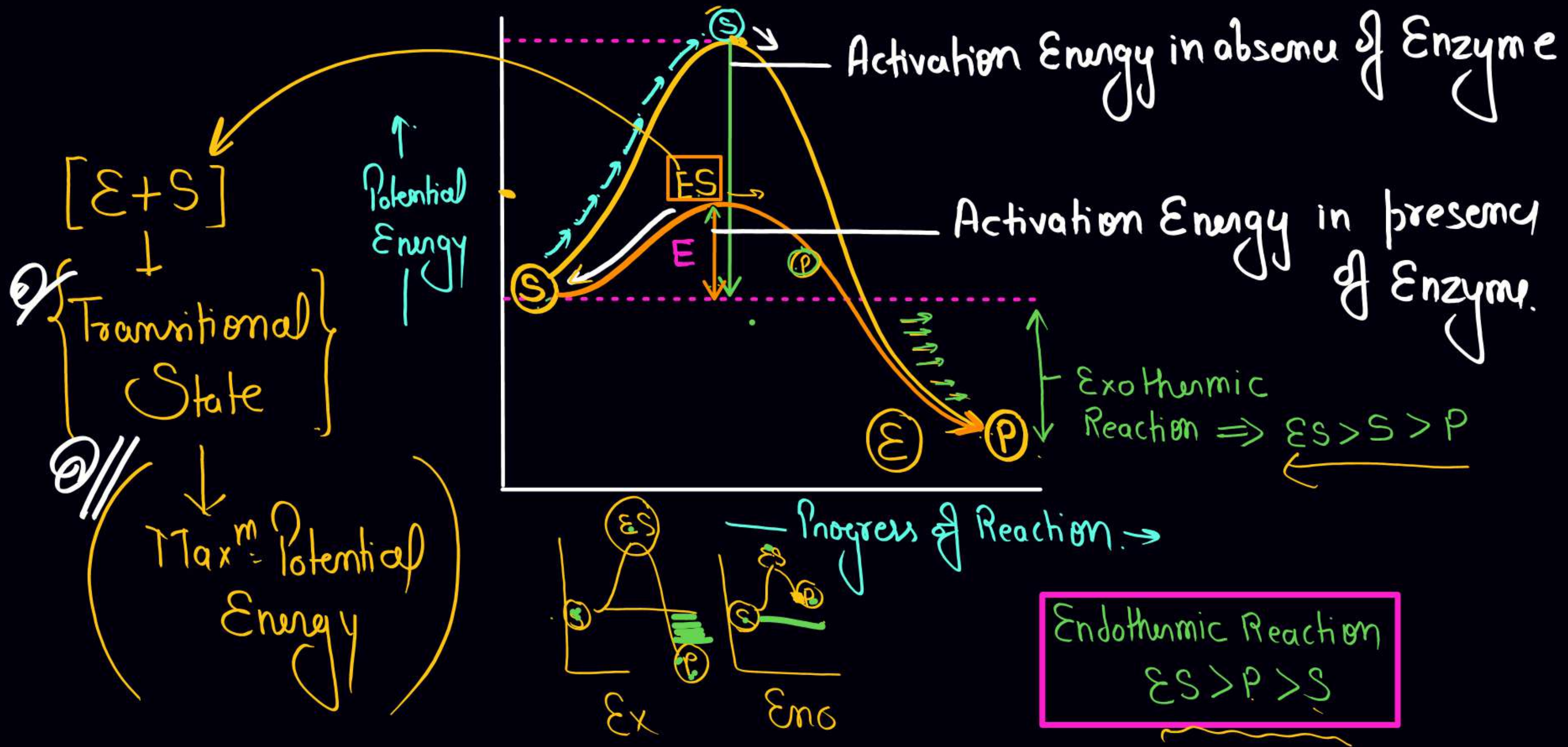
[Reaction 10 million Times]

How do Enzymes bring about such High Rates of Chemical Conversions?

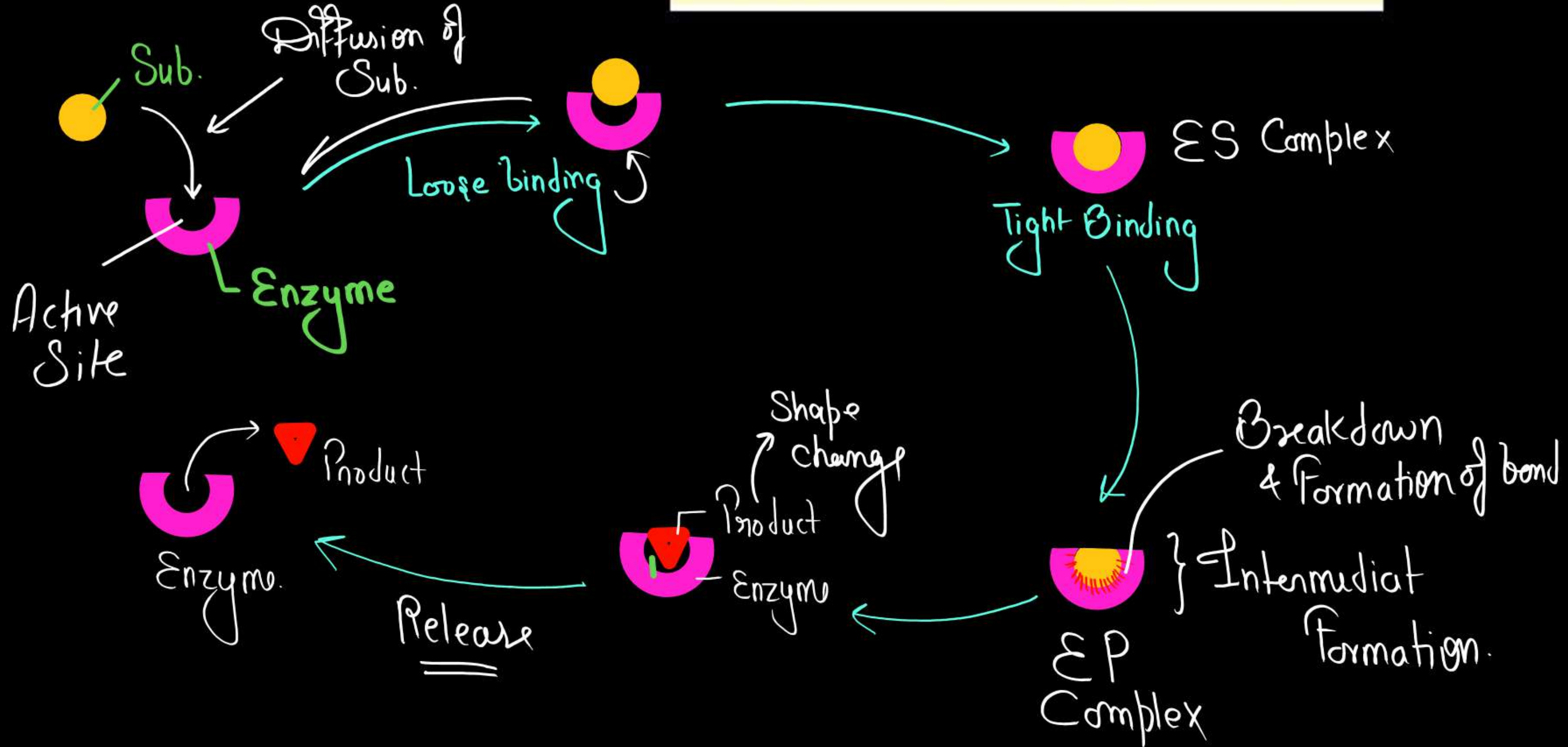
* Enzyme → Lower the activation Energy
Which require for reaction



Activation Energy } Energy require for
initiation of Chemical Reaction

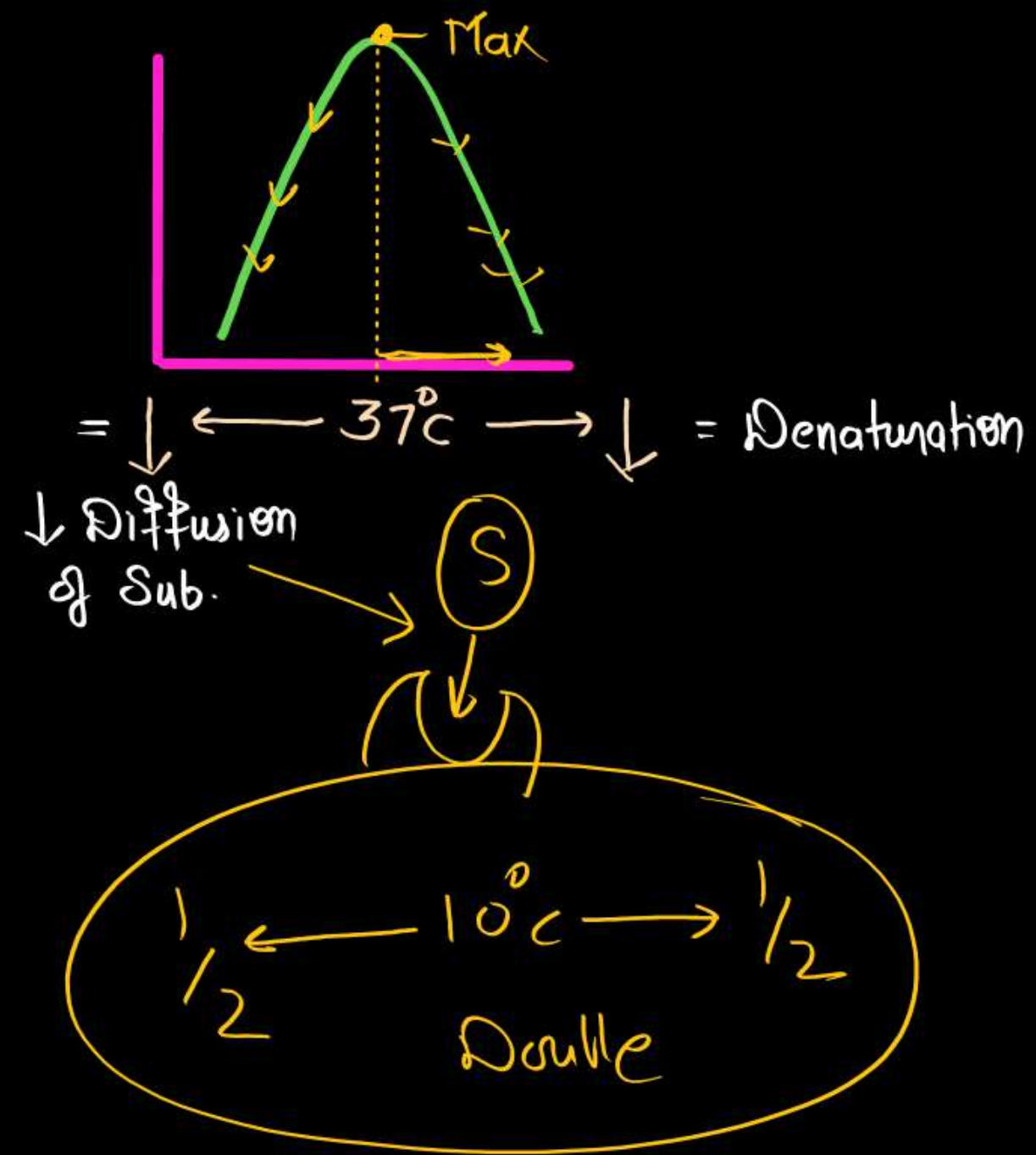
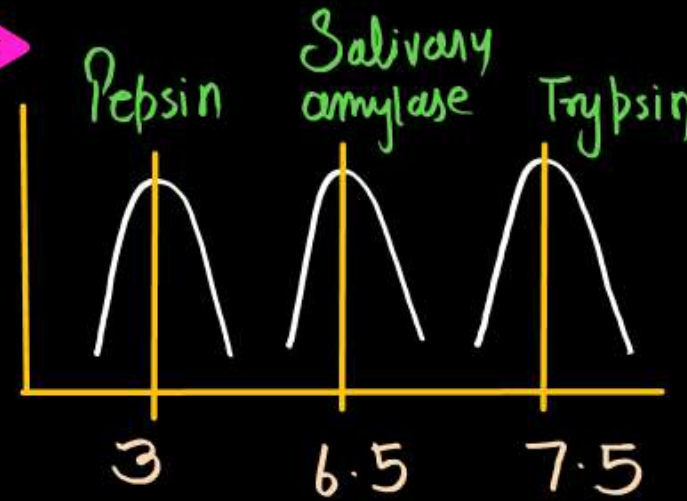


Nature of Enzyme Action -

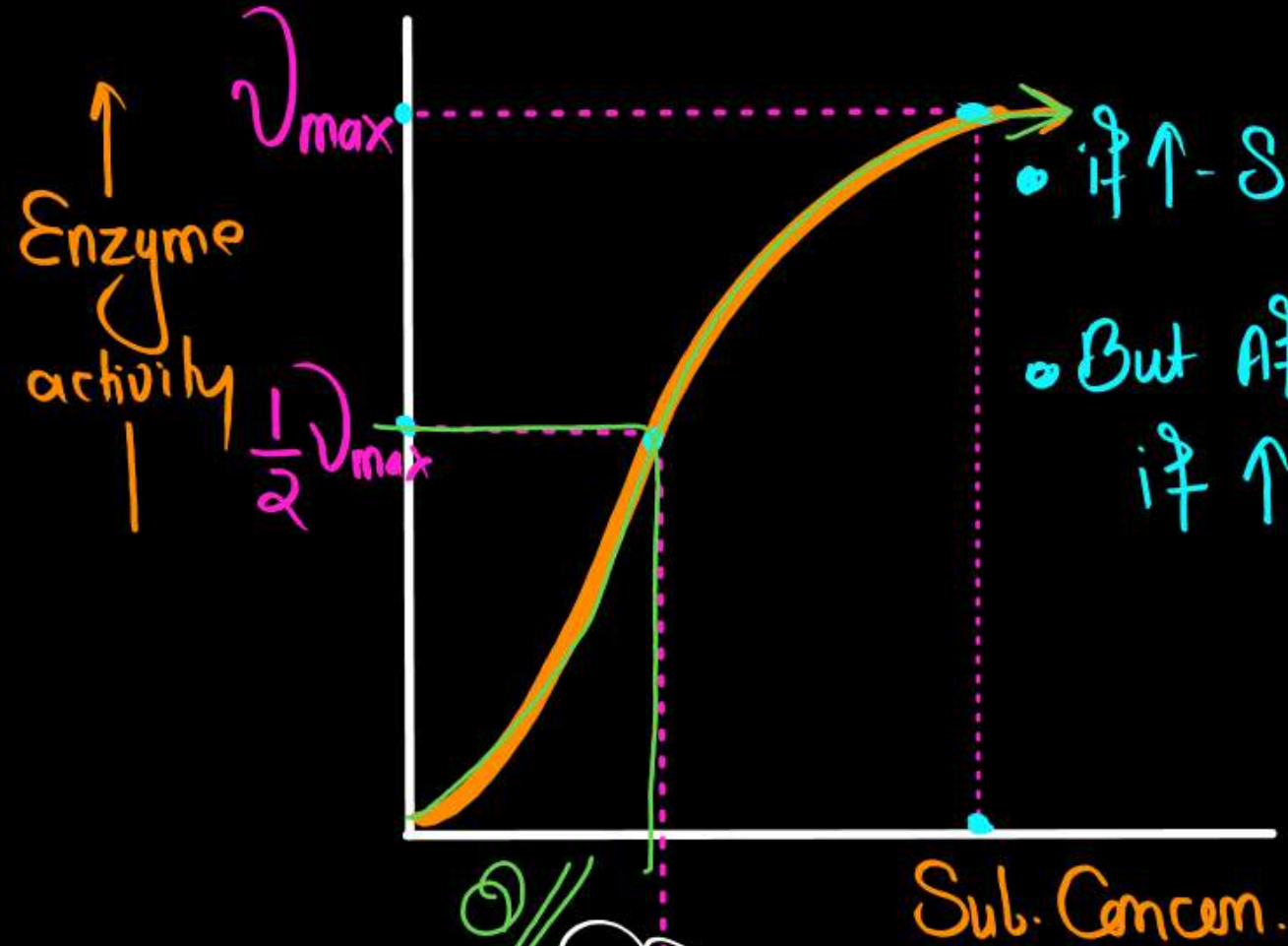


Factors Affecting Enzyme Activity

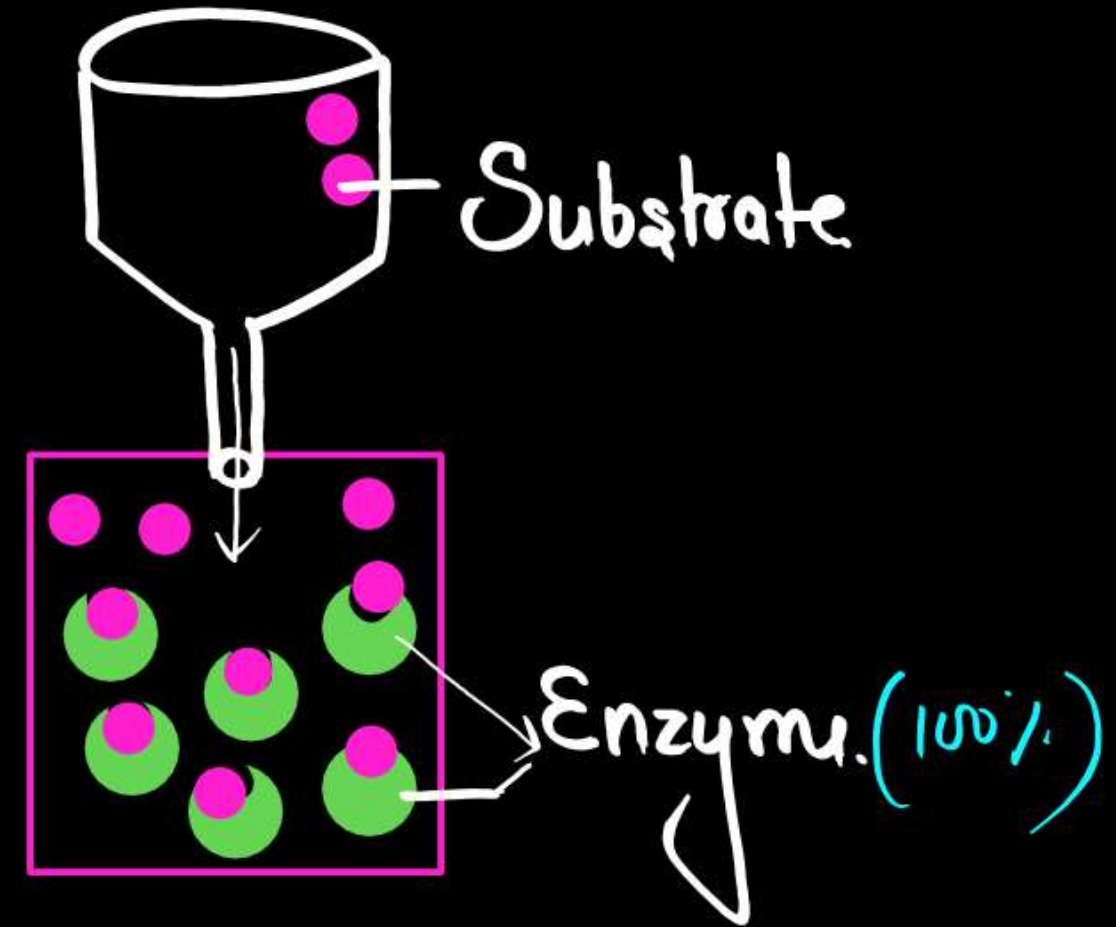
- Temperature →
- pH →
- Substrate concentration ✓
- Inhibitors



(3) Substrate concentration



- if $\uparrow$ Sub. Cons = $\uparrow$ Rate of Reaction.
- But After 100% Saturation
if $\uparrow$ Sub. Cons = Rate Cons.



K_m = Concentration of Substrate
at Velocity of Enzyme Reaction = $\frac{1}{2}$ of V_{max}

Competitive

Reversible Binding
Common binding Site $\leftarrow \begin{matrix} S \\ I \end{matrix}$

In Presence of Inhibitor -

V_{max} = Same
Sub Cons. $= \uparrow = K_m = \uparrow$

eg - Methanol $\leftrightarrow$ Ethanol vvv
Succinic acid $\leftrightarrow$ Malonate

Non competitive

Irreversible Binding
Seperate binding Site $\leftarrow \begin{matrix} S \\ I \end{matrix}$

In Presence of Inhibitor

$V_{max} = \downarrow$
 $K_m = \text{Cons.}$

eg - Cyanide.

Allosteric modulation

Reversible binding
Seperate binding Site

K_m = Not follow

Product act as Inhibitor

eg - In Glycolysis
Hexokinase.

Classification and Nomenclature of Enzymes

Enzymes are divided into 6 classes each with 4-13 subclasses and named accordingly by a four-digit number

- 1) - oxidoreductase
- 2) - Transferase.
- 3) - Hydrolase
- 4) - Lyase
- 5) - Isomerase
- 6) - Ligase

OTH-LIL

1st-digit
Class
(6)

6
1:1:1:1

E.C. No.

1st = Class

2nd = Sub class

3rd = Sub-Sub Class

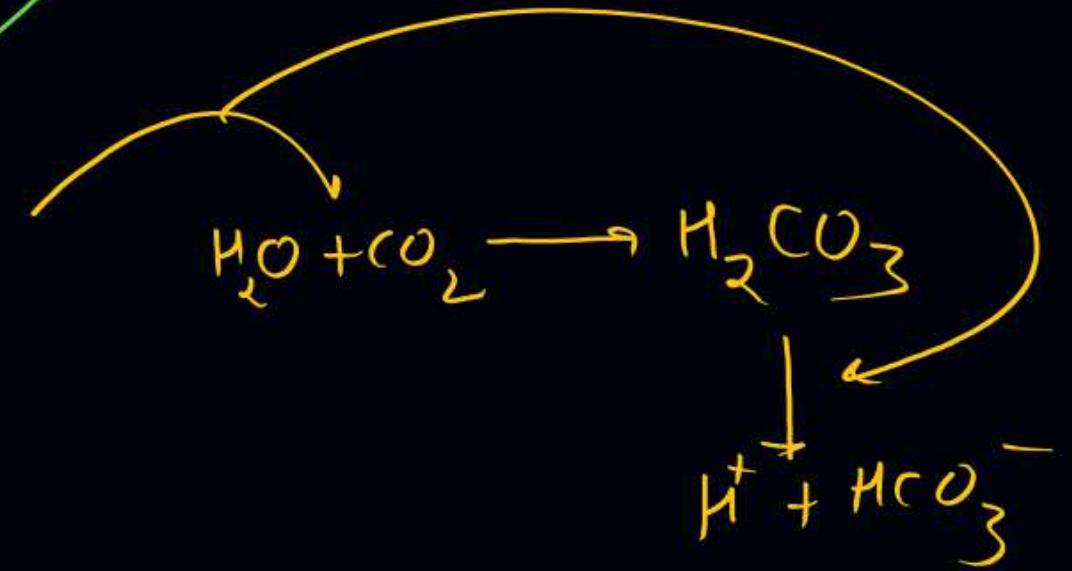
4th = Ser. No.

Class-1 Oxidoreductase/Dehydrogenase = Dehydrogenase, Reductase, Peroxydase

Class-2 Transferases = Transaminase, Hexokinase

Class-3 Hydrolases = All Digestive Enzymes ✓

Class-4 Lyases = Carbonic anhydrase



Class-5 Isomerases = Isomerase ✓

Class-6 Lygase = Ty-ligase

Types of Enzyme

Simple Enzyme

↓
Formed by only
Protein
[Active form]

② Conjugate Enzyme. Imp

Protein
[Apo enzyme] + ③ Non Protein
[Inactive] [Co-factor]

(Holoenzyme)

Co-factors

Inorganic [Metal ions]
[Bind with - Coordination bond]

Zn^{+2} → Carbonic anhydrase
→ Carboxy peptidase } Prob

Cl^- → Salivary amylase

Fe^{2+} → Cytochrome oxidase
→ Peroxydase

Mg^{2+} → ATPase
→ Hexokinase

Cu^{++} → Cytochrome oxidase

Organic

② Co-Enzyme [Non-Covalent] (Loose)
Prosthetic group [Covalent] (Strong)

Organic

Co-enzyme (Reversible binding)

[loosely bind $\rightarrow$ Non Covalent Interaction $\rightarrow$ Detach after Reaction]

* Derivative of Vit.

Riboflavin Vit.
(B₂)

F.A.D (Flavin adenine
dinucleotide)

Succinic Dehydrogenase

Thiamine Vit
B₁

T.P.P (Thiamine Pyro-
phosphate)

Pyruvic Dehydrogenase

Niacin Vit
B₃

NAD⁺ (Nicotinamide
Adenosine Dinucleotide)

Lactate Dehydrogenase

Co-Enzyme

Enzymes

~~v/vmp~~

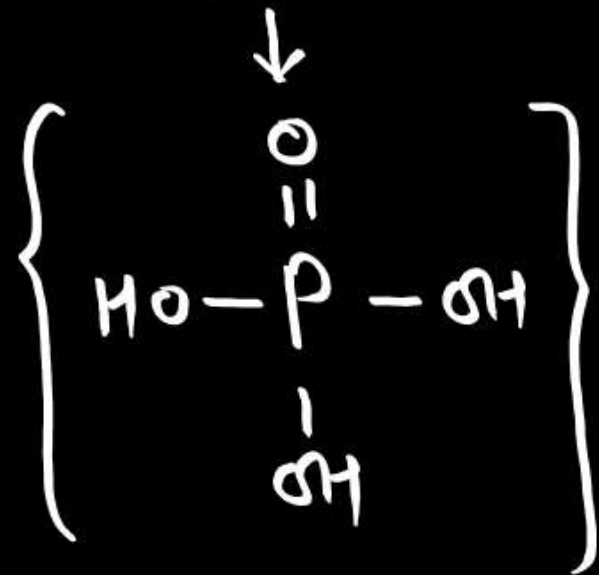
Prosthetic group

[Bind with - Covalent Interaction]

eg. Heme in Catalase & Peroxy-dase

Nucleic acid → Polymer of Nucleotides = unit of DNA/RNA

① Phosphate
(H_3PO_4)



② Pentose Sugar (5-C)

Ribose
↓
(RNA)

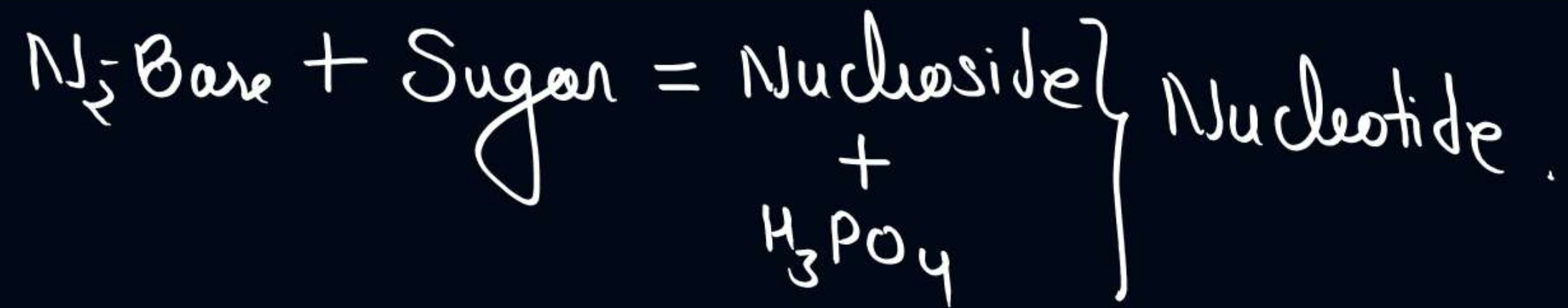
Deoxyribose
↓
(DNA)

③ N₂-Base

Purine
↓
(A+G)
Double Ring

Pyrimidine
↓
(C T U)
Single Ring

Difference b/w Nucleotide & Nucleoside.



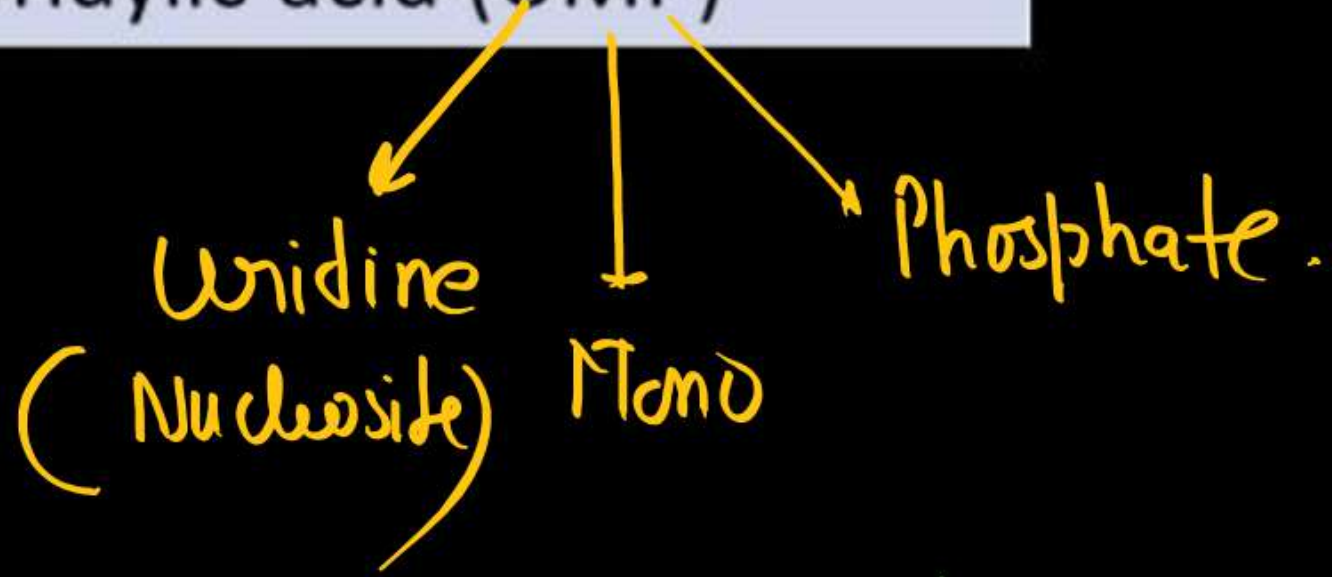
8/11/2022

Nitrogenous Base	Nucleoside	Nucleotide
Adenine (A) + S =	Adenosine + P =	Adenylic acid (AMP)
Guanine (G) + S =	Guanosine + P =	Guanylic acid (GMP)
Cytosine (C) + S =	Cytidine + P =	Cytidylic acid (CMP)
Thymine (T) + S =	Thymidine + P =	Thymidylic acid (TMP)
Uracil (U) + S =	Uridine + P =	Uridylic acid (UMP)

} Purines

} Pyrimidines

UMP



Types of Fatty Acids

Saturated

- ✓ Double bond -nt
- ✓ Solid at room temp.
- ✓ High M.P.

Mostly +nt in animal fat

eg - Palmitic acid
Stearic acid

Non-Essential

unsaturated

- ✓ = +nt
- ✓ = Liquid.
- ✓ = Low.

= Mostly +nt in Plant.

eg - oleic acid } MUFA

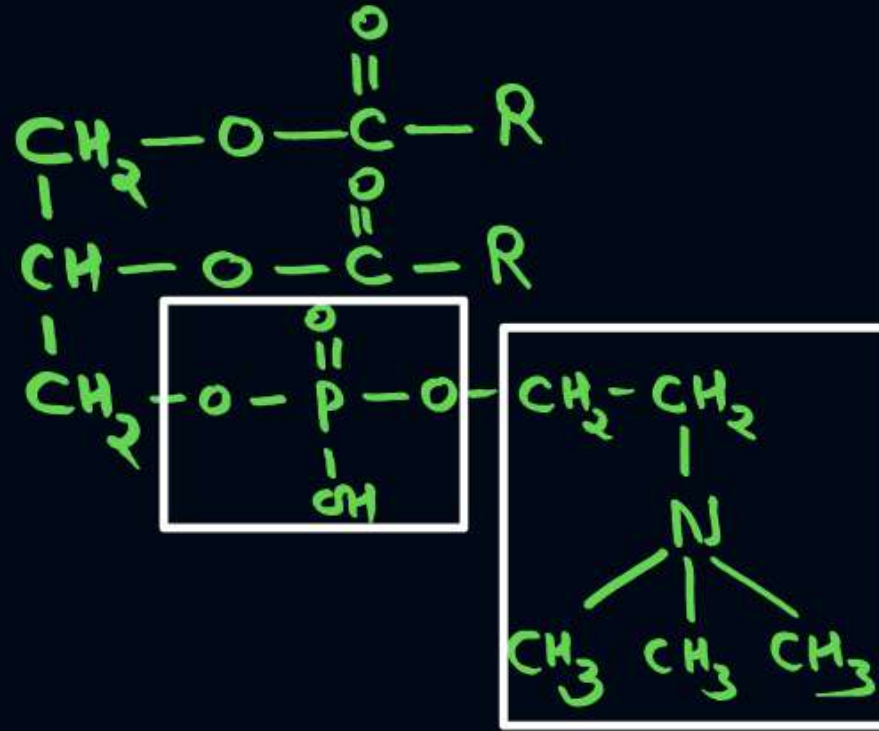
linoleic acid
linolenic acid
Arachidonic acid } PUFA (Essential)

~~high temp~~

Phospholipid

(Leathine)

vvvvfmb



Glycolipid

Carbohydrate

eg → Glycosphingo lipids
(+nt in Neurons)

- Fatty acids ✓
- Alcohol = Sphingosin
- Sugar = Galactose

Lipoprotein

→ Chylomicron [Protein + lipid]

Chromolipid

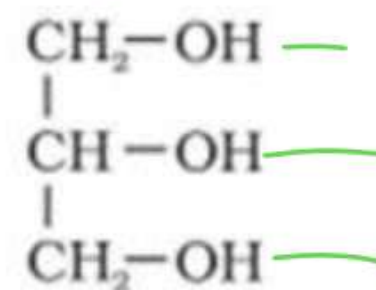
Pigment Containing Lipid

→ Carotene (Vit-A)

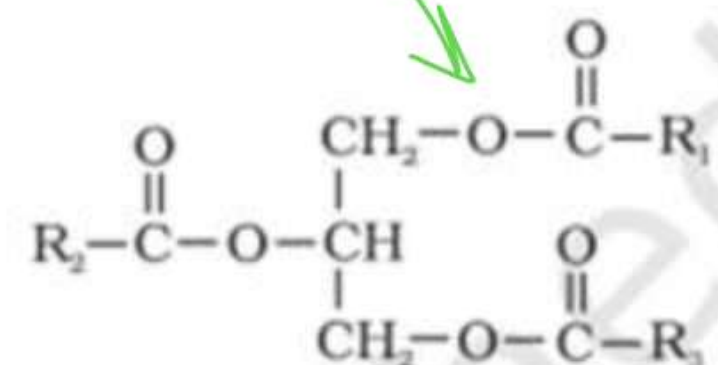
vvv imp



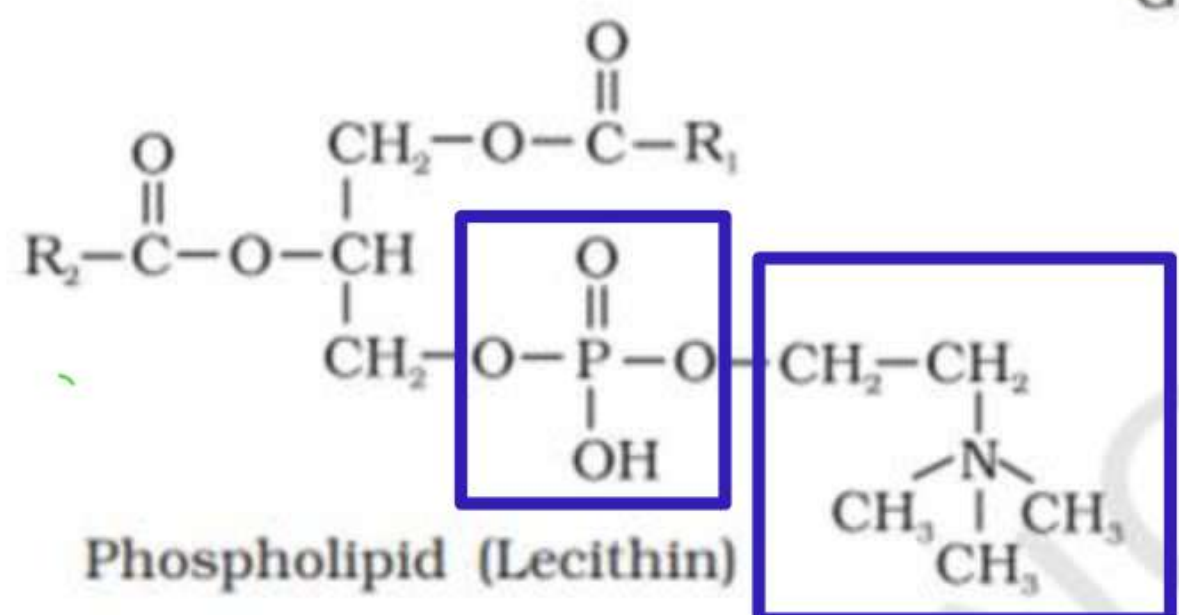
Fatty acid
(Palmitic acid)



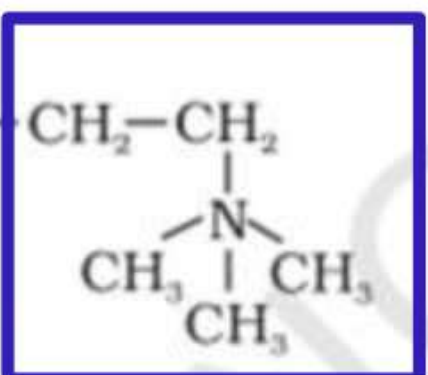
Glycerol



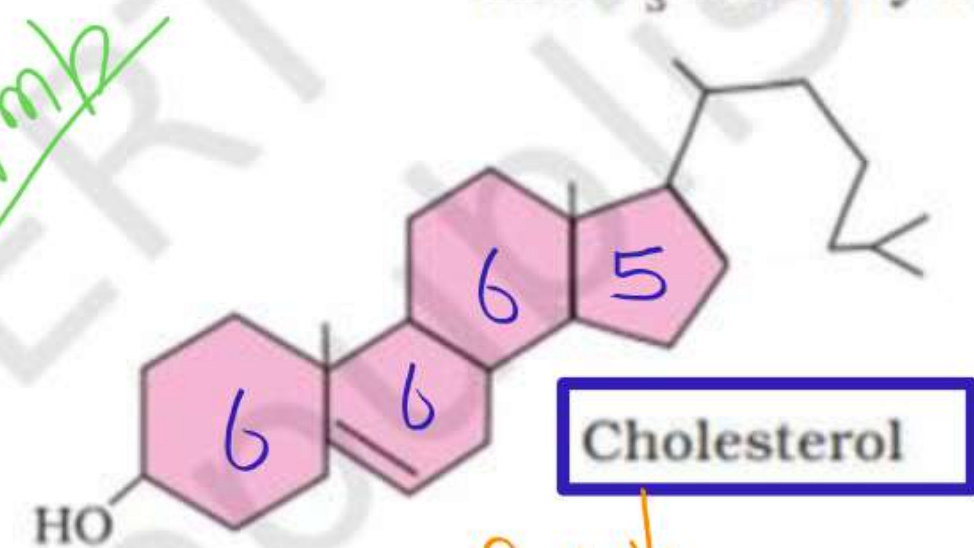
Triglyceride (R_1 , R_2
and R_3 are fatty acids)



Phospholipid (Lecithin)



Fats and oils (lipids)



- Produce in Liver
- Most abundant **Sterooids** in Animals
- Precursor of Hormones
- Fat like Property